CAT MATHEMATICS CHAPTER-1

TOPIC: NUMBER SYSTEM

(1) $i^{14}+i^{22}+i^{30}+i^{38}=?$

(a) 1
(b) -1
(c) -4
(d) 0

(2) $\sqrt{n}$ is an irrational number for $n \in$ _____

(a) N
(b) O
(c) P
(d) E

(3) $\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3=x+iy$, then $(x,y)=$

(a) $(0,2)$
(b) $(-2,0)$
(c) $(0,-2)$
(d) $(2,0)$

(4) **Which one must be false for two complex numbers Z_1 and Z_2?**

(a) $|Z_1|<|Z_2|$
(b) $Z_1<Z_2$
(c) $|Z_1-Z_2|\leq|Z_1|+|Z_2|$
(d) $|Z_1+Z_2|>0$

(5) **Which of the following does not obey closure law w.r.t to addition?**

(a) R
(b) N
(c) Q
(d) E

TAHIR

(6) $\text{img}\left[(2,3)(i,0)\right]=$

(a) 0
(b) 1
(c) 2
(d) 3

If $b<a<0$, then;

(a) $-b<-a$
(b) $b>a$
(c) $\frac{-1}{a}>\frac{-1}{b}$
(d) $\frac{-1}{a}<\frac{-1}{b}$

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(8) Conjugate of i lies on;

(a) $\overrightarrow{OX}$ (b) $\overrightarrow{OX'}$

(c) $\overrightarrow{OY}$ (d) $\overrightarrow{OY'}$

(9) Which of the following can be argument of a pure imaginary complex number?

(a) 0° (b) 240°

(c) 180° (d) 270°

(10) $\left[\cos\left(\frac{2n\pi}{3}\right)+i\sin\left(\frac{2n\pi}{3}\right)\right]^3 =$

(a) 0 (b) 1

(c) i (d) $-i$

(11) If $\frac{1}{z}=i-7$, then $|Z|=$

(a) $\sqrt{7}$ (b) $\sqrt{50}$

(c) $\frac{1}{\sqrt{7}}$ (d) $\frac{1}{\sqrt{50}}$

(12) If $Z_1=9+12i$ and $|Z_2|=5$, then least value of $|Z_1+Z_2|$ is;

(a) 5 (b) 10

(c) 15 (d) 20

(13) $(-1)^{\frac{-51}{2}} =$

(a) 1 (b) -1

(c) i (d) $-i$

(14) For any real number $a>0$, which one is not true?

(a) $\frac{1}{a}<0$ (b) $\frac{1}{a}>0$

(c) $-\frac{1}{a}<0$ (d) $\frac{1}{a}\geq 0$

(15) For any complex number Z, if $|Z+1|\leq 4$, then maximum value of $|Z+3|$ is;

(a) 2 (b) 4

(c) 6 (d) 8

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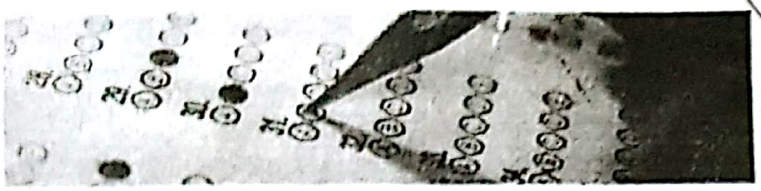

(16) $\frac{22}{\sqrt{207}}$ is

(a) Rational number
(b) Irrational number
(c) Imaginary
(d) All of these

(17) $(\cos 80° + i\sin 80°)^2 \div (\cos 10° + i\sin 10°)^7 =$

(a) 1
(b) -1
(c) i
(d) $-i$

(18) $\cos 20° + i\sin 20° =$

(a) $\cos 80° + i\sin 80°$
(b) $\sin 70° + i\cos 70°$
(c) $\cos 40° + i\sin 40°$
(d) $\sin 40° + i\cos 40°$

(19) If $\arg(Z) = 30°$, then $\arg(|Z|) =$

(a) $-30°$
(b) $30°$
(c) $60°$
(d) $0°$

arg $\sqrt{30^2}$
arg
arg $180° = 0$

(20) **Polar form of $z = -\sqrt{3} + i$ is;**

(a) $2\text{cis}\left(\frac{5\pi}{6}\right)$
(b) $2\text{cis}\left(\frac{\pi}{6}\right)$
(c) $2\text{cis}\left(\frac{11\pi}{6}\right)$
(d) $\text{cis}\left(\frac{7\pi}{6}\right)$

(21) **The locus of the complex number Z in complex plane, which satisfies the equation $|z-1| = |z|$ is;**

(a) A circle
(b) A horizontal straight line
(c) A vertical straight line
(d) An ellipse

(22) $\left|1 - \frac{1}{i}\right| =$

(a) 2
(b) $\sqrt{2}$
(c) 1
(d) $\sqrt{3}$

(23) **Argument** $\left[\left(1+\sqrt{3}i\right)(1-i)\right] =$

(a) $0°$
(b) $15°$
(c) $30°$
(d) $60°$

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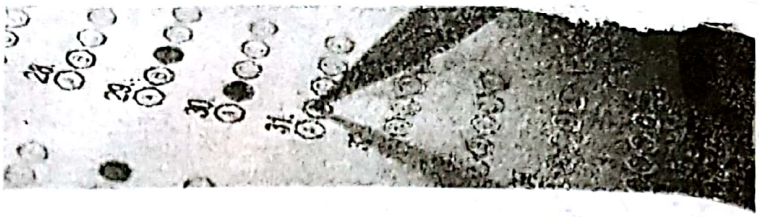

(24) $i^2 \times i^4 \times i^6 \times i^8 \times i^{10} \times \ldots \times i^{30} =$

(a) 1 (b) -1

(c) i (d) $-i$

(25) If $Z_1.\bar{Z}_2 = 2+3i, |Z_2| = 2$, then $\frac{Z_1}{Z_2} =$

(a) $\frac{2+3i}{2}$ (b) $\frac{2-3i}{2}$

(c) $\frac{2+3i}{4}$ (d) $\frac{2-3i}{4}$

(26) $\forall a,b \in \mathbb{R}^-$ if $a < b$, then which one is true;

(a) $ab < b^2$ (b) $a^2 < ab$

(c) $\frac{b}{a} < 1$ (d) $\frac{a}{b} < 1$

(27) $(a,b).(c,d) =$

(a) $(ac+bd, ad-bc)$ (b) $(ac-bd, ad+bc)$

(c) $(ad-bc, ac+bd)$ (d) $(ad+bc, ac-bd)$

(28) For any non-zero complex number Z, which of the following is true?

(I) $\text{Re}(Z) \leq |Z|$ (II) $\text{Im}\,g(Z) \leq |Z|$ (III) $Z + \bar{Z} = 2\text{Re}(Z)$

(a) Only I (b) Only II

(c) Only I, II (d) Only I, II, III

(29) The least positive integer n such that $\left(\frac{2i}{1+i}\right)^n$ is a positive integer is;

(a) 16 (b) 8

(c) 4 (d) 2

(30) The conjugate of a complex number is $\frac{1}{i-1}$. Then that complex number is;

(a) $\frac{-1}{i-1}$ (b) $\frac{1}{i+3}$

(c) $\frac{-1}{i+1}$ (d) $\frac{1}{i-1}$

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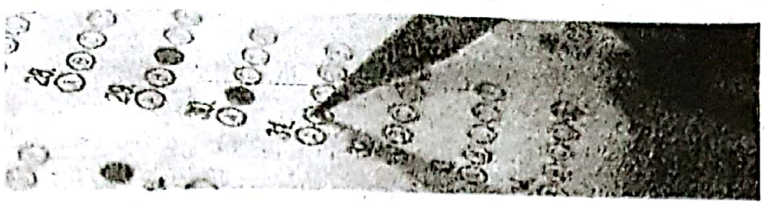

ECAT MATHEMATICS CHAPTER-2

TOPIC: SETS, FUNCTIONS & GROUPS

(1) The set $S=\{1,-1,0\}$ is;

(a) Closed for addition as well as multiplication

(b) Closed for addition but not for multiplication

(c) Closed for multiplication but not for addition

(d) Closed for division but not for addition

(2) Let R be the set of all real numbers then;

(a) $(R,+)$ as well as $(R,\times)$ is a group

(b) $(R,+)$ is a group but $(R,\times)$ is not a group

(c) $(R,\times)$ is a group but $(R,+)$ is not a group

(d) None of these

(3) If set P and Q contain five and six elements respectively with two elements in common, then the number of elements in $P\cup Q$ is;

(a) Seven

(b) Nine

(c) Eleven

(d) Thirteen

(4) Which of the following is a true statement?

(a) Every subset of an infinite set is an infinite set

(b) Every set has a proper subset

(c) For any sets A and B, either $A\subseteq B$ or $B\subseteq A$

(d) $\{x\mid x\in\mathbb{N}\wedge x+6=6\}=\phi$

(5) Let R^+ be the set of all positive real numbers. Then, the function $f:R^+\to R^+, f(x)=a^x$ is;

(a) Many – one, onto

(b) Many – one, into

(c) One – one, onto

(d) One – one, into

(6) Given $A=\{x\mid x \text{ is a root of } x^2-1=0\}, B=\{x\mid x \text{ is a root of } x^2-2x+1=0\}$, then;

$x^2=1$, $x=\pm1$; $(x-1)^2=0$, $(x-1)=0$, $x=1$

(a) $A\cap B=\phi$

(b) $A\cup B=\phi$

(c) $A\cap B=A$

(d) $A\cup B=A$

(7) If $A\cap B=\phi$, then $A\cup B'$ is;

(a) A

(b) B

(c) A'

(d) B'

(8) In a survey, it was found that 63% Pakistani like apples and 76% like orange. How many Pakistani like both?

(a) 13%

(b) 6.5%

(c) 39%

(d) 19.5%

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(9) In a group of 52 persons, 16 take tea but not coffee, while 33 take tea. How many persons take coffee but not tea:

(a) 17 (b) 19

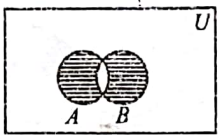
(c) 36 (d) 23

(10) The order of group $\left(\{1,\omega,\omega^2\},\times\right)$ is:

(a) 0 (b) 1

(c) 2 (d) 3

(11) The shaded region in the figure represents:



(a) $(A-B)\cup(B-A)$ (b) $B-A$

(c) $A\cup B$ (d) $(A\cap B)'$

(12) The number of all possible subsets of $\{1,\{2,3\}\}$ is:

(a) 2 (b) 4

(c) 6 (d) 8

(13) Which of the following is a true statement?

(a) $\{a\}\in\{a,b,c\}$ (b) $\{a\}\subseteq\{a,b,c\}$

(c) $\phi\in\{a,b,c\}$ (d) $a\in\{\{a\},b\}$

(14) R is a relation on N given by $R=\{(x,y)\mid 4x+3y=20\}$ which of the following belongs to R?

(a) $(-4,12)$ (b) $(2,4)$

(c) $(5,0)$ (d) $(3,4)$

(15) $\forall x\in\mathbb{N}$ minimum value of $|x-3|+|x-2|+|x-5|$ is:

(a) 3 (b) 7

(c) 9 (d) 15

(16) The converse of inverse of $\sim q\rightarrow\sim p$ is:

(a) $p\rightarrow q$ (b) $\sim p\vee q$

(c) $\sim p\rightarrow\sim q$ (d) Both (a) and (b)

(17) $S=\{\omega^4,\omega^6,\omega^8\}$ is a group w.r.t $\times$, ω,ω^2 are cube roots of then inverse of ω^4 is:

(a) ω^4 (b) ω

(c) $\frac{1}{\omega}$ (d) None

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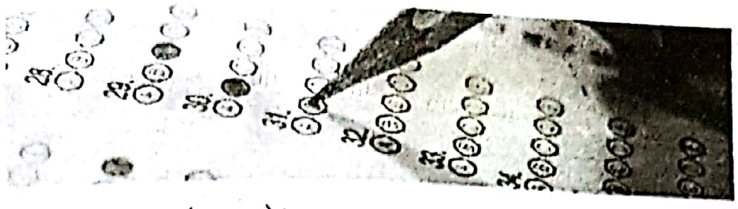

(18) If $A=\{1,2,3,4,5\}, B=\mathbb{R}$, and $C=\mathbb{Z}$ then $A\cup(A\cap B)$ is:

(a) $A\cup B$.
(b) $A\cap B$
(c) A
(d) $A\cup C$

(19) If x and y are elements in the set of real numbers. Which is not a function?

(a) $f=\{(x,y)\mid y=x^2+1\}$
(b) $f=\{(x,y)\mid y=2x^{\frac{3}{2}}\}$
(c) $f=\{(x,y)\mid y=9-x^{\frac{1}{3}}\}$
(d) $f=\{(x,y)\mid y=x^2-|x|\}$

(20) If $A\subset B$, then $A'\cup B'=?$

(a) A'
(b) B'
(c) $A'\cap B'$
(d) None of these

(21) If A and B are subsets of a set X such that $n(X)=900, n(A)=300, n(B)=400$ and $n(A\cap B)=150$, then, $n(A'\cap B')=?$

(a) 250
(b) 350
(c) 400
(d) 450

(22) If A and B are subsets of a set X then:

(a) $A\cup B=X\Rightarrow A'=B$
(b) $A\cup B=X\Rightarrow A\cap B=\phi$
(c) $A\cup B=X\Rightarrow B'=A$
(d) $A\cup B=X$ and $A'=B\Rightarrow A\cap B=\phi$

(23) The domain inverse of $f=\{(1,2),(2,3),(3,4),(4,1),(5,2)\}$ is;

(a) $\{1,3,5\}$
(b) $\{1,2,3,4\}$
(c) $\{1,5\}$
(d) $\{1,2,4,5\}$

(24) Which of the following statement cannot be true?

(a) Inverse of a function is always function
(b) Inverse of a line is line
(c) Inverse of injective function is function
(d) Both (a) and (c)

(25) The truth value of the proposition 3 is a +ve number or $2+2=7$ is:

(a) True
(b) False
(c) Either true or false
(d) Cannot be predicted

(26) $\{x\mid x^2+1=0, \forall x\in\mathbb{R}\}=$ _____

(a) Empty set
(b) Singleton set
(c) Infinite set
(d) Finite set

(27) Two finite sets have m and n elements. The total number of subsets of the first set is 56 more than the total number of subsets of the second set. The values of m and n are:

(a) 7, 6
(b) 6, 3
(c) 5, 1
(d) 8, 7

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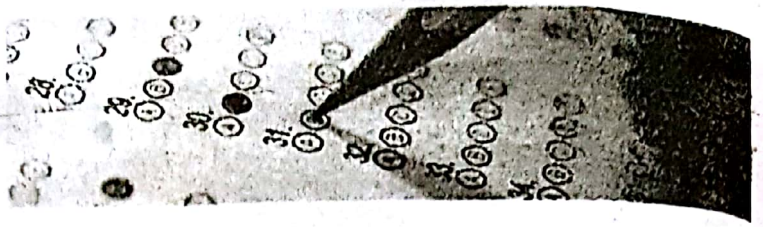

(28) **If P implies Q, an equivalent statement is:**

(a) Q implies P (b) Q is a necessary condition for P

(c) P is a necessary condition for Q (d) Not P implies Q

(29) **If $f(x)=ax+b$, which of the following makes $f(x)=f^{-1}(x)$?**

(I) $a=-1, b=$ any real number

(II) $a=-1, b=0$

(III) $a=$ any real number, $b=0$

(a) Only I (b) Only I and II

(c) Only II (d) Only I and III

(30) **If a set A has 3 elements and another set B has 6 elements. Then:**

(a) $0\le n(A\cup B)\le 9$ (b) $3\le n(A\cup B)\le 6$

(c) $6\le n(A\cup B)\le 9$ (d) $3\le n(A\cup B)\le 9$

(31) **The collection of intelligent student in a class is:**

(a) A null set (b) A singleton set

(c) A finite set (d) Not a set

(32) **The set $S=\{1,2,3\}$ is:**

(a) Closed for addition modulo 4 as well as for multiplication modulo 4

(b) Closed for addition modulo 4 but not closed for multiplication modulo 4

(c) Closed for multiplication modulo 4 but not closed for addition modulo 4

(d) None of these

(33) **If A is the set of numbers obtained by adding 1 to each of the even numbers, then its set builder notation is:**

(a) $A=\{x: x \text{ odd and } x>1\}$ (b) $A=\{x: x \text{ odd and } x\in Z\}$

(c) $A=\{x: x \text{ is even}\}$ (d) $A=\{x: x \text{ is an integer}\}$

(34) **Set M and N have 20 and 15 elements respectively. What can be the minimum number of element in $M\cup N$?**

(a) 15 (b) 20

(c) 35 (d) 5

(35) **A bag of chintoo candies has a total mass of 180g. Excluding the mass of the bag. Each candy has a mass of 6g. As a candy is taken out of the bag and eaten, the mass of the remaining candies is plotted versus how many candies are left in the bag determine the range for this relation:**

(a) $\{0,1,2,3,...30\}$ (b) $\{0,6,12,...130\}$

(c) $\{0,1,2,3,4,5,6\}$ (d) $\{6,12,18,...,60\}$

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TOPIC: MATRICES AND DETERMINANTS

(1) If $A=[a_{ij}]_{m\times n}$ and mn is prime number then matrix A could be;

(a) Column matrix
(b) Rectangular matrix
(c) Square matrix
(d) No such matrix

(2) If A and B are two matrices and $O(A+B)=3\times 4$ then $O(AB)=?$

(a) 3×4
(b) 4×3
(c) 4×4
(d) Not defined

(3) If A and B are square matrices and $AB=O$ then which of the following must be true;

(a) Both A and B are null matrices
(b) Either A or B are null matrix
(c) Either A is singular or B is singular
(d) Both A and B are not null matrices

4) If $A=\begin{bmatrix}1 & 2\\ 5 & 8\end{bmatrix}$ and $B=\begin{bmatrix}2 & 1\\ 6 & 3\end{bmatrix}$ then $A+2B'=?$

(a) $\begin{bmatrix}5 & 14\\ 14 & 7\end{bmatrix}$
(b) $\begin{bmatrix}5 & 17\\ 4 & 14\end{bmatrix}$
(c) $\begin{bmatrix}5 & 14\\ 7 & 14\end{bmatrix}$
(d) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

5) If $B=\begin{bmatrix}5 & 2\\ -2 & 1\end{bmatrix}$ and $C=\begin{bmatrix}-1 & 5\\ 12 & 3\end{bmatrix}$ such that $AB=C$ then $A=?$

(a) $\begin{bmatrix}1 & 2\\ 3 & -1\end{bmatrix}$
(b) $\begin{bmatrix}5 & 2\\ -2 & 1\end{bmatrix}$
(c) $\begin{bmatrix}1 & 3\\ 2 & 1\end{bmatrix}$
(d) $\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix}$

(6) If $A=\begin{bmatrix}-3 & 0\\ 0 & -3\end{bmatrix}$ and $A+A^2+A^3=3kI$ then $k=?$

(a) -1
(b) -4
(c) -7
(d) 1

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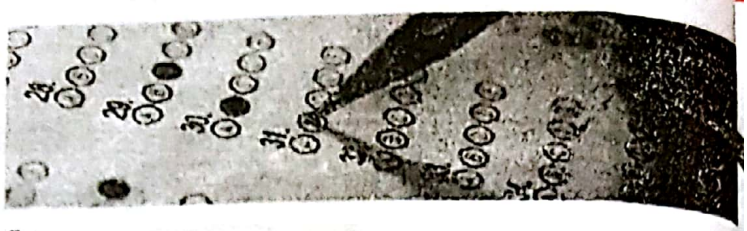

(7) If $A=\begin{bmatrix}3 & 2\\ 7 & 5\end{bmatrix}$ and $B=\begin{bmatrix}a & -2\\ b & 3\end{bmatrix}$ such that $(A+B)^2=A^2+B^2+2AB$ then values of a and b are;

(a) $a=5,b=-7$

(b) $a=-5,b=-7$

(c) $a=7,b=5$

(d) $a=-5,b=7$

(8) If $A=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$ and $(A)(Adj\ of\ A)=?$

(a) $\begin{bmatrix}10 & 1\\ 1 & 10\end{bmatrix}$

(b) $\begin{bmatrix}10 & 0\\ 0 & 10\end{bmatrix}$

(c) $\begin{bmatrix}0 & 10\\ 10 & 0\end{bmatrix}$

(d) None of above

(9) If $A=\begin{bmatrix}3 & -1\\ 2 & 1\end{bmatrix}$ and $5B=\begin{bmatrix}1 & 1\\ -2 & 3\end{bmatrix}$ then which of the following is true;

(I) A and B are multiplicative inverse of each other

(II) A and B are commutative with each other with respect to addition

(III) A and B are commutative with each other with respect to multiplication

(a) I only

(b) II only

(c) I and III only

(d) I, II and III

(10) If $A=\begin{bmatrix}0 & 2 & 3\\ -2 & 0 & 1\\ -3 & -1 & 0\end{bmatrix}$ trace$(A^3)=?$

0+0+0

(a) 3

(b) 7

(c) 9

(d) 0

(11) If $X=[3]$ then $|X|=$

(a) 1

(b) 0

(c) 3

(d) Not defined

(12) For what value of λ, $\frac{1}{\lambda-7}=\begin{vmatrix}1 & 2\\ 0 & 4\end{vmatrix}$;

(a) 11

(b) 4

(c) $\frac{29}{7}$

(d) $\frac{29}{4}$

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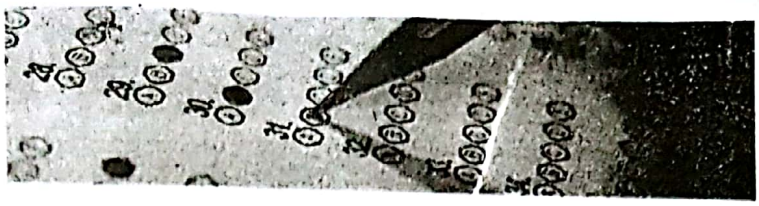

(13) If $T^{-1}=\begin{bmatrix}3 & 1\\ 11 & 4\end{bmatrix}$ then adj of $T=?$

(a) $\begin{bmatrix}4 & -1\\ -11 & 3\end{bmatrix}$

(b) $\begin{bmatrix}3 & 1\\ 11 & 4\end{bmatrix}$

(c) $\begin{bmatrix}4 & -2\\ 7 & 3\end{bmatrix}$

(d) Cannot determine

(14) If $A=(a_{ij})_{2\times 2}$ and $a_{ij}=i+j$ then $A=$

(a) $\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}$

(b) $\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix}$

(c) $\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$

(d) $\begin{bmatrix}2 & 3\\ 3 & 4\end{bmatrix}$

(15) If $A=\begin{bmatrix}4 & 2 & -1\\ 3 & 1 & 9\\ 1 & 6 & 4\end{bmatrix}$ then $a_{21}A_{11}+a_{22}A_{12}+a_{23}A_{13}=?$

(a) 4

(b) -6

(c) 1

(d) 0

(16) If $A=\begin{bmatrix}0 & 0\\ 0 & m\end{bmatrix}$ then for what value of m and n, $A+A^{-1}=nI$ is valid;

(a) 1 and 2

(b) 2 and 1

(c) 1 and $\frac{1}{2}$

(d) Not valid for any value

(17) $\begin{vmatrix}\sin\frac{\pi}{6} & \sin\frac{\pi}{3} & \cos\frac{\pi}{4}\\ \cos\frac{\pi}{2} & \cos\pi & \sin\frac{\pi}{3}\\ \sin\pi & \cos\frac{3\pi}{2} & \cos\frac{\pi}{6}\end{vmatrix}=?$

(a) $\frac{-\sqrt{3}}{4}$

(b) $\frac{-1}{4}$

(c) 0

(d) $\frac{-1}{\sqrt{2}}$

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(18) $\begin{vmatrix} 0 & 3 & 4 & -1 & 7 \\ -3 & 0 & 5 & -2 & -3 \\ -4 & -5 & 0 & 1 & 9 \\ 1 & 2 & -1 & 0 & -6 \\ -7 & 3 & -9 & 6 & 0 \end{vmatrix} = ?$

(a) 4
(b) –24
(c) 0
(d) –29

(19) $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 0 \\ 1 & 1 & 1+y \end{vmatrix}$ **is divisible by;**

(a) x only
(b) y only
(c) Both x and y
(d) Neither x nor y

(20) **If** $A = \begin{bmatrix} 1 & -5 \\ 2 & 8 \end{bmatrix}$ **and** $B = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ **then** $\left|(BA)^{-1}\right| = ?$

(a) $\frac{-1}{13}$
(b) $\frac{-1}{9}$
(c) $\frac{-1}{12}$
(d) $\frac{-1}{18}$

(21) $\begin{vmatrix} \sqrt{2} & 2 & 2\sqrt{2} \\ 4 & 4\sqrt{2} & 8 \\ 8\sqrt{2} & 16 & 16\sqrt{2} \end{vmatrix} =$

(a) $128\sqrt{2}$
(b) $48\sqrt{2}$
(c) $18\sqrt{2}$
(d) 0

(22) **If B is Non–singular matrix and A is square matrix then** $det\left(B^{-1}AB\right) =$ **;**

(a) $Det\left(B\right)$
(b) $Det\left(A\right)$
(c) 1
(d) 0

(23) $\begin{vmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{vmatrix} =$

(a) 1
(b) 2
(c) 0
(d) None of these

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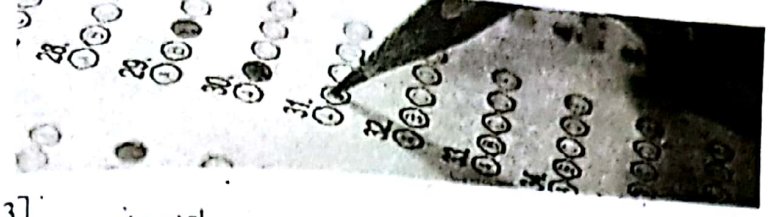

(24) If $K=\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix}$ and $L=\begin{bmatrix}1 & 3\\ -1 & 3\end{bmatrix}$ then $K(LK)^{-1}=$

(a) $\frac{1}{4}\begin{bmatrix}3 & -3\\ 1 & 1\end{bmatrix}$

(b) $\frac{1}{6}\begin{bmatrix}3 & -3\\ 1 & 1\end{bmatrix}$

(c) $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

(d) None of these

(25) If $A=\begin{bmatrix}1 & -3 & 2\\ 2 & 1 & 0\\ 0 & -1 & 1\end{bmatrix}$ then inverse of A^{-1} is;

(a) $\frac{1}{3}\begin{bmatrix}1 & 1 & -2\\ -2 & 1 & 4\\ -2 & 1 & 7\end{bmatrix}$

(b) $\frac{1}{3}\begin{bmatrix}-3 & 1 & -1\\ 2 & 0 & 1\\ 1 & 1 & 2\end{bmatrix}$

(c) $\begin{bmatrix}3 & 4 & 5\\ 8 & 6 & 8\\ 1 & 4 & 2\end{bmatrix}$

(d) None of these

(26) If $|T|=-1, |U|=3$ where T and U are square matrices of order 2 then $|3TU|=$

(a) -9

(b) -27

(c) 9

(d) 27

(27) If A is a square matrix such that $\det(A)=1$ then A must be;

(a) Identity matrix

(b) Diagonal matrix

(c) Singular matrix

(d) None of these

(28) If A is skew symmetric matrix then which of the following is symmetric?

(a) A^9

(b) A^{12}

(c) A^{21}

(d) None of these

(29) If $A=[a_{ij}]_{3\times3}$ is real valued matrix which of the following is Hermition matrix?

(I) $A(\overline{A})'$ (II) $A+(\overline{A})'$ (III) $A+A'$

(a) (I) only

(b) (II) only

(c) (I), (II) only

(d) (I), (II) and (III)

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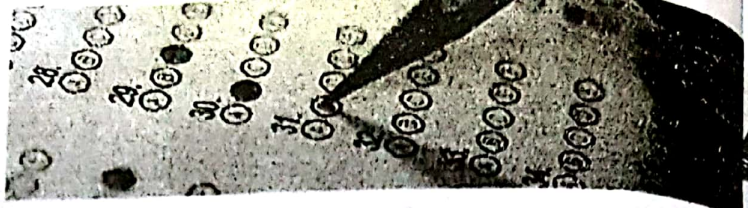

(30) **Which of the following is Hermition matrix? Where ω and ω^2 are complex cube roots of unity;**

(a) $\begin{bmatrix} \omega^3 & \omega & 2\omega^2 \\ 2\omega^2 & 2\omega^4 & 3\omega^4 \\ 2\omega & \omega^5 & \omega^5 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & \omega^5 & \omega^2 \\ \omega & 0 & \omega^7 \\ \omega & \omega^5 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega^2 & \omega & 1 \\ \omega & 1 & \omega^2 \end{bmatrix}$

(d) None of these

(31) **If A and B are skew symmetric matrices of same order and $AB = BA$ then AB is;**

(a) Square matrix

(b) Skew symmetric matrix

(c) Symmetric matrix

(d) Not define

(32) **The matrix** $\begin{bmatrix} 0 & 1 & -3 & 0 & 4 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ **is in;**

(a) Reduce echelon form

(b) Echelon form

(c) Not echelon form

(d) Diagonal matrix form

(33) **Which of the following is true if matrix** $\begin{bmatrix} 1 & x & y & z \\ 0 & 0 & 1 & t \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **is in reduce echelon form?**

(a) $x \in \mathbb{R}$

(b) $y \in \mathbb{R}$

(c) $y, z \in \mathbb{R}$

(d) $t \in \mathbb{R}$

(34) **The system** $\begin{cases} x-2y=4 \\ 3x-6y=5 \end{cases}$ **has;**

For Non-homogeneous $|A| \neq 0 \rightarrow$ Non-trivial then it has No solution

(a) Unique solution

(b) Infinite many solution

(c) No solution

(d) None of these

(35) **Which of the following system has unique solution?**

(a) $\begin{matrix} 2x+3y+4z=5 \\ x+y+z=12 \\ 5x+7y+9z=22 \end{matrix}$

(b) $\begin{matrix} x+y+z=4 \\ x+2y-z=9 \\ 2x+2y+2z=8 \end{matrix}$

(c) $\begin{matrix} x-2y+z=-4 \\ 2x-3y+2z=-6 \\ 2x+2y+z=5 \end{matrix}$ $|A| \neq 0$

(d) None of these

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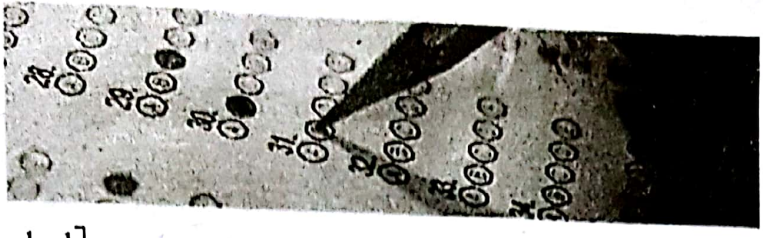

(36) If $A=\begin{bmatrix}1 & 2 & 3\\ 4 & 2 & 1\end{bmatrix}$ and $B=\begin{bmatrix}2 & 1 & 1\\ -1 & 3 & 1\end{bmatrix}$ then which of the following is NOT false?

(a) $(A+B)^t = B^t + A^t$

(b) $(AB)^t = B^t A^t$

(c) $(AB)^t = A^t B^t$

(d) Both (a) and (b)

(37) Order of a matrix C is 40 x 20 and order of another matrix D is 20 x 30 then the number of elements in CD = _____

(a) 1200

(b) 600

(c) 800

(d) 900

(38) If $\begin{bmatrix}1 & 2\\ Z & t\end{bmatrix}^2 = O_2$ then $2Z + t^2 = ?$

(a) 1

(b) $\frac{1}{2}$

(c) 0

(d) $\frac{-1}{2}$

(39) If $Z=[Z_{ij}]_{4\times3}$ then _____

(a) $I_3Z = Z$

(b) $ZI_3 = Z$

(c) $I_4Z = Z$

(d) Both (b) and (c)

(40) The solution of the system of linear equations $\begin{matrix}2x-3y=5\\ 5x+y=4\end{matrix}$ is _____

(a) x = 1, y = 1

(b) x = -1, y = 1

(c) x = 1, y = -1

(d) x = -1, y = -1

(41) If $B=\begin{bmatrix}1 & 0 & 2\\ 0 & 2 & 1\\ 1 & -1 & 1\end{bmatrix}$ then $10B_{32} = ?$

(a) 2

(b) 20

(c) -10

(d) -1

(42) $\begin{vmatrix}3 & 4 & 6x\\ 4 & 5 & 8x\\ 5 & 6 & 10x\end{vmatrix} = 0, if$

(a) x = 0

(b) x = 2

(c) $\forall x \in Z$

(d) $\forall x \in \mathbb{R}$

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(43) $\begin{vmatrix} 0.5 & 1.5 & 2. \\ 3.5 & 4.5 & 5. \\ 6.5 & 7.5 & 8.5 \end{vmatrix} = ?$

(a) 100.5

(b) 0

(c) 180.5

(d) 0.005

(44) If $A = \begin{bmatrix} 1 & 2 \\ 4 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$ then

(a) $|AB| = |A||B|$

(b) $|A+B| = |A|+|B|$

(c) $|A-B| = |A'|-|B|$

(d) All of these

(45) **For a non-Hc mogeneous system of linear equations, $|A| = 0$ where A represents co-efficient matrix then it represents:**

(a) Consistent system

(b) Inconsistent system

(c) Either consistent or inconsistent system

(d) None of these

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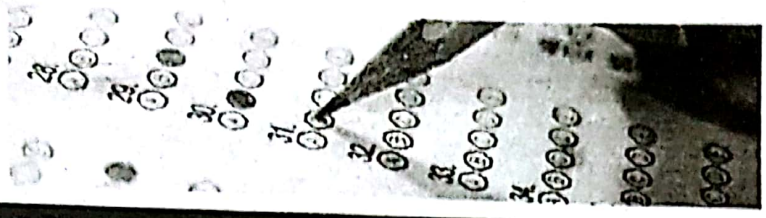

ECAT MATHEMATICS CHAPTER-4

TOPIC: QUADRATIC EQUATIONS

(1) $3x^3 + 5x + 9 + 8x^{-1} + 3x^{-2} = 0$ **is a;**

(a) Polynomial of degree zero
(b) Polynomial of degree 2
(c) Quadratic equation
(d) Not a polynomial

(2) **If the roots of** $3x^2 + kx + 12 = 0$ **are equal then** k **=**

(a) 36
(b) ± 6
(c) 4
(d) ± 12

(3) **If** α, β **are irrational roots of the equation** $ax^2 + bx + c = 0, (\forall a, b, c \in Q)$**, then;**

(a) $\alpha = \beta$
(b) $\alpha\beta = 1$
(c) α and β are conjugate surds
(d) $\alpha^2 + \beta^2 = 1$

(4) **If the product of the roots of the equation** $\alpha x^2 + bx + \alpha^2 + 1 = 0$ **is** -2**, then** α **equals;**

(a) 2
(b) 1
(c) -2
(d) -1

(5) **If** ω **is a cube root of unity, then** $(2 + \omega)(2 + \omega^2) =$

(a) 1
(b) 2
(c) 3
(d) 0

(6) **The equation of the smallest degree with real coefficients having** $2+3i$ **as one of the roots is**

(a) $x^2 + 4x + 13 = 0$
(b) $x^2 + 4x - 13 = 0$
(c) $x^2 - 4x + 13 = 0$
(d) $x^2 - 4x - 13 = 0$

(7) **If** $g(x) = x^3 - 7x^2 + 3x + a$ **has a remainder 1 when divided by** $x + 1$**, then** a **=**

(a) -7
(b) 7
(c) 0
(d) 12

(8) **If one root of** $5x^2 + 13x + k = 0$ **is reciprocal of the other, then** k **=**

(a) 5
(b) -5
(c) 1
(d) -1

(9) **If the roots of the equation** $\frac{a}{x-a} + \frac{b}{x-b} = 1$ **are equal in magnitude and opposite in sign, then;**

(a) $a + b = 1$
(b) $a = b$
(c) $a = -b$
(d) $a - b = 1$

(10) **Sum of cube roots of 19683 is;**

(a) 0
(b) 1
(c) 19683
(d) -19683

(11) **Product of fourth roots of 130321are;**

(a) 1
(b) -1
(c) 130321
(d) -130321

(12) **The numerical difference of the roots of** $x^2 - 7x - 9 = 0$ **is;**

(a) $\sqrt{85}$
(b) 85
(c) $\sqrt{77}$
(d) $\sqrt{99}$

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(13) If one root of quadratic equation is $2-\sqrt{3}$, then other root is;

(a) $\sqrt{3}-2$
(b) $\sqrt{3}+2$
(c) $2-\sqrt{3}$
(d) $-2-\sqrt{3}$

(14) If the roots of the equation $x^2+px-q=0$ differ by 1,then;

(a) $p^2=4q+1$
(b) $p^2=4q$
(c) $p^2=4q-1$
(d) $p^2=-4q+1$

(15) Sum of roots of $(x^2-9)(x^2-4)=0$

(a) 5
(b) –5
(c) 10
(d) 0

(16) Remainder of $x^{24}+x^{23}+1$ when divided by $x+1$ is;

(a) 0
(b) 1
(c) 2
(d) 3

(17) The geometric mean of the roots of the equation $x^2-18x+9=0$ is;

(a) 3
(b) $3\sqrt{2}$
(c) 9
(d) $9\sqrt{2}$

(18) If the roots of $x^2+ax+9=0$ are complex then;

(a) $a<-6$
(b) $a<6$
(c) $|a|<6$
(d) $|a|>6$

(19) If the equation $mx^2+4x+2(m+1)=0$ has real roots, then m lies in the interval;

(a) $[-2,1]$
(b) $[-1,1]$
(c) $(2,3)$
(d) None of these

(20) If the Sum of roots of the equation $\lambda x^2+6x+(2\lambda-1)=0$ is –1, then the value of λ is;

(a) –1
(b) $-\frac{1}{3}$
(c) $\frac{1}{3}$
(d) 6

(21) If α,β are the roots of the equation $6x^2-5x+1=0$, then the value of $\tan^{-1}\alpha+\tan^{-1}\beta$ is;

(a) 0
(b) $\frac{\pi}{4}$
(c) $\frac{\pi}{2}$
(d) 1

(22) If a and b are the roots of the equation $x^2+ax+b=0, a\neq 0, b\neq 0$ then the values of a and b are respectively;

(a) –1 and 2
(b) –1 and 1
(c) 1 and –2
(d) 2 and –2

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(23) If α and β are the roots of $4x^2+3x+7=0$ then the value of $\frac{1}{\alpha}+\frac{1}{\beta}$ is;

(a) $-\frac{3}{7}$ (b) $-\frac{3}{4}$

(c) $\frac{3}{7}$ (d) $\frac{4}{7}$

(24) If one of the roots of the equation $x^2+bx+3=0$ is thrice the other, then b is equal to;

(a) ± 1 (b) ± 2

(c) ± 3 (d) ± 4

(25) The quadratic equation whose roots are twice the roots of $2x^2-5x+2=0$ is;

(a) $x^2-10x+6=0$ (b) $x^2-5x+4=0$

(c) $2x^2-5x+2=0$ (d) $8x^2-10x+2=0$

(26) If α and β are the real roots of a quadratic equation such that $\alpha+\beta=3$ and $\alpha^2+\beta^2=5$, then the equation is;

(a) $x^2+3x+2=0$ (b) $x^2-3x+2=0$

(c) $x^2-2x+16=0$ (d) $x^2-2x+8=0$

(27) The roots of the equation $x^3-3x-2=0$ are;

(a) $-1,-1,-2$ (b) $-1,2,-3$

(c) $-1,-1,2$ (d) $-1,1,-2$

(28) The quadratic equation whose roots are three times the roots of $3ax^2+3bx+c=0$ is;

(a) $ax^2+bx+3c=0$ (b) $ax^2+3bx+c=0$

(c) $ax^2+3bx+3c=0$ (d) $9ax^2+9bx+c=0$

(29) If both $x-1$ and $x-2$ are the factors of $x^3-3x^2-2x+4p$, then p must be;

(a) 0 (b) 2

(c) 3 (d) 4

(30) The real roots of equation $x^{\frac{2}{3}}+x^{\frac{1}{3}}-2=0$ are;

(a) $-8,-1$ (b) $-8,1$

(c) $8,-1$ (d) $8,1$

(31) The solution set of $\sqrt{2x-6}+\sqrt{x+4}=5$ is;

(a) $\{5\}$ (b) $\{0,5\}$

(c) $\{1,3\}$ (d) $\{3,5\}$

(32) If $x^2+y^2=25, xy=12$, then x is equal to;

(a) $-3,-3$ (b) $-3,3$

(c) $3,4$ (d) $3,-3,4,-4$

(33) The roots of the equation $|x^2-x-6|=x+2$ are;

(a) $-2,2,4$ (b) $-2,1,4$

(c) $0,1,4$ (d) $0,2,4$

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(34) The value of 't' which satisfies the equation $2^{2t}-3.2^{t+2}+32=0$ is;

(a) 2 (b) 4

(c) -3 (d) -2

(35) $2x^4-3x^3-x^2-3x+2=0$ is_____ equation;

(a) Exponential (b) Reciprocal

(c) Radical (d) Quadratic

(36) $\left(-1+\sqrt{-3}\right)^5+\left(-1-\sqrt{-3}\right)^5=$

(a) -16 (b) 16

(c) 32 (d) -32

(37) $\left(1+\omega-\omega^2\right)^4=$

(a) 0 (b) 16ω

(c) $16\omega^2$ (d) $16\omega^3$

(38) When $x-1$ divides x^3+4x^2-2x+5 then the remainder is;

(a) 8 (b) -8

(c) 4 (d) -4

(39) If -1 and 2 are the roots of $x^4-5x^2=-4$ then the depressed equation is;

(a) $x^2+x-2=0$ (b) $x^2-x+2=0$

(c) $x^2-x-2=0$ (d) $x^2+x+2=0$

(40) Sum of roots of the equation $4x+x^{-1}+7=0$ is;

(a) -4 (b) 4

(c) $-\frac{7}{4}$ (d) $\frac{7}{4}$

(41) If p and q are the roots of the equation $3x^2-2x+4=0$ then $\frac{p}{q}+\frac{q}{p}=$

(a) $\frac{5}{3}$ (b) $-\frac{5}{3}$

(c) $\frac{3}{5}$ (d) $-\frac{3}{5}$

(42) An equation whose roots are i and -1 is;

(a) $x^3-1=0$ (b) $x^3+1=0$

(c) $x^4-1=0$ (d) $x^4+1=0$

(43) The graph of $f(x)=4x^2+6x+1$ crosses the $x-axis$

(a) 2 times (b) 1 time

(c) 0 times (d) 3 times

(44) The vertex of the parabola $y=x^2+x+k$ is tangent to $x-axis$ if;

(a) $k=\frac{1}{2}$ (b) $k=\frac{1}{4}$

(c) $k=-\frac{1}{2}$ (d) $k=-\frac{1}{4}$

(45) The product of the digits of a two digit number is 8. If the digits are interchanged the number will exceed the original one by 18, the original number is;

(a) 42 (b) 24

(c) 18 (d) 39

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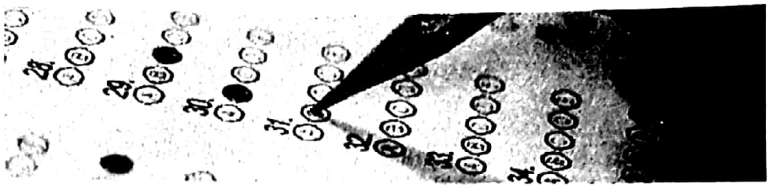

ECAT MATHEMATICS CHAPTER-5

TOPIC: PARTIAL FRACTIONS

(1) $(x+3)(x-4)=x^2-x-12$ **is a/an:**

(a) Identity
(b) Conditional equation
(c) Improper rational fraction
(d) Proper rational fraction

(2) $\frac{4}{x+3}$ **is a/an:**

(a) Identity
(b) Improper rational fraction
(c) Proper rational fraction
(d) Polynomial

(3) **A rational fraction** $\frac{P(x)}{Q(x)}$ **is called improper rational fraction if:**

(a) Degree of $P(x) \geq$ Degree of $Q(x)$
(b) Degree of $Q(x) >$ Degree of $P(x)$
(c) Degree of $P(x) <$ Degree $Q(x)$
(d) Degree of $P(x) \leq$ Degree of $Q(x)$

(4) **If** $ax^4+bx^3+cx^2+dx+e=9x^3+8x^2-7x+x^2+5$ **is an identity then** $a+c=?$

(a) 9
(b) 8
(c) 17
(d) None of these

(5) **The mixed form of** $\frac{3x^2+1}{x-2}$ **is:**

(a) $(3x+6)\frac{13}{x-2}$
(b) $3x+6+\frac{13}{x-2}$
(c) $3x+1+\frac{1}{x-2}$
(d) $3x+5+\frac{11}{x-2}$

(6) **If** $\frac{3x-11}{(x^2+1)(x+3)}=\frac{Ax-3}{x^2+1}-\frac{2}{x+3}$ **then** $A=?$

(a) -2
(b) 1
(c) 3
(d) 2

(7) **Partial fractions of** $\frac{x^2-2x+3}{x^4+x^2+1}$ **are**

(a) $\frac{x+1}{x^2+x+1}+\frac{2x+5}{x^2+x-1}$
(b) $\frac{x-2}{x^2+2x-1}+\frac{3x+1}{2(x^2-2x-1)}$
(c) $\frac{-2x+1}{2(x^2-x+1)}+\frac{2x+5}{2(x^2+x+1)}$
(d) $\frac{-x+5}{x^2-x-1}+\frac{5x-6}{x^2+x+1}$

(8) **Partial fractions of** $\frac{x^4}{x^4-1}$ **are of the form:**

(a) $1+\frac{A}{1-x}+\frac{B}{1+x}+\frac{Cx+D}{1-x^2}$
(b) $\frac{Ax+B}{x^2+1}+\frac{Cx+D}{x^2-1}$
(c) $1+\frac{Ax+B}{x^2+1}+\frac{Cx+D}{x^2-1}$
(d) $1+\frac{A}{x-1}+\frac{B}{x+1}+\frac{Cx+D}{x^2+1}$

(9) **If** $\frac{2x^2-3x+4}{(x+1)^3}=\frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{(x+1)^3}$ **then** $A+B+C=?$

(a) 3
(b) 4
(c) 9
(d) $\frac{3}{8}$

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(10) $\frac{2}{x-1}+\frac{1}{x+2}$ **are the partial fractions of:**

(a) $\frac{3x+3}{(x-1)(x+2)}$ (b) $\frac{x+1}{(x-1)(x+2)}$

(c) $\frac{3x+2}{(x-1)(x+2)}$ (d) $\frac{3x-1}{(x-1)(x+2)}$

(11) **If** $\frac{-x+6}{2x^2-7x+6}=\frac{A}{x-2}+\frac{B}{2x-3}$ **then** $A=?$

(a) -9 (b) 6

(c) 4 (d) 2

(12) $2x-3>0$ **is a/an:**

(a) Identity (b) Conditional equation

(c) Equation (d) Open sentence

(13) **The expression** $\frac{3}{x^3-1}=\frac{1}{x-1}-\frac{x+2}{x^2+x+1}$ **obtained after partial fraction resolution is called a/an:**

(a) Improper rational fraction (b) Identity

(c) Conditional equation (d) Proper rational fraction

(14) x^2+2x+5 **has** _____ **with real co-efficients:**

(a) One linear factor (b) Two linear factors

(c) Three linear factors (d) No linear factor

(15) **Resolution of proper rational fraction** $\frac{P(x)}{Q(x)}$ **will result in at the most** _____ **partial fractions if** $Q(x)=(x-2)^3$**:**

(a) One (b) Two

(c) Three (d) Four

(16) **The rational fraction** $\frac{x^{n-2}+3^{n+1}}{x^2+2}$ **is improper if** $n=?$

(a) 2 (b) 3

(c) 4 (d) None of these

(17) **If** $\frac{x+11}{x^2-2x-15}=\frac{-1}{p(x+3)}+\frac{2}{x-5}$**, then** $p=?$

(a) -1 (b) 1

(c) $\frac{-1}{x+3}$ (d) 2

(18) **If** $\frac{2x-1}{(x^2+1)^2}=\frac{Ax+B}{x^2+1}+\frac{Cx+D}{(x^2+1)^2}$**, then** $A=?$

(a) 2 (b) 1

(c) 0 (d) $\frac{1}{2}$

(19) **Partial fractions of** $\frac{(x-1)(x-2)}{(x-3)(x-4)}$ **are of the form:**

(a) $1+\frac{A}{x-3}+\frac{B}{x-4}$ (b) $\frac{A}{x-3}+\frac{B}{x-4}$

(c) $\frac{A}{x-1}+\frac{B}{x-2}$ (d) $1+\frac{A}{x-1}+\frac{B}{x-2}$

(20) **If** $\frac{1}{(x^2+1)(x^2+2)}=\frac{Ax+B}{x^2+1}+\frac{Cx+D}{x^2+2}$**, then** $A+C=?$

(a) 1 (b) 2

(c) 3 (d) 0

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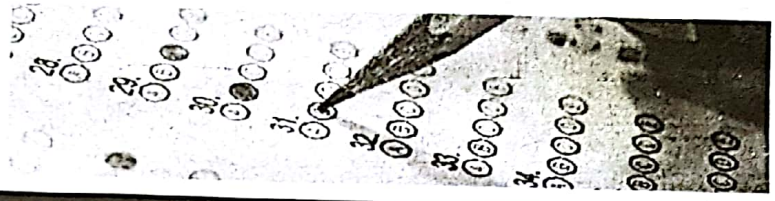

ECAT MATHEMATICS CHAPTER-6

TOPIC: SEQUENCE AND SERIES

(1) Recursive formula satisfied by terms of sequence $1,3,6,10,15,\ldots$;

(a) $a_{n+1}=a_n+n$ (b) $a_{n+1}=a_n+(n-1)$

(c) $a_{n+1}=a_n+(n+1)$ (d) $a_{n+1}=a_n+(n+2)$

(2) Sequence with the general term $a_n=(-1)^{n-1}+(-1)^{n+1}$;

(a) $-2,0,2,-2,0,2,\ldots$ (b) $-2,2,-2,2,\ldots$

(c) $2,-2,2,-2,\ldots$ (d) None of these

(3) If first term of an A.P is "a", then $(n+1)^{th}$ is;

(a) $a+(n-1)d$ (b) $a+nd$

(c) $a+(n+1)d$ (d) $a+(n+2)d$

(4) What is $(p-q)^{th}$ term of the sequence $9,17,25,\ldots$?

(a) $8p-q+1$ (b) $8p-8q+1$

(c) $p-8q+8$ (d) $8(p-q)-1$

First find general term
$a_n = a_1+(n-1)d$
$a_n = 9+(n-1)8$
$a_n = 1+8n$ Now for $n = p-q$
$a_{p-q} = 1+8(p-q)$
$= 1+8p-8q$
$a_{p-q} = 8p-8q+1$

(5) If $a_5=15, a_{10}=35$ are the terms of A.P then 15^{th} term of that A.P will be;

(a) 40 (b) 45

(c) 50 (d) 55

(6) If there are 10 A.M's between 8 and 30 then A_8 =

(a) 24 (b) 28

(c) 26 (d) 22

(7) Arithmetic mean of complex cube root of unity is:

(a) 1 (b) ω

(c) ω^2 (d) $\frac{-1}{2}$

(8) If $a_9=50, a_5=14$ are two terms of A.P if there are 25 A.M's between 1^{st} and last term then A_9 =

(a) 55 (b) 59

(c) 60 (d) 65

(9) Five belongs to which A.P?

(a) $100,85,70,\ldots$ (b) $125,120,115,\ldots$

(c) $-90,-80,-70,\ldots$ (d) None of these

$5 = 100+(n-1)(-15)$
$5 = 100-15n+15$
$5 = 115-15n$
$-110 = -15n$
$\frac{-110}{-15} = n \Rightarrow n$

$5 = 125+(n-1)(-5)$
$5 = 125-5n+5$
$5 = 130-5n$
$-125 = -5n$
$25 = n$

(10) Sum of the series $1+3+5+7+\ldots$ upto 50 terms is;

(a) 2000 (b) 2500

(c) 5000 (d) 1500

(11) If $a_1=10$ and $a_{11}=150$ then S_{11} = fraction

(a) 770 (b) 880

(c) 990 (d) 1010

$S_n = \frac{n}{2}[a_1+a_n]$
$= \frac{11}{2}[10+150]$
$= \frac{11}{2}[160]$
$= 11 \times 80$
$= 880$

$150 = a_1+(n-1)d$
$150 = 10+10d$

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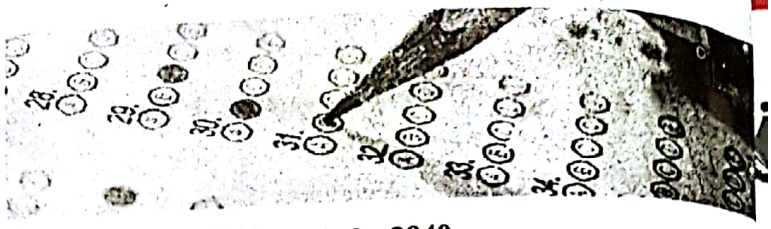

(12) How many terms of the series $11+17+23+\ldots$, amount to be 2940;

(a) 29 (b) 30

(c) 31 (d) 35

(13) Sum of 10 A.M's between roots of equation $\begin{vmatrix} (x-3) & 0 & 0 \\ 0 & (x-5) & 0 \\ 0 & 0 & 1 \end{vmatrix} = 0$;

(a) 40 (b) 80

(c) 50 (d) 30

(14) What is the sum of first 50 whole numbers?

(a) 1225 (b) 1075

(c) 2025 (d) 2075

(15) In a certain A.P $S_{50} - S_{48} = 50$ then A.M of a_{49} and a_{50} will be;

(a) 50 (b) 25

(c) $\frac{25}{2}$ (d) 100

(16) If $2+3i$ is one root of the quadratic equation then A.M of roots of such quadratic equatio will be;

(a) $3i$ (b) 4

(c) 2 (d) $6i$

(17) Which of the following is a G.P;

(a) $\frac{1}{2},\frac{1}{4},\frac{1}{6},\frac{1}{8},\ldots$ (b) $3,6,9,12,\ldots$

(c) $1,i,-1,-i,\ldots$ (d) None of these

(18) 11^{th} term of the sequence $1+i,2,\frac{4}{1+i},\ldots$;

(a) $32(1-i)$ (b) $32(1+i)$

(c) $16(1-i)$ (d) $16(1+i)$

(19) For what value of n, $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is an G.M between a and b?

(a) $n=0$ (b) $n=1$

(c) $n=-1$ (d) None of these

(20) If $a_{10}=50$ is 10^{th} term of A.P then $S_{19}=$

(a) 190 (b) 95

(c) 950 (d) In sufficient data

(21) If three numbers $m+2, 4m-6$ and $3m-2$ are in A.P then $m=$

(a) Zero (b) 1

(c) 2 (d) 3

(22) How many terms of the A.P 18,15,12,...are needed to give the sum 45;

(a) 3 (b) 10

(c) 15 (d) Both (a) and (b)

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(23) If there are 5 G.M's between 1 and $\frac{729}{4096}$ then $G_3 =$

(a) $\frac{9}{16}$ (b) $\frac{27}{64}$

(c) $\frac{81}{256}$ (d) $\frac{243}{1024}$

(24) In a G.P if $a_4 = -i, a_7 = -1$ then $a_{13} =$

(a) i (b) $-i$

(c) -1 (d) 1

(25) If there are 50 G.M's between 1 and ω where ω is cube root of unity then 31st term from the end will be;

(a) $\omega^{\frac{19}{52}}$ (b) $\omega^{\frac{19}{51}}$

(c) $\omega^{\frac{1}{3}}$ (d) None of these

(26) If $a_1, a_2, a_3, \ldots$ are in G.P then $\log a_1, \log a_2, \log a_3, \ldots$ will be in;

(a) G.P (b) A.P

(c) H.P (d) All of these

(27) If there are 100 G.M's between 1 and i then product of these G.M's;

(a) 1 (b) -1

(c) i (d) ± 1

(28) Product of first 9 terms of G.P is 512 then a_5 is;

(a) 128 (b) 64

(c) 16 (d) 2

(29) A man buy a pair of rabbits after each month they become double, so at the end of a year, how many rabbits are there?

(a) 4096 (b) 8192

(c) 8190 (d) None of these

(30) The vulgar fraction of $1.00\overline{5}$ is;

(a) $\frac{179}{180}$ (b) $\frac{181}{180}$

(c) $\frac{125}{130}$ (d) $\frac{1005}{900}$

(31) If $x = 9^{\frac{1}{2}}.9^{\frac{1}{4}}.9^{\frac{1}{8}}.9^{\frac{1}{16}}\ldots$, then $x^2 =$?

(a) 3 (b) $\frac{1}{9}$

(c) $\frac{1}{3}$ (d) 81

(32) In which interval sum of the series $x + \frac{x^2}{3} + \frac{x^3}{9} + \frac{x^4}{27} + \ldots$ is not possible?

(a) $\frac{-1}{3} < x < \frac{1}{3}$ (b) $-3 < x < 3$

(c) $-1 < x < 1$ (d) $-9 < x < 9$

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(33) **Which of the following is H.P?**

(a) 1,3,5,7,...

(b) $\frac{1}{2},\frac{1}{4},\frac{1}{8},\frac{1}{16},\cdots$

(c) $\frac{1}{8},\frac{1}{13},\frac{1}{18},\frac{1}{23},\cdots$

(d) $\frac{1}{\pi},\frac{1}{\sqrt{\pi}},\frac{1}{\pi\sqrt{\pi}},\cdots$

(34) **If 8^{th} and 15^{th} term of H.P respectively $\frac{1}{29},\frac{1}{50}$ then 12^{th} term of H.P will be;**

(a) $\frac{1}{31}$

(b) $\frac{1}{35}$

(c) $\frac{1}{41}$

(d) $\frac{1}{44}$

(35) **The value of H.M between -1 and 1 is;**

(a) Zero

(b) $\frac{1}{2}$

(c) 2

(d) undefined

(36) **For what value of n, $\frac{a^{5n}+b^{5n}}{a^{5n-1}+b^{5n-1}}$ is H.M between a and b?**

(a) $n=-1$

(b) $n=\frac{1}{5}$

(c) $n=5$

(d) $n=0$

(37) **If A.M, G.M and H.M of distinct positive real numbers are 15,18,20 then $H.M=$**

(a) 20

(b) 18

(c) 15

(d) Cannot be determined

(38) $\sum_{i=1}^{\infty}\left(\frac{5}{4}\right)^{i}=$

(a) -5

(b) $\frac{1}{5}$

(c) $-\frac{4}{5}$

(d) Cannot be determine

(39) **If a and b are positive distinct real numbers and A,G and H have usual meaning then which of the relation could be true?**

(a) $A<G<H$

(b) $A>G>H$

(c) $A=G=H$

(d) $A<G>H$

(40) **Sum of the series upto n-terms $1+(1+2)+(1+2+3)+\ldots=$**

(a) $\frac{n(n+1)(2n+1)}{6}$

(b) $\frac{n(n+1)}{2}$

(c) $\frac{n(n+1)(n+2)}{6}$

(d) $\frac{2n(n+1)}{3}$

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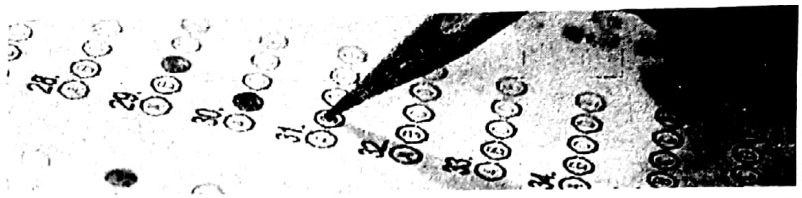

ECAT MATHEMATICS CHAPTER-6

TOPIC: SEQUENCE AND SERIES

(1) **Recursive formula satisfied by terms of sequence $1,3,6,10,15,\ldots$;**

(a) $a_{n+1}=a_n+n$ (b) $a_{n+1}=a_n+(n-1)$

(c) $a_{n+1}=a_n+(n+1)$ (d) $a_{n+1}=a_n+(n+2)$

(2) **Sequence with the general term $a_n=(-1)^{n-1}+(-1)^{n+1}$;**

(a) $-2,0,2,-2,0,2,\ldots$ (b) $-2,2,-2,2,\ldots$

(c) $2,-2,2,-2,\ldots$ (d) None of these

(3) **If first term of an A.P is "a", then $(n+1)^{th}$ is;**

(a) $a+(n-1)d$ (b) $a+nd$

(c) $a+(n+1)d$ (d) $a+(n+2)d$

(4) **What is $(p-q)^{th}$ term of the sequence $9,17,25,\ldots$?**

(a) $8p-q+1$ (b) $8p-8q+1$

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(a) 40 (b) 45

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(6) **If there are 10 A.M's between 8 and 30 then A_8 =**

(a) 24 (b) 28

(c) 26 (d) 22

(7) **Arithmetic mean of complex cube root of unity is:**

(a) 1 (b) ω

(c) ω^2 (d) $\frac{-1}{2}$

(8) **If $a_9=50, a_5=14$ are two terms of A.P if there are 25 A.M's between 1^{st} and last term then A_9 =**

$a_5=14$ $14=a_1+4d$ $d=\frac{50-14}{4}=\frac{36}{4}=9$

$a_1=$ $14=a_1+36$ $A_9=\frac{a+nb}{n+1}=\frac{-22+25(-13)}{26}$

$a_1=14-36$

$a_1=-22$

(a) 55 (b) 59

(c) 60 (d) 65

(9) **Five belongs to which A.P?**

(a) $100,85,70,\ldots$ (b) $125,120,115,\ldots$

(c) $-90,-80,-70,\ldots$ (d) None of these

(10) **Sum of the series $1+3+5+7+\ldots$ upto 50 terms is;**

(a) 2000 (b) 2500

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(c) H.P (d) All of these

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(32) In which interval sum of the series $x + \frac{x^2}{3} + \frac{x^3}{9} + \frac{x^4}{27} + \ldots$ is not possible?

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(33) **Which of the following is H.P?**

(a) $1,3,5,7,...$

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(c) $\frac{1}{8},\frac{1}{13},\frac{1}{18},\frac{1}{23},...$

(d) $\frac{1}{\pi},\frac{1}{\sqrt{\pi}},\frac{1}{\pi\sqrt{\pi}},...$

(34) **If 8^{th} and 15^{th} term of H.P respectively $\frac{1}{29},\frac{1}{50}$ then 12^{th} term of H.P will be;**

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(d) $\frac{1}{44}$

(35) **The value of H.M between -1 and 1 is;**

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(b) $\frac{1}{2}$

(c) 2

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(d) $n=0$

(37) **If A.M, G.M and H.M of distinct positive real numbers are $15,18,20$ then $H.M=$**

(a) 20

(b) 18

(c) 15

(d) Cannot be determined

(38) $\sum_{i=1}^{\infty}\left(\frac{5}{4}\right)^{i}=$

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(a) $\frac{n(n+1)(2n+1)}{6}$

(b) $\frac{n(n+1)}{2}$

(c) $\frac{n(n+1)(n+2)}{6}$

(d) $\frac{2n(n+1)}{3}$

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ECAT MATHEMATICS CHAPTER-7

TOPIC: PERMUTATION, COMBINATION AND PROBABILITY

(1) **Factorial form of $n(n^2-1)$ is: $n \in \mathbb{N}$;**

(a) $\frac{n!}{(n-2)!}$ (b) $\frac{(n+1)!}{(n-2)!}$

(c) $\frac{n!}{(n-1)!}$ (d) Cannot be determine

(2) **Product of $n!$ and $(n-1)!$ is not divisible by;**

(a) $n!$ (b) $(n-1)!$

(c) $(n+1)!$ (d) n

(3) **An amusement park has 3 entrance and 4 exit gates and 5 rides. In how many ways a kid can enter in park, take a ride and exit from it;**

(a) 12 (b) 60

(c) 5 (d) 15

(4) **In an Examination the question paper consist of 10 questions. Out of these there are 3 questions having 4 choices and 4 questions having 5 choices and remaining are true false type. In how many ways a student can attempt all questions in given sequence;**

(a) $3^3.4^3.2^3$ (b) $4^3.5^4.2^3$

(c) $4^5.3^4.3^2$ (d) $4^3.5^4.3^2$

(5) **In a restaurant, there are 4 types of meat and 3 types of vegetable dishes and 2 types of sweet dishes. In how many ways a person can take a meat or vegetable dish and then sweet dish;**

(a) 24 (b) 9

(c) 14 (d) 10

(6) **How many different words can be formed from the letters of the word "KIPS" if each word starts or end with vowel;**

(a) 9 (b) 6

(c) 12 (d) 18

(7) **The number of different words that can be formed from the letters of the word "SPECIAL" when vowels are always on Even places;**

(a) 144 (b) 5040

(c) 120 (d) 178

(8) **How many 3 digit numbers can be formed from the digits 1,2,3,5 if every digit used at least once and no more than once;**

(a) 12 (b) 24

(c) 3 (d) No number formed

(9) **The domain of function $f(x) = {}^{7-x}P_{x-3}$ is;**

(a) $\{1,2,3\}$ (b) $\{3,4,5\}$

(c) $\{2,3,4\}$ (d) $\{4,5,6\}$

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(10) How many 4-digit numbers can be formed by using 9,8,7,6 if numbers with all distinct digits are excluded;

(a) 256 (b) 232

(c) 24 (d) 4

(11) How many 3 letters word can be formed if no letter is repeated;

(a) 15600 (b) 17576

(c) 900 (d) 1000

(12) The number of 3 digit numbers in which digit of unit place and tens place are always equal

(a) 900 (b) 100

(c) 90 (d) 1000

(13) The number of ways in which 4 books of different subjects and 2 different books of English can be arranged so that books of English are never together;

(a) 720 (b) 360

(c) 420 (d) 480

(14) Number of words that can be formed from the letters of the word "BANANA" if every letter start and end with *N*;

(a) 720 (b) 60

(c) 4 (d) 240

(15) How many arrangements are possible from the letters of the word "PIEAS" if each vowel comes on prime numbered positions;

(a) 12 (b) 120

(c) 24 (d) 60

(16) How many 4 digit number can be formed by using first five whole numbers without repeating any digit;

(a) 120 (b) 96

(c) 48 (d) 60

(17) If ${}^{n}C_{r} = 7$ then $(n,r) = ?$

(a) $(1,7)$ (b) $(6,7)$

(c) $(7,6)$ (d) $(1,1)$

(18) ${}^{11}C_4 + {}^{11}C_5 + {}^{11}C_6 + {}^{11}C_7 = ?$

(a) $4.{}^{12}C_5$ (b) $4.{}^{12}C_4$

(c) $2.{}^{12}C_7$ (d) $2.{}^{12}C_4$

(19) How many circles can be drawn from six points lying on a unit circle;

(a) 6 (b) 1

(c) 20 (d) 10

(20) By joining the vertices of a polygon, 35 triangles can be formed, then the polygon is;

(a) Heptagon (b) Octagon

(c) Hexagon (d) Pentagon

(21) If $A = \{1,2,3,...,10\}$, then number of subsets having 4 elements are;

(a) 210 (b) 150

(c) 4^{10} (d) 10^4

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(22) How many committees of 2 members can be formed from 4 Boys and 4 Girls if 2 particular boys and 1 particular girl can never be selected;

(a) 10 (b) 6

(c) 120 (d) 2

(23) Out of n objects, there are 2 ways to select m objects and 12 ways to arrange these m objects then $m = ?$

(a) 3 (b) 6

(c) 4 (d) 5

(24) From 8 consonants and 5 vowels the number of words containing 2 consonants and 2 vowels that can be made if all letters are different;

(a) 6720 (b) 6520

(c) 6580 (d) 6780

(25) From a deck of 52 playing cards, a card is drawn at random. What is probability that, it is red or king;

(a) $\frac{1}{13}$ (b) $\frac{9}{13}$

(c) $\frac{7}{13}$ (d) $\frac{3}{26}$

(26) Two candidates A and B went for an interview probability of A's selection is $\frac{1}{9}$ and B's selection is $\frac{1}{4}$ then the probability that only one of them is selected;

(a) $\frac{11}{36}$ (b) $\frac{13}{36}$

(c) $\frac{5}{13}$ (d) $\frac{1}{18}$

(27) A word is selected from the letters of the word "ENGLISH" the probability of selecting a word that has consecutive vowels is;

(a) $\frac{1}{7}$ (b) $\frac{2}{7}$

(c) $\frac{3}{7}$ (d) $\frac{1}{6}$

(28) A coin is tossed 5 times the probability of getting at least 3 heads is;

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{1}{4}$ (d) $\frac{1}{5}$

(29) If probability that a player scores 3 goals is $\frac{2}{3}$ and team wins is $\frac{1}{4}$ then probability that player scores 3 goals and team does not win is;

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{3}{4}$ (d) $\frac{1}{6}$

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(30) 3 friends entered in a bus and finds 5 vacant seats. In how many ways they can be seated
(a) 15 (b) 60
(c) 240 (d) 360

(31) In how many ways 3 coins can be put into four pockets;
(a) 81 (b) 64
(c) 12 (d) 24

(32) In how many ways 6 persons can be seated at round table if two particular persons never sit together;
(a) 48 (b) 72
(c) 120 (d) 24

(33) $\frac{(n-1)!(n-2)!}{(n!)^2} = ?$
(a) n^2 (b) $\frac{1}{n^2(n-1)}$
(c) $\frac{n-1}{n^2}$ (d) $\frac{1}{n^2}$

(34) How many numbers can be formed by using 0, 1, 1, 2, 2, 2;
(a) 60 (b) 120
(c) 100 (d) 50

(35) Atif bought 7 beads of different colours. How many necklaces can be made by using all them?
(a) 5040 (b) 720
(c) 360 (d) 120

(36) A T-20 tournament is playing round-robin and 8 teams are playing. How many matches will be played in the tournament
(a) 16 (b) 7
(c) 28 (d) 56

(37) What is the probability of an odd prime number from first 10 primes?
(a) $\frac{7}{10}$ (b) 1
(c) $\frac{9}{10}$ (d) $\frac{1}{10}$

(38) Three cards are drawn at random from a deck of 52 – cards what is the probability that all cards are red?
(a) $\frac{2}{17}$ (b) $\frac{1}{17}$
(c) $\frac{3}{17}$ (d) $\frac{26}{37}$

(39) A bag contains 8 red, 5 white and 7 black balls. 3 balls are drawn at random what is the probability that first ball is red, 2nd is white and 3rd is black?
(a) $\frac{35}{100}$ (b) 0.7
(c) $\frac{7}{171}$ (d) .007

(40) From a box containing red colour balls ALI takes out one ball at random, the probability that it is of blue color is?
(a) 1 (b) 0
(c) 0.5 (d) None of these

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ECAT MATHEMATICS CHAPTER-8

TOPIC: MATHEMATICAL INDUCTION AND BINOMIAL THEOREM

(1) **There is no integer n for which 3^n is;**

(a) Divisible by 81 (b) Prime
(c) Odd (d) Even

(2) **Sum of first n even positive integers is equal to;**

(a) $n+2$ (b) n^2+3
(c) $n(n+1)$ (d) $\frac{n(n+2)}{2}$

(3) **If $S(n)=n^2-n+5$ then which of the following is not prime?**

(a) $S(1)$ (b) $S(2)$
(c) $S(4)$ (d) $S(5)$

(4) **If the number of terms in the expansion of $(x+y)^{n^2-n}$ is 7 then $n=$**

(a) 6 (b) 3
(c) 8 (d) –2

(5) **Which of the following represents an integer for all $n \in \mathbb{N}$?**

(a) $\frac{n^3+2}{3}$ (b) $\frac{3^{2n}-2^{2n}}{5}$
(c) $\frac{n(n+4)}{5}$ (d) $\frac{n+5}{2}$

(6) $2^n < \binom{2n}{n}$ **is true for integral values of;**

(a) $n \geq 1$ (b) $n > 2$
(c) $n \geq 2$ (d) $n > 3$

(7) **The term independent of y in the expansion of $\left(\sqrt{y}-\frac{p}{y^2}\right)^{10}$ is 405, then $p=$**

(a) 3 (b) 4
(c) 5 (d) –1

(8) **In the expansion of $(l+m)^{20}$, the sum of exponents of l and m in the 10^{th} term from end is;**

(a) 21 (b) 2^{20}
(c) 20 (d) 2^{10}

(9) **If the coefficients of $(2r+1)^{th}$ term and $(r+1)^{th}$ term in the expansion of $(x+1)^{30}$ are equal, then value of r is;**

(a) 15 (b) 10
(c) 31 (d) $\frac{28}{3}$

(10) **In the expansion of $\left(x^4-\frac{1}{x^3}\right)^6$, if x^{-4} occurs in the r^{th} term then $r=$**

(a) 5 (b) 4
(c) 3 (d) 2

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(11) If the co-efficient of x^{30} is 1 in the expansion of $\left(ax^5+\frac{1}{x^3}\right)^6$ then $a=$

(a) 6
(b) 1
(c) 5
(d) $\frac{1}{6}$

(12) $(2n+1)^{th}$ term from the end in the expansion of $\left(x-\frac{1}{2x}\right)^{3n}$ is _____ term of the expansion from the beginning;

(a) nth
(b) $(n+2)th$
(c) $(n+1)th$
(d) $(n-1)th$

(13) $(0.8)^{\frac{1}{3}}=?$

(a) 0.9283
(b) 0.6616
(c) 0.7126
(d) 0.5315

(14) The number of non-zero terms in the expansion of $\left(1+\sqrt{2}x\right)^9-\left(1-\sqrt{2}x\right)^9$ is;

(a) 10
(b) 18
(c) 4
(d) 5

(15) $2+6+18+\ldots+2\times3^{n-1}=3^n-1$ is true for;

(a) $n=1$
(b) $n=2$
(c) $n=3$
(d) All of these

(16) The co-efficient of x^4 in the expansion of $\frac{1+x^2}{(1-x)^2}$ is;

(a) 5
(b) 6
(c) 7
(d) 8

(17) Constant term in the expansion of $\left(\sqrt{x}+\frac{1}{2x^2}\right)^{10}$ is;

(a) ${}^{10}C_2$
(b) $\frac{1}{4}\,{}^{10}C_2$
(c) $\frac{1}{8}\,{}^{10}C_3$
(d) ${}^{10}C_3$

(18) Middle term(s) in the expansion of $\left(2x-\frac{1}{2x}\right)^{2m+1}$, $m\in\mathbb{N}$ is/are;

(a) $(2m+1)^{th}$ and $(2m+2)^{th}$ terms
(b) $\left(\frac{2m+1}{2}+1\right)^{th}$ term
(c) $(m+1)^{th}$ and $(m+2)^{th}$ terms
(d) None of these

(19) In the expansion of $(2-3x)^{10}$, the sum of binomial coefficients of even powers of x is;

(a) 2^9
(b) 2^{10}
(c) 1
(d) -1

(20) The middle term in the expansion of $\left(x^2+\frac{1}{x^2}\right)^{10}$;

(a) $\binom{10}{6}x^{-4}$
(b) $\binom{10}{5}$
(c) $x^4\binom{10}{4}$
(d) $x^{-8}\binom{10}{7}$

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(21) If n is a positive even integer, then ${}^{n+2}C_1 + {}^{n+2}C_3 + {}^{n+2}C_5 + \ldots + {}^{n+2}C_{n+1} =$

(a) 2^{n-1}
(b) $2^{n+2} - 1$
(c) 2^{n+1}
(d) 2^{n+2}

(22) For all integral values of $n \geq -1$, $3 + 5 + 7 + \ldots + (2n+5) =$

(a) $n^2 + 2$
(b) $(n+2)(n+4)$
(c) n^3
(d) $n^2 + 6$

(23) $\forall n \in \mathbb{N}, 2^{2n} - 1$ is a multiple of;

(a) 5
(b) 3
(c) 2
(d) 15

(24) If n is a negative integer, then last term in the expansion of $(1+x)^n$ is;

(a) x^n
(b) 1
(c) nx
(d) None of these

(25) r^{th} term in the expansion of $(1+x)^n$, where n is negative fractions is;

(a) $\dfrac{n(n-1)(n-2)\ldots(n-r)}{(r-1)!}x^{r-1}$
(b) $\dfrac{n(n-1)(n-2)\ldots(n-r+1)}{r!}x^r$
(c) $\dfrac{n(n-1)(n-2)\ldots(n-r)}{r!}x^r$
(d) $\dfrac{n(n-1)(n-2)\ldots(n-r-1)}{(r-1)!}x^{r-1}$

(26) $1 + 2x + 4x^2 + 8x^3 + \ldots$ is the expansion of;

(a) $(1-x)^{-2}$
(b) $(1-2x)^{-1}$
(c) $(1+4x)^{\frac{-1}{2}}$
(d) $(1-4x)^{\frac{-1}{2}}$

(27) The expansion of $(1 + x - 2x^2)^{\frac{1}{2}}$ is valid if;

(a) $-\dfrac{1}{2} < x < 1$
(b) $|x| < \dfrac{3}{2}$
(c) $|x| < \dfrac{2}{3}$
(d) $|x| < \dfrac{1}{2}$

(28) If x is so small that its square and higher powers can be neglected and $\dfrac{\sqrt{1+2x}}{\sqrt{1-x}} \approx 1 + mx$ then $m =$

(a) $\dfrac{-3}{2}$
(b) $\dfrac{2}{3}$
(c) $\dfrac{3}{2}$
(d) $\dfrac{-2}{3}$

(29) Coefficient of x^n in the expansion of $(1-x)^{-3}$ is;

(a) $\dfrac{(n+1)(n+2)}{2}$
(b) $\dfrac{n+1}{2}$
(c) $n+1$
(d) $(-1)^n \dfrac{(n+1)(n+2)}{2}$

(30) If sum of all coefficients in the expansion of $(ax-8)^6$ is zero, then $a =$

(a) 6
(b) 8
(c) 7
(d) None of these

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(31) 6^{th} **term in the expansion of** $\left(x^2-\frac{1}{x}\right)^{10}$ **is;**

(a) ${}^{10}C_6 x^2$ (b) ${}^{10}C_5 x^5$

(c) ${}^{10}C_5\left(-x^5\right)$ (d) $-{}^{10}C_6 x^2$

(32) **Two consecutive terms with equal coefficients in the expansion of** $(1+x)^{13}$ **are;**

(a) 7^{th} and 8^{th} terms (b) 6^{th} and 7^{th} terms

(c) 5^{th} and 6^{th} terms (d) 8^{th} and 9^{th} terms

(33) $\binom{n}{0}-\binom{n}{1}+\binom{n}{2}-\binom{n}{3}+\binom{n}{4}-\binom{n}{5}+\ldots+(-1)^n\binom{n}{n}=?$

(a) 2^n (b) 2^{n-1}

(c) 0 (d) $n+!$

(34) **If** $(1+x)^n=1+\frac{2}{3}.\frac{1}{2}+\frac{2.5}{3.6}.\frac{1}{2^2}+\frac{2.5.8}{3.6.9}.\frac{1}{2^3}+\ldots$ **then value of** x **is;**

(a) $\frac{-1}{2}$ (b) $\frac{-2}{3}$

(c) $\frac{2}{3}$ (d) $\frac{1}{2}$

(35) **For any positive integer** n, $(1+x)^n=$

(a) $\sum_{r=0}^{\infty}\binom{n}{r}x^r$ (b) $\sum_{r=1}^{n}\binom{n}{r}x^r$

(c) $1+nx+\frac{n(n-1)}{2!}x^2+\ldots$ (d) $1+nx+\frac{n(n-1)}{2!}x^2+\ldots+x^n$

(36) **Coefficient of** x^{10} **in the expansion of** $(1+x)^{-3}$ **is;**

(a) –66 (b) 66

(c) 33 (d) –33

(37) **The middle term in the expansion of** $(4+2x)^{-7}$ **is;**

(a) 8^{th} term (b) 11^{th} term

(c) 20^{th} term (d) None of these

(38) **Let** $(a+b)^n=1+\frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+\ldots$ **If** $a=1$, **then;**

(a) $b=\frac{-2}{3}, n=\frac{1}{2}$ (b) $b=\frac{2}{3}, n=\frac{-1}{2}$

(c) $b=\frac{-2}{3}, n=\frac{-1}{2}$ (d) $b=\frac{2}{3}, n=\frac{1}{2}$

(39) **The term involving** t^3 **in the expansion of** $\left(s-\sqrt{t}\right)^{11}$ **is;**

(a) $463t^3$ (b) $-462s^5t^3$

(c) $462s^5t^3$ (d) $412t^3$

(40) **If** 5^{th} **term in the expansion of** $\left(y+\frac{2}{y^2}\right)^n$ **is independent of y then the index is;**

(a) 12 (b) 11

(c) 13 (d) 10

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ECAT MATHEMATICS CHAPTER-9

TOPIC: FUNDAMENTALS OF TRIGONOMETRY

(1) $10°=$

(a) 0.01745 radian
(b) 0.17450 radian
(c) 1.7450 radian
(d) 7.450 radian

(2) **0.1 radian =**

(a) 57.29577°
(b) 5.729577°
(c) 0.5729577°
(d) 0.05728577°

(3) $330°=$

(a) $3\frac{\pi}{4}$ radian
(b) $5\frac{\pi}{6}$ radian
(c) $7\frac{\pi}{6}$ radian
(d) $11\frac{\pi}{6}$ radian

(4) **If** $\ell = 56cm, \theta = 45°$ **then** $r =$

(a) $71.3cm$
(b) $68.3cm$
(c) $73.3cm$
(d) $75.5cm$

(5) **For a circle if** $\ell = 32$ **m and r = 2m then Area of the sector is** _____.

(a) 3.2 m^2
(b) 32m^2
(c) 320 m^2
(d) 3200m^2

(6) **(sin160°)(cos 80°)**_____.

(a) < 0
(b) > 0
(c) < -1
(d) > 1

(7) **The terminal side of -1035° lies in** _____ **quadrant**

(a) 1st
(b) 3rd
(c) 2nd
(d) 4th

(8) **If** $\cos\alpha = -\frac{\sqrt{3}}{2}$ **then which of the following must be true?**

(a) $\sec^2\alpha - \tan^2\alpha = 1$
(b) $\sin^2\alpha + \cos^2\alpha = 1$
(c) $\operatorname{cosec}^2\alpha - \cot^2\alpha = 1$
(d) All of these

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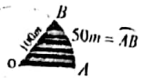
(9) **Following figure shows a circular field and sector OAB is reserved for fertilization of wheat, cost of fertilization land per square meter is Rs 5, then find total cost for the sector;**



(a) Rs.2500 (b) Rs.12500

(c) Rs.25000 (d) Rs.10,000

(10) $0.256°$ **in** $D°M'S''$ **form is;**

(a) $15.36'$ (b) $21.6''$

(c) $0°15'22''$ (d) $1°15'21''$

(11) **The pendulum of a clock is** $18cm$ **long and it swings through on angle of** $20°$ **each seconds. How for does the tip of the pendulum move in 1 second?**

(a) $360cm$ (b) $6.28cm$

(c) $628cm$ (d) $62.8cm$

(12) **Angle between hour hand and minute hand of a clock at** $2:55$ **is;**

(a) $240°$ (b) $60.5°$

(c) $90.5°$ (d) $117.5°$

(13) **If** θ=45° **then angle conterminal to** θ **after two revolution in the anticlockwise direction is;**

(a) $405°$ (b) $765°$

(c) $315°$ (d) $-315°$

(14) **If** $\tan^2 45° - \cos^2 60° = A \sin 45° \cos 45° \tan 60°$ **then A = ?**

(a) $\sin 60°$ (b) $\sin 30°$

(c) $\tan 60°$ (d) $\tan 45°$

(15) $\cos^4 \alpha - \sin^4 \alpha = ?$

(a) $\sin^2 \alpha + \cos 2\alpha$ (b) $\cos^2 \alpha - \sin^2 \alpha$

(c) $2\cos^2 \alpha$ (d) $\sin^2 \alpha$

(16) $\operatorname{cosec} \alpha = \dfrac{1}{\sin \alpha}$, **when**

(a) $\forall \alpha \in R$ (b) $\forall \alpha \in R, \alpha \neq k\pi, k \in Z$

(c) $\forall \alpha \in R, \alpha = \dfrac{(2k+1)}{2}, k \in z$ (d) $\forall \alpha \in R, \alpha \neq 0$

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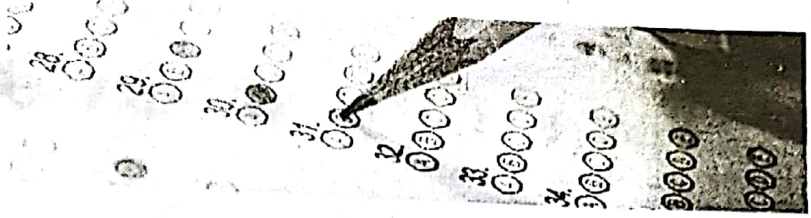

(17) **In term of** $\sec\theta$, $\sin\theta = ?$

(a) $\sqrt{1-\sec^2\theta}$

(b) $\dfrac{\sec\theta}{\sqrt{1-\sec^2\theta}}$

(c) $\dfrac{\sqrt{\sec\theta}}{1-\sec^2\theta}$

(d) $\dfrac{\sqrt{\sec^2\theta-1}}{\sec\theta}$

(18) **Which of the following is fundamental trigonometric identity for all real value of** θ**;**

(a) $\sin^2\theta+\cos^2\theta=1$

(b) $1+\tan^2\theta=\sec^2\theta$

(c) $1+\cot^2\theta=\operatorname{cosec}^2\theta$

(d) All of these

(19) **If** $\tan\theta=\dfrac{8}{15}$ **and the terminal arm of the angle is in the III quadrant then** $\operatorname{cosec}\theta=$

(a) $\dfrac{-17}{8}$

(b) $\dfrac{-8}{17}$

(c) $\dfrac{-15}{17}$

(d) $\dfrac{15}{8}$

(20) **If** $x=45°$ **then which of the following is true?**

(a) $\sin(x-15°)>0, \cos(2x-180°)<0$

(b) $\sin(2x-90°)=0, \operatorname{cosec}(x-75°)>0$

(c) $\tan(4x-90°)=\infty, \sec(x-180°)>0$

(d) $\operatorname{cosec}(x+30°)>0, \cot(x+30°)>0$

(21) $\cot\left(\dfrac{19\pi}{3}\right)=$

(a) $\sqrt{3}$

(b) $\dfrac{1}{\sqrt{3}}$

(c) 2

(d) $\dfrac{-1}{\sqrt{3}}$

(22) **If** $\sec^2\theta+\operatorname{cosec}^2\theta=\sec^2\theta\operatorname{cosec}^2\theta$ **then** $\forall n\in Z, \theta\neq$

(a) $n\dfrac{\pi}{2}$

(b) $(2n+1)\dfrac{\pi}{2}$

(c) $n\pi$

(d) $(n+1)\dfrac{\pi}{2}$

(23) **If** $\sin\theta+\operatorname{cosec}\theta=\sqrt{3}$ **then** $\sin^3\theta+\operatorname{cosec}^3\theta=$

(a) $3\sqrt{3}$

(b) $2\sqrt{3}$

(c) $\sqrt{3}$

(d) Zero

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(24) $\sin^2\frac{\pi}{6}+\sin^2\frac{\pi}{3}+\tan^2\frac{\pi}{4}=$

(a) 0

(b) 1

(c) 2

(d) $\frac{3}{4}$

(25) $\sec(216°).\cos\left(\frac{1}{216°}\right)$

(a) $=1$

(b) <0

(c) >0

(d) $=0$

(26) **The terminal arm of angle $\theta=27\frac{\pi}{2}$ falls;**

(a) On positive $x-axis$

(b) On negative $x-axis$

(c) On positive $y-axis$

(d) On negative $y-axis$

(27) **Terminal side of standard angle lies in/at;**

(a) Positive $y-axis$

(b) 2nd quadrant

(c) Positive $x-axis$

(d) Any where in the plane

(28) **If $\sin(x-30°)=\cos(2x-60°)$ then $x=$**

(a) 30°

(b) 60°

(c) 90°

(d) 120°

(29) **Value of $\cot 5°\cot 10°\cot 15°...\cot 80°\cot 85°$;**

(a) 1

(b) $\frac{1}{\sqrt{2}}$

(c) $\sqrt{3}$

(d) Zero

(30) $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}}=$

(a) $\sec\theta+\tan\theta$

(b) $\cos\theta-\tan\theta$

(c) $\sec\theta-\tan\theta$

(d) None of these

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ECAT MATHEMATICS CHAPTER-10

TOPIC: TRIGONOMETRIC IDENTITIES, SUM AND DIFFERENCE OF ANGLES

(1) $\sin 15 =$

(a) $\frac{2\sqrt{2}}{\sqrt{3}-1}$

(b) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

(c) $\frac{\sqrt{3}+1}{2\sqrt{2}}$

(d) $\frac{2\sqrt{2}}{\sqrt{3}+1}$

(2) **Range of** $y=\cos\theta+\sin\theta$ **is;**

(a) $(-1,1)$

(b) $[-1,1]$

(c) $[-2,2]$

(d) $\left[-\sqrt{2},\sqrt{2}\right]$

(3) $\sin(\alpha+\beta).\sin(\alpha-\beta)=$

(a) $\cos^2\beta-\sin^2\beta$

(b) $\sin^2\alpha-\sin^2\beta$

(c) $\cos^2\alpha\cos^2\beta$

(d) $\sin^2\alpha\sin^2\beta$

(4) $\sin\left(5\frac{\pi}{2}+\alpha\right)=$

(a) $\sin\alpha$

(b) $-\sin\alpha$

(c) $\cos\alpha$

(d) $-\cos\alpha$

(5) **If** $\sin\left(\frac{\phi}{2}\right)=-\frac{1}{4}$, **then** $\cos\phi=$

(a) $\frac{1}{8}$

(b) $\frac{3}{8}$

(c) $\frac{5}{8}$

(d) $\frac{7}{8}$

(6) $\cos 147^{\circ} =$

(a) $\cos 33^{\circ}$

(b) $-\sin 33^{\circ}$

(c) $-\cos 33^{\circ}$

(d) $\sin 33^{\circ}$

(7) **If** a,β,γ **are the angles of triangles, then** $\cos\left(\frac{a+\beta}{2}\right)=?$

(a) $\cos\frac{\gamma}{2}$

(b) $\sin\frac{\gamma}{2}$

(c) $-\sin\frac{\gamma}{2}$

(d) $-\cos\frac{\gamma}{2}$

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(8) **The value of** $\sin 20^\circ \cos 10^\circ + \cos 20^\circ \sin 10^\circ =$

(a) $\cos 30^\circ$ (b) $\cos 45^\circ$

(c) $\cos 60$ (d) $\cos 90^\circ$

(9) **The value of** $\cos^2 52^\circ + \cos^2 38^\circ =$

(a) $\cos 45^\circ$ (b) $\sin 45^\circ$

(c) $\tan 45^\circ$ (d) $\cos ec 45^\circ$

(10) $\cos 23^\circ =$

(a) $\pm\sqrt{\frac{1-\cos 46^\circ}{2}}$ (b) $\sqrt{\frac{1-\cos 46^\circ}{2}}$

(c) $\pm\sqrt{\frac{1+\cos 46^\circ}{2}}$ (d) $\sqrt{\frac{1+\cos 46^\circ}{2}}$

(11) **If** $\cos(\alpha+\beta)=-1$, **then** $\sin(2\alpha+\beta)=$

(a) $\sin\alpha$ (b) $-\sin\alpha$

(c) $\cos\alpha$ (d) $-\cos\alpha$

(12) **The value of** $\tan 1^\circ . \tan 2^\circ . \tan 3^\circ ..... \tan 89^\circ$ **is;**

(a) 0 (b) 1

(c) -1 (d) -2

(13) **The value of** $\cos 100^\circ + \sin 10^\circ =$

(a) 0 (b) 1

(c) -1 (d) 2

(14) **If** α **is the angle of a** ΔABC, **then which one is true?**

(a) $\cos\alpha$ is +ve (b) $\sin\alpha$ is −ve

(c) $\cos\frac{\alpha}{2}$ is +ve (d) $\sin\alpha = 0$

(15) $\frac{2\tan 15^\circ}{1+\tan^2 15^\circ} =$

(a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$

(c) 1 (d) 0

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(16) If $\sin(x+20^\circ)=\cos y$, then $x+y=$

(a) 70° (b) 90°

(c) 20° (d) Can't be determine

(17) If α,β,γ are the angles of ΔABC and $\tan\alpha+\tan\beta+\tan\gamma=6$, then $\cot\alpha\cot\beta\cot\gamma=$

(a) 6 (b) $\frac{1}{6}$

(c) 1 (d) Can't be calculated

(18) $\cos 20^\circ+\cos 100^\circ+\cos 140^\circ=$

(a) 0 (b) 1

(c) 2 (d) 3

(19) $\frac{\sin 3000^\circ}{\sin 1000^\circ}-\frac{\cos 3000^\circ}{\cos 1000^\circ}=$

(a) 0 (b) 1

(c) 2 (d) -1

(20) $\frac{\cot 11^\circ-1}{\cot 11^\circ+1}=$

(a) $\cot 34^\circ$ (b) $\tan 34^\circ$

(c) $\cot 56^\circ$ (d) $\tan 56^\circ$

(21) $\frac{\tan\left(\frac{\theta}{2}\right)}{\cot\theta}=$

(a) $1-\cos\theta$ (b) $1+\cos\theta$

(c) $1+\sec\theta$ (d) $\sec\theta-1$

(22) In any ΔABC, $\cos B\cos C-\sin B\sin C=$

(a) $\sin A$ (b) $-\sin A$

(c) $\cos A$ (d) $-\cos A$

(23) $\cot\left(\frac{\pi}{4}-\theta\right)=$

(a) $\frac{\cot\theta-1}{\cot\theta+1}$ (b) $\frac{\cot\theta+1}{\cot\theta-1}$

(c) $\frac{1-\cot\theta}{1+\cot\theta}$ (d) $\frac{1+\cot\theta}{1-\cot\theta}$

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(24) $\sin 3\alpha =$

(a) $4\sin\alpha - 3\sin^3\alpha$

(b) $3\cos\alpha - 4\cos^3\alpha$

(c) $3\sin\alpha - 4\sin^3\alpha$

(d) $3\cos\alpha - 4\cos^3\alpha$

(25) $\sin 10 \sin 30 \sin 50 \sin 70 =$

(a) $\frac{1}{16}$

(b) $\frac{3}{16}$

(c) $\frac{5}{16}$

(d) $\frac{7}{16}$

(26) **If** $3\sin p + 4\cos p = r\sin(p+\theta)$, **the** $r =$

(a) 5

(b) -5

(c) $\sqrt{5}$

(d) $-\sqrt{5}$

(27) $\sqrt{\frac{1+\cos 200^\circ}{1-\cos 200^\circ}} =$

(a) $\tan 100^\circ$

(b) $-\tan 100^\circ$

(c) $\cot 100^\circ$

(d) $-\cot 100^\circ$

(28) **Which one is the fundamental law of trigonometry?**

(a) $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

(b) $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

(c) $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

(d) $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

(29) **Which one is not an allied angle?**

(a) $3\frac{\pi}{2} + \theta$

(b) $3\frac{\pi}{4} + \theta$

(c) $3\frac{\pi}{6} + \theta$

(d) $\frac{\pi}{2} + \theta$

(30) $\frac{1}{8}[3 + 4\cos 2q + \cos 4q] =$

(a) $\cos q$

(b) $\cos^2 q$

(c) $\cos^3 q$

(d) $\cos^4 q$

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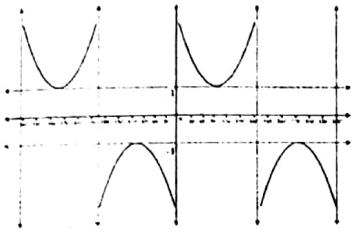
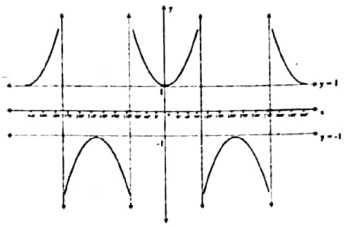
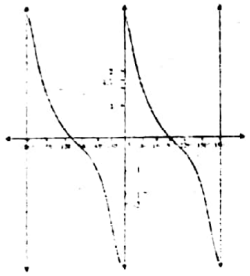
ECAT MATHEMATICS CHAPTER-11

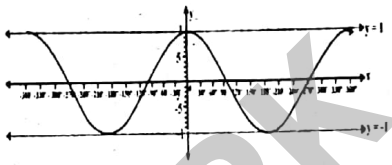
TOPIC: TRIGONOMETRIC FUNCTIONS AND THEIR GRAPHS

(1) Which of the following graph represents the co-ratio of secant function;

(a) 

(b) 

(c) 

(d) 

(2) Which of the following graph cannot crosses $y-axis$;

(a) $y=\cos x$

(b) $y=\cos ecx$

(c) $y=\tan x$

(d) None of these

(3) Minimum distance between the graph of $y=\sec x$ and $x-axis$ is;

(a) 0

(b) 1

(c) $\frac{\pi}{2}$

(d) Cannot be determine

(4) Domain of the function $y=3\sec\frac{x}{4}$ is;

(a) $\mathbb{R}-\left\{(2n+1)\frac{\pi}{2}.n\in\mathbb{Z}\right\}$

(b) $\mathbb{R}-\{n\pi,n\in E,n\neq 0\}$

(c) $\mathbb{R}-\{(2n+1)\pi;n\in\mathbb{Z}\}$

(d) None of these

(5) Range of $y=-2\sin\frac{3x}{2}$ is;

(a) $[-1.1]$

(b) $[2,-2]$

(c) $[-2.2]$

(d) $[-3,3]$

(6) If the maximum value of function $y=\alpha\cos(3x+4)$ is $\frac{5}{2}$ then which one is true?

(a) $\alpha=\frac{1}{2}$

(b) $\alpha=1$

(c) $\alpha=\frac{5}{2}$

(d) $\alpha=\frac{4}{3}$

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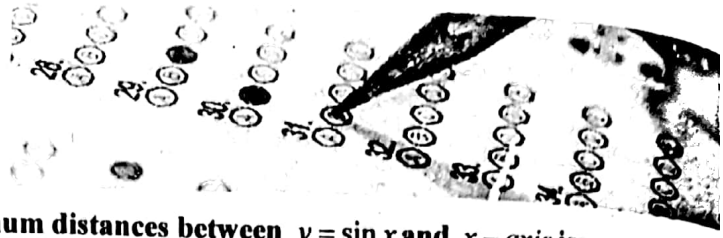

(7) **Sum of maximum and minimum distances between $y=\sin x$ and $x-axis$ is;**

(a) 0 (b) 1

(c) 2 (d) $\sqrt{2}$

(8) **Which of the following functions have same range?**

(a) $y=\sin x$ and $y=\cos x$ (b) $y=\tan x$ and $y=\cot x$

(c) $y=\sec x$ and $y=\operatorname{cosec} x$ (d) All of these

(9) **Graph of $y=\sin x$ and $y=\cos x$ intersect at;**

(a) $x=\frac{\pi}{2}$ (b) $x=\frac{\pi}{4}$

(c) $x=\frac{\pi}{3}$ (d) $x=\frac{\pi}{6}$

(10) **Which of the following is true?**

(a) $\sin 18^\circ > \cos 12^\circ$ (b) $\sin 48^\circ < \cos 50^\circ$

(c) $\sin 28^\circ < \cos 23^\circ$ (d) $\sin 45^\circ > \cos 44^\circ$

(11) **Amplitude of the function $y=3\sin x+\cos x$ is;**

(a) $\sqrt{2}$ (b) 2

(c) $\sqrt{10}$ (d) $\sqrt{5}$

(12) **If $y=3+5\cos(4x+9)$ then which of the following is true?**

(a) $-5\le y\le 5$ (b) $-2\le y\le 8$

(c) $-1\le y\le 1$ (d) $-3\le y\le 5$

(13) **Period of $f(\theta)=\pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$ is;**

(a) 2π (b) $\frac{\pi}{2}$

(c) π (d) Cannot be determine

(14) **The smallest interval after which function $f(\alpha)=\tan\alpha+\cot\alpha$ repeat it's values is;**

(a) $[0,2\pi]$ (b) $[0,\pi]$

(c) $\left[0,\frac{\pi}{2}\right]$ (d) None

(15) **Which of the following functions are symmetric about origin?**

(I) $y=\sec x$ (II) $y=\tan x$ (III) $y=\sin x$ (IV) $y=\cos x$

(a) (I) and (IV) only (b) (II),(III) and (IV) only

(c) (II) and (III) only (d) (I),(II),(III) and (IV)

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(16) If $\alpha+\beta=90°$, then minimum value of the function $y=\sin\alpha\sin\beta$ is;

(a) -1

(b) $\frac{-1}{2}$

(c) 0

(d) 1

(17) Frequency of $y=3+2\tan(x-9)$ is;

(a) $\frac{\pi}{9}$

(b) $\frac{1}{\pi}$

(c) $\frac{9}{\pi}$

(d) $\frac{2}{\pi}$

(18) Period of $f(x)=3\sin^2 5x$ is;

(a) $\frac{2\pi}{5}$

(b) $\frac{\pi}{5}$

(c) $\frac{5}{\pi}$

(d) 2π

(19) Phase shift of $y=2+3\cos(4x+9)$;

(a) $-\frac{9}{4}$

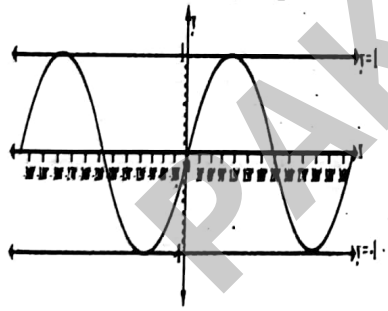
(b) 2

(c) $-\frac{4}{9}$

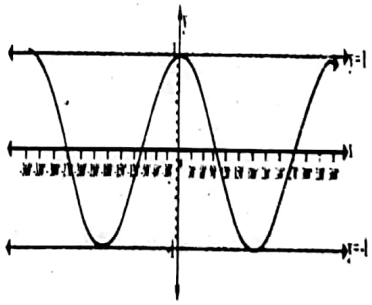
(d) $-\frac{27}{4}$

(20) Which of the following represents graph of $y=\sin^2 x$?

(a)



(b)



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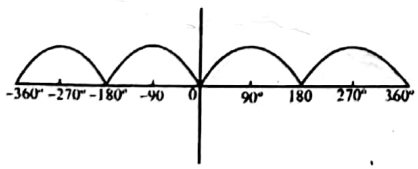

(d) None of these

(21) Which of the following function has same period as $y=\tan 3x$?

(a) $y=\sin\frac{2x}{3}$

(b) $y=\cos\frac{3x}{2}$

(c) $y=\cos ec3x$

(d) $y=\sec 6x$

(22) Domain of $f(\varphi)=\frac{1-\tan^2\varphi}{1+\tan^2\varphi}$ is;

(a) $\mathbb{R}$

(b) $\mathbb{R}-\left\{(2n+1)\frac{\pi}{2}, n\in Z\right\}$

(c) $\mathbb{R}-\{n\pi, n\in Z\}$

(d) $(-\infty,-1)\cup(1,\infty)$

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(23) **Which of the following trigonometric identity has the same domain as a function $y = \cos ecx$?**

(a) $\sin^2\theta + \cos^2\theta = 1$

(b) $1 + \tan^2\theta = \sec^2\theta$

(c) $1 + \cot^2\theta = \cos ec^2\theta$

(d) All of these

(24) **Domain of $f(x) = \sin\sqrt{x}$ is;**

(a) $(-\infty, \infty)$

(b) $(0, \infty)$

(c) $[0, \infty)$

(d) $(1, \infty)$

(25) **If graph of $f(x) = \sin x$ and $g(x) = \sin(x+3)$ cut the $x-axis$ at point p and q respectively, distance between p and q is:**

(a) 0

(b) 1

(c) 3

(d) 2

(26) **Range of the function defined by y = 2 cosec $\frac{x}{3}$ is;**

(a) $y \in (-1, 1)'$

(b) $y \in (-3, 3)'$

(c) $y \in (-2, 2)'$

(d) None of these

(27) **How many times the graph of $y = \tan x$, crosses $x-axis$ between -180° and 180°?**

(a) 1

(b) 2

(c) 3

(d) 4

(28) **The graph of $y = \cos 2x$, crosses $x-axis$ at $x =$**

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{8}$

(d) $\frac{\pi}{16}$

(29) **Which of the following can be a solution of the equation $x = \sin x$?**

(a) $\frac{\pi}{4}$

(b) 0

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{6}$

(30) **$y = \cot x$ has a vertical asymptote at $x =$?**

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{4}$

(c) 0

(d) $\frac{\pi}{3}$

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ECAT MATHEMATICS CHAPTER-12

TOPIC: APPLICATION OF TRIGONOMETRY

(1) **In a triangle ABC if $\sin\gamma = \frac{\sqrt{3}}{2}$ then $\gamma =$**

(a) 30° (b) 120°
(c) 60° (d) Both (b) and (c)

(2) **In a triangle ABC if $\alpha = 57^\circ 37'$ then $\beta+\gamma =$**

(a) $32^\circ 23'$ (b) $122^\circ 23'$
(c) $123^\circ 23'$ (d) $122^\circ 33'$

(3) **In a right triangle ABC if $a = 5, b = 5\sqrt{3}$ and $c = 10$ then $\beta =$**

(a) 30° (b) 45°
(c) 60° (d) 120°

(4) **Which of the following is the only rigid rectilinear figure?**

(a) Triangle (b) Circle
(c) Square (d) Rectangle

(5) **At a distance of 100 m from a light house the angle of elevation from a ship to the top of a light house is 30°. The height of the light house is;**

(a) $100\sqrt{3}$ (b) $\frac{100}{\sqrt{3}}$
(c) $50\sqrt{3}$ (d) $\frac{50}{\sqrt{3}}$

(6) **A man standing on the bank of a canal observes that the measure of the angle of elevation of a tree on the other side of the canal is 60°. On retreating 60 m from the bank, he finds the measure of the angle of elevation of the tree as 30°. The width of the canal is;**

(a) 20 m (b) 60 m
(c) 30 m (d) $40\sqrt{3}$ m

(7) **In ΔABC, which of the following is/are true?**

(a) $\cos\gamma = \frac{a^2+b^2-c^2}{2ab}$ (b) $b^2+c^2 = a^2+2bc\cos\alpha$
(c) $a\sin\beta = b\sin\alpha$ (d) All of these

(8) **In ΔABC if $a = b$, then $\cos\beta =$**

(a) $\frac{c}{a}$ (b) $\frac{c}{2a}$
(c) $\frac{b}{c}$ (d) $\frac{a}{c}$

(9) **In ΔABC if $\gamma = 90^\circ$, then $\tan\left(\frac{\alpha-\beta}{2}\right) =$**

(a) $\frac{a-b}{a+b}$ (b) $\frac{1}{\sqrt{2}}\left(\frac{a-b}{a+b}\right)$
(c) $a-b$ (d) $\tan(\alpha+\beta)\frac{a-b}{a+b}$

(10) **In ΔABC, if $3\sin\alpha = 6\sin\beta = 2\sqrt{3}\sin\gamma$ then $a:b:c =$**

(a) $2:1:\sqrt{3}$ (b) $1:2:3$
(c) $4:1:3$ (d) $3:6:2$

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(11) In ΔABC, if $\alpha = 90°$ then $\sin^2\beta + \sin^2\gamma =$

(a) $b^2 + c^2$ (b) 1

(c) $\sqrt{2}$ (d) 2

(12) In any ΔABC with usual notations, which of the following is always positive?

(a) $\sin\alpha$ (b) $\cos\frac{\beta}{2}$

(c) $\tan\frac{\alpha}{2}$ (d) All of these

(13) In ΔABC with usual notations, $\sqrt{\frac{ac}{s(s-b)}} =$

(a) $\cos\frac{\beta}{2}$ (b) $\sin\frac{\beta}{2}$

(c) $\sin\frac{\gamma}{2}$ (d) $\sec\frac{\beta}{2}$

(14) In ΔABC, if $\beta = 60°, \gamma = 30°, b = \sqrt{6}$ then $c =$

(a) $\sqrt{6}$ (b) $\sqrt{2}$

(c) $3\sqrt{2}$ (d) $2\sqrt{2}$

(15) In ΔABC, if $b = 3, c = 5$ and $\alpha = 120°$ then $a =$

(a) 4 (b) $\sqrt{34 + 15\sqrt{3}}$

(c) 7 (d) 6

(16) Sides of a triangle are $7, 4\sqrt{3}$ and $\sqrt{13}$. The smallest angle is;

(a) 30° (b) 45°

(c) 15° (d) None of these

(17) If area of ΔABC with $a = 5$ and $b = 6$ is $\frac{15\sqrt{3}}{2}$ then $\gamma =$

(a) 30° (b) 45°

(c) 120° (d) None of these

(18) Area of the triangle ABC is;

(a) $\frac{\sin\beta\sin\gamma}{2a^2\sin\alpha}$ (b) $\frac{a^2\sin\beta\sin\gamma}{2\sin\alpha}$

(c) $\frac{a\sin\beta\sin\gamma}{2\sin\alpha}$ (d) $\frac{a^2\sin\alpha\sin\beta}{2\sin\gamma}$

(19) Angles in the same segment of a circle are always;

(a) Equal (b) Complementary angles

(c) Supplementary angles (d) Right angles

(20) In-centre is the point of intersection of _____ the triangle;

(a) Altitudes (b) Medians

(c) Right bisectors (d) Angle bisectors

(21) In ΔABC, if $b = 2, \beta = 30°$ then area of circum circle is;

(a) π (b) 2π

(c) 4π (d) 6π

(22) A piece of art is in the shape of an equilateral triangle with area $8\sqrt{3}$. The length of each side of the triangle is;

(a) 3 (b) 4

(c) $4\sqrt{3}$ (d) $4\sqrt{2}$

(23) A forest ranger spots a fire from a 21 feet high tower. The angle of depression from the tower to the fire is 30°. The distance of fire from the base is;

(a) $\frac{21}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{21}$

(c) 42 (d) $21\sqrt{3}$

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(24) In ΔABC, with usual notations, $r_1 r_2 r_3 =$

(a) rs (b) Δ^2

(c) Δs (d) s^2

(25) In ΔABC, $\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}=$

(a) r^2 (b) $\frac{1}{r}$

(c) r (d) $\frac{2s}{\Delta}$

(26) In an equilateral triangle $r_1 : 2R : r =$

(a) 3:4:1 (b) 1:4:3

(c) 1:2:3 (d) 3:4:2

(27) The area of ΔABC is $72cm^2$ and its perimeter is $36cm$. The radius of its in-circle is;

(a) $2cm$ (b) $4m$

(c) $\frac{1}{2}cm$ (d) $4cm$

(28) If measures of the sides of a triangle ABC are 5,6,7 then $r_2 =$

(a) $\frac{2\sqrt{6}}{3}$ (b) $3\sqrt{6}$

(c) $2\sqrt{6}$ (d) $\frac{3\sqrt{6}}{2}$

(29) In ΔABC, $(s-a)\tan\frac{\alpha}{2}=$

(a) r (b) r_1

(c) r_2 (d) r_3

(30) When measures of two sides of a triangle and their included angle are given, the length of third side can be found by using;

(a) The law of sines (b) The law of cosine

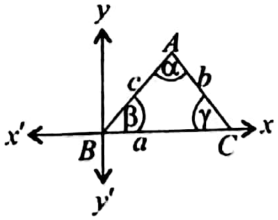
(c) The law of tangents (d) Half angle formulas

(31) In a triangle ABC, if $2(s-a)(s-c)=ac$ then $\beta =$

(a) 30° (b) 45°

c) 60° (d) 90°

(32) In the given triangle ABC, the co-ordinates of the vertex A are;



(a) $(c\cos\alpha, c\sin\alpha)$ (b) $(c\cos\beta, c\sin\beta)$

(c) $(a,0)$ (d) $\left(\frac{a}{2},\frac{c}{2}\right)$

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(33) **Which of the following data can form two triangles?**

(a) $\gamma = 30°, c = 8, b = 12$

(b) $\beta = 45°, a = 12\sqrt{2}, b = 15\sqrt{2}$

(c) $\gamma = 60°, a = 6, b = 4$

(d) $a = \sqrt{7}, b = \sqrt{3}, c = 2$

(34) **Angles of a ΔABC are $\alpha = 120°, \beta = 30°, \gamma = 30°$. The value of $\frac{a}{b}$ is;**

(a) $\frac{1}{\sqrt{3}}$

(b) $\sqrt{3}$

(c) 1

(d) 4

(35) **For a ΔABC, $\frac{(a+b+c)(b+c-a)}{4bc} =$**

(a) $\frac{s(s-a)}{bc}$

(b) $\cos^2\frac{\alpha}{2}$

(c) $\frac{1+\cos\alpha}{2}$

(d) All of these

(36) **If the sides of a ΔABC are $a = 17, b = 10, c = 21$ then $r_1 : r_2 =$**

(a) 12:5

(b) 2:1

(c) 1:2

(d) 4:3

(37) **Length of circum radius of an equilateral triangle of sides 6cm is;**

(a) 16

(b) $6\sqrt{3}$

(c) $2\sqrt{3}$

(d) $14\sqrt{3}$

(38) **Area of a triangle whose sides are $5, 5\sqrt{3}$ and 10 is;**

(a) $25\sqrt{3}$

(b) $\frac{25\sqrt{3}}{2}$

(c) $\frac{25\sqrt{3}}{4}$

(d) $\frac{2\sqrt{3}}{5}$

(39) **If the sides of triangle are $m^2 + m + 1, 2m + 1$ and $m^2 - 1$ then the largest angle is;**

(a) 90°

(b) 120°

(c) 150°

(d) 45°

(40) **The measures of the angles of elevation of the top of a tower as observed from the top and the foot of a tree 300m high are 30° and 60° respectively. Find the height of the tower.**

(a) 200m

(b) 150m

(c) 300m

(d) 450m

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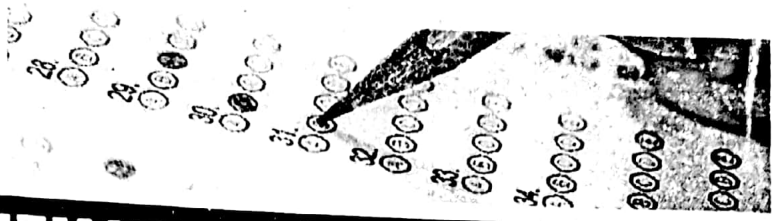

ECAT MATHEMATICS CHAPTER-13+14

TOPIC: SOLUTIONS OF TRIGONOMETRIC EQUATIONS

(1) $\sin^{-1}\left(\frac{1}{2}\right)=?$

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

(2) **Inverse of the function** $y=\cos x$ **exists if;**

(a) $-\infty < x < \infty$ (b) $\frac{-\pi}{2} \le x \le \frac{\pi}{2}$

(c) $0 \le x \le \pi$ (d) $-1 \le x \le 1$

(3) $\cos^{-1}(-A)=?$

(a) $\cos^{-1}(A)$ (b) $-\cos^{-1}(A)$

(c) $\pi-\cos^{-1}(A)$ (d) None of these

(4) $\tan^{-1}\left(\frac{5}{12}\right)=?$

(a) $\cos^{-1}\left(\frac{5}{13}\right)$ (b) $\sin^{-1}\left(\frac{5}{13}\right)$

(c) $\sin^{-1}\left(\frac{4}{13}\right)$ (d) $\cos^{-1}\left(\frac{4}{13}\right)$

(5) $\tan\left(\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\right)=?$

(a) 45° (b) 30°

(c) 60° (d) None of these

(6) $\tan\left(\tan^{-1}\left(-\frac{3}{5}\right)\right)=?$

(a) $\frac{-3}{5}$ (b) $\frac{-5}{3}$

(c) Undefined (d) 1

(7) $\sin^{-1}\left(\frac{1}{2}\right)+\cos^{-1}\left(\frac{1}{2}\right)=?$

(a) 1 (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{3}$

(8) $\cos^{-1}(1)+\cos^{-1}(2)=\cos^{-1}x$, **then** $x=?$

(a) 2 (b) –2

(c) 0 (d) Does not exist

(9) $\tan^{-1}\left(\frac{2x}{1-x^2}\right)=$

(a) $\tan^{-1}(2x)$ (b) $2\tan^{-1}(x)$

(c) $2\tan^{-1}(2x)$ (d) $2\tan^{-1}(4x)$

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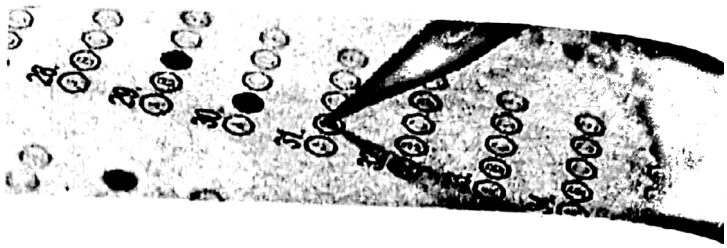

(10) $\cos\left(\sin^{-1}(4x)\right) =$

(a) $\sqrt{1-x^2}$

(b) $\sqrt{1-2x^2}$

(c) $\sqrt{1-4x^2}$

(d) $\sqrt{1-16x^2}$

(11) $2\cos^{-1}(x) =$

(a) $\sin^{-1}\left(\sqrt{1-x^2}\right)$

(b) $\sin^{-1}\left(x\sqrt{1-x^2}\right)$

(c) $\sin^{-1}\left(2x\sqrt{1-x^2}\right)$

(d) $\cos^{-1}\left(1-x^2\right)$

(12) $\sec^{-1}\left(\frac{1}{2}\right)+\operatorname{cosec}^{-1}\left(\frac{1}{2}\right)=$

(a) $\frac{\pi}{2}$

(b) 0

(c) $\frac{\pi}{6}$

(d) None of these

(13) **If** $\alpha = \tan^{-1}\left(\frac{2}{3}\right)$ **then** $\sin 2\alpha =$ **?**

(a) $\frac{2}{13}$

(b) $\frac{12}{13}$

(c) $\frac{5}{13}$

(d) $\frac{17}{25}$

(14) **If** $\sin^2 x+\cos x=1$ **then** $x =$ **?**

(a) $\left\{\frac{\pi}{2}\right\}$

(b) $\left\{\frac{3\pi}{2}\right\}$

(c) $\{2\pi\}$

(d) All of these

(15) **If** $\operatorname{cosec}^2\theta = \frac{4}{3}$ **then** θ**=?**

(a) $\frac{\pi}{3}$

(b) $\frac{2\pi}{3}$

(c) $\frac{4\pi}{3}$

(d) All of these

(16) **Solution of the equation** $\tan^{-1}(x-1)+\tan^{-1}x+\tan^{-1}(x+1)=\tan^{-1}3x$ **is;**

(a) $x=0$

(b) $x=\pm\frac{1}{2}$

(c) $x=\pm\frac{1}{3}$

(d) None of these

(17) $\cos\left(\frac{\pi}{2}-2\cot^{-1}3\right)=$

(a) $\frac{3}{5}$

(b) $\frac{4}{5}$

(c) $\frac{3}{4}$

(d) None of these

(18) $\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\left(\frac{x-y}{x+y}\right)=?$

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{4}$ ro $-\frac{3\pi}{4}$

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(19) The value of $\cos\left(2\cos^{-1}x+\sin^{-1}x\right)$ at $x=\frac{1}{5}$ is;

(a) $\frac{2\sqrt{6}}{5}$

(b) $-\frac{2\sqrt{6}}{5}$

(c) $\frac{3\sqrt{6}}{5}$

(d) None of these

(20) The value of $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right]$ is;

(a) 1

(b) -1

(c) 0

(d) $\frac{1}{2}$

(21) $\cot^{-1}(1)-\tan^{-1}(2)=x$, then $x=$

(a) $\tan^{-1}\left(\frac{1}{3}\right)$

(b) $\tan^{-1}3$

(c) $-\tan^{-1}\left(\frac{1}{3}\right)$

(d) $-\tan^{-1}3$

(22) $2\tan^{-1}\frac{2}{3}=x$ then $x=$

(a) $\sin^{-1}\frac{12}{13}$

(b) $\sin^{-1}\frac{13}{12}$

(c) $2\cos^{-1}\frac{12}{13}$

(d) $\tan^{-1}\frac{1}{3}$

(23) The value of $\tan^{-1}\left[\tan\left(\frac{3\pi}{4}\right)\right]$ is;

(a) $\frac{\pi}{3}$

(b) $-\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) $-\frac{\pi}{4}$

(24) If $\sin\left[\cot^{-1}(2x+1)\right]=\cos\left(\tan^{-1}x\right)$, then $x=$

(a) $-\frac{1}{2}$

(b) $\frac{1}{2}$

(c) 0

(d) -1

(25) The general solution of the equation $2\sin^2x+\sqrt{3}\cos x+1=0$ is;

(a) $x=2n\pi\pm\frac{7\pi}{6}$

(b) $x=2n\pi\pm\frac{5\pi}{3}$

(c) $x=n\pi\pm\frac{\pi}{3}$

(d) None of these

(26) The solution of the equation $\sin 2x+\sin 4x=2\sin 3x$ is;

(a) $x=\frac{n\pi}{3}$

(b) $x=n\pi$

(c) $x=2n\pi$

(d) None of these

(27) $\sec\left(\cos^{-1}\left(\frac{1}{2}\right)\right)=?$

(a) $\frac{1}{2}$

(b) 2

(c) 1

(d) 0

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(28) **How many values of "x" satisfies the equation $\sin^2 x - 1 = 0$, if $0 \le x \le 2\pi$**

(a) 1
(b) 2
(c) 3
(d) 4

(29) **Which of the following is not defined?**

(a) $\arcsin\left(\frac{1}{9}\right)$
(b) $\arccos\left(\frac{-4}{3}\right)$
(c) $\arctan\left(\frac{11}{12}\right)$
(d) $arc\cot(-4)$

(30) $\sin x + \cos x = 0, x = ?$

(a) $\left\{\frac{\pi}{4} + 2n\pi\right\}$
(b) $\left\{\frac{5\pi}{4} + 2n\pi\right\}$
(c) $\left\{\frac{3\pi}{4} + n\pi\right\}$
(d) All of these

(31) **$4\cos^2 x - 3 = 0$, then x =?**

(a) $\left\{\frac{\pi}{3} + 2n\pi\right\} \cup \left\{\frac{2\pi}{3} + 2n\pi\right\} \cup \left\{\frac{4\pi}{3} + 2n\pi\right\} \cup \left\{\frac{5\pi}{3} + 2n\pi\right\}$

(b) $\left\{\frac{\pi}{4} + 2n\pi\right\} \cup \left\{\frac{3\pi}{4} + 2n\pi\right\} \cup \left\{\frac{5\pi}{4} + 2n\pi\right\} \cup \left\{\frac{7\pi}{4} + 2n\pi\right\}$

(c) $\left\{\frac{\pi}{2} + 2n\pi\right\} \cup \left\{\frac{3\pi}{2} + 2n\pi\right\} \cup \left\{\frac{5\pi}{2} + 2n\pi\right\} \cup \left\{\frac{7\pi}{2} + 2n\pi\right\}$

(d) $\left\{\frac{\pi}{6} + 2n\pi\right\} \cup \left\{\frac{5\pi}{6} + 2n\pi\right\} \cup \left\{\frac{7\pi}{6} + 2n\pi\right\} \cup \left\{\frac{11\pi}{6} + 2n\pi\right\}$

(32) $\sqrt{3}\tan x - \sec x - 1 = 0, x = ?$

(a) $\left\{\frac{\pi}{2} + 2n\pi\right\} \cup \left\{\frac{\pi}{3} + 2n\pi\right\}$
(b) $\left\{\frac{\pi}{4} + 2n\pi\right\} \cup \left\{\frac{\pi}{6} + 2n\pi\right\}$
(c) $\{\pi + 2n\pi\} \cup \left\{\frac{\pi}{3} + 2n\pi\right\}$
(d) $\left\{\frac{\pi}{4} + n\pi\right\} \cup \left\{\frac{\pi}{6} + n\pi\right\}$

(33) **The quadratic equation $8\sec^2\theta - 6\sec\theta + 1 = 0$ has;**

(a) Exactly two roots
(b) Exactly four roots
(c) Infinitely many roots
(d) No roots

(34) **$\operatorname{cosec}^{-1} x$ in terms of $\cos^{-1}$;**

(a) $\cos^{-1}\left(\frac{1}{x}\right)$
(b) $\cos^{-1}\left(\frac{x^2 - 1}{x}\right)$
(c) $\cos^{-1}\left(\frac{\sqrt{x^2 - 1}}{x}\right)$
(d) $\cos^{-1}\left(\frac{x}{x^2 + 1}\right)$

(35) $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{1}{5}\right) = ?$

(a) $\tan^{-1}\left(\frac{9}{19}\right)$
(b) $\frac{9}{19}$
(c) $\tan^{-1}\left(\frac{5}{9}\right)$
(d) $\frac{5}{9}$

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ANSWER KEY

CHAPTER-1

1	c	12	b	23	b
2	c	13	c	24	a
3	c	14	a	25	c
4	b	15	c	26	c
5	c	16	a	27	b
6	d	17	c	28	d
7	c	18	b	29	c
8	b	19	d	30	c
9	d	20	a		
10	b	21	c		
11	d	22	b		

CHAPTER-2

1	c	12	b	23	b	34	a
2	b	13	b	24d		35	b
3	b	14	a	25	a		
4	d	15	b	26	a		
5	c	16	a	27	b		
6	d	17	c	28	b		
7	d	18	c	29	c		
8	c	19	b	30	c		
9	b	20	a	31	d		
10	d	21	b	32	a		
11	a	22	d	33	b		

CHAPTER-3

1	a	12	d	23	c	34	c	45	c
2	d	13	b	24	b	35	b		
3	c	14	d	25	a	36	a		
4	c	15	d	26	b	37	a		
5	d	16	d	27	d	38	c		
6	c	17	a	28	b	39	d		
7	a	18	c	29	d	40	c		
8	b	19	c	30	b	41	c		
9	d	20	d	31	c	42	d		
10	d	21	d	32	a	43	b		
11	c	22	b	33	a	44	a		

CHAPTER-4

1	d	12	a	23	a	34	a	45	b
2	d	13	b	24	d	35	b		
3	c	14	d	25	b	36	d		
4	d	15	d	26	b	37	c		
5	c	16	b	27	d	38	a		
6	c	17	a	28	c	39	a		
7	d	18	c	29	d	40	c		
8	a	19	d	30	b	41	b		
9	c	20	d	31	a	42	d		
10	a	21	b	32	c	43	a		
11	d	22	c	33	a	44	b		

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CHAPTER-5

1	a	12	d
2	c	13	b
3	a	14	d
4	b	15	c
5	b	16	c
6	d	17	b
7	c	18	c
8	d	19	a
9	b	20	d
10	a		
11	c		

CHAPTER-6

1	c	12	b	23	b	34	c
2	c	13	a	24	d	35	d
3	b	14	a	25	d	36	d
4	b	15	b	26	b	37	c
5	d	16	c	27	b	38	d
6	a	17	c	28	d	39	b
7	d	18	a	29	c	40	c
8	b	19	d	30	b		
9	b	20	c	31	d		
10	b	21	d	32	b		
11	b	22	d	33	c		

CHAPTER-7

1	b	12	c	23	a	34	c
2	c	13	d	24	a	35	c
3	b	14	c	25	c	36	c
4	b	15	a	26	a	37	c
5	c	16	b	27	b	38	a
6	c	17	c	28	b	39	c
7	a	18	c	29	b	40	b
8	b	19	c	30	b		
9	b	20	a	31	b		
10	b	21	a	32	b		
11	a	22	a	33	b		

CHAPTER-8

1	d	12	c	23	b	34	a
2	c	13	a	24	d	35	d
3	d	14	d	25	a	36	b
4	b	15	d	26	b	37	d
5	b	16	d	27	d	38	c
6	c	17	b	28	c	39	c
7	a	18	c	29	a	40	a
8	c	19	a	30	b		
9	b	20	b	31	c		
10	a	21	c	32	a		
11	b	22	b	33	c		

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CHAPTER-9

1	b	13	b	25	b
2	b	14	a	26	d
3	d	15	b	27	d
4	a	16	b	28	b
5	b	17	d	29	a
6	b	18	a	30	c
7	a	19	a		
8	d	20	d		
9	b	21	b		
10	a	22	a		
11	b	23	d		
12	d	24	c		

CHAPTER-10

1	b	12	b	23	b
2	d	13	a	24	c
3	b	14	c	25	a
4	c	15	a	26	a
5	d	16	a	27	c
6	c	17	b	28	d
7	b	18	a	29	b
8	c	19	c	30	d
9	c	20	b		
10	d	21	d		
11	b	22	d		

CHAPTER-11

1	a	12	b	23	c
2	b	13	a	24	c
3	b	14	b	25	c
4	b	15	c	26	c
5	c	16	b	27	a
6	c	17	b	28	b
7	b	18	b	29	b
8	d	19	a	30	c
9	b	20	c		
10	c	21	d		
11	c	22	a		

CHAPTER-12

1	d	12	d	23	d	34	b
2	b	13	d	24	c	35	d
3	c	14	b	25	b	36	b
4	a	15	c	26	a	37	c
5	b	16	a	27	d	38	b
6	c	17	c	28	c	39	b
7	d	18	b	29	a	40	d
8	b	19	a	30	b		
9	a	20	d	31	d		
10	a	21	c	32	b		
11	b	22	d	33	a		

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CHAPTER-13+14

1	a	12	d	23	d	34	c
2	c	13	b	24	d	35	a
3	c	14	d	25	a		
4	b	15	d	26	c		
5	d	16	a	27	b		
6	a	17	a	28	b		
7	b	18	c	29	b		
8	d	19	b	30	c		
9	b	20	a	31	d		
10	d	21	c	32	c		
11	c	22	a	33	d		

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ECAT MATHEMATICS CHAPTER-1

TOPIC: FUNCTIONS AND LIMITS

(1) Given that $f(x)=7$ for all x then $f(x+3)=$

(a) 0
(b) $x+3$
(c) 7
(d) 10

(2) Domain of $f(x)=\sqrt{x^2-9}$ is;

(a) $|x|\geq 3$
(b) $|x|<3$
(c) $|x|>3$
(d) $|x|\leq 3$

(3) Range of the function $\sqrt{x^2-4}$ is;

(a) $[2,2]$
(b) $[-\infty,\infty]$
(c) $[0,2]$
(d) $[0,\infty)$

(4) $\dfrac{2}{e^x+e^{-x}}=$

(a) $\cosh x$
(b) $\sinh x$
(c) $\operatorname{sech} x$
(d) $\cosh x$

(5) Range of $f(x)=\sinh x$ is ;

(a) Positive real numbers
(b) All real numbers
(c) Non-negative real numbers
(d) Integers

(6) $\dfrac{1}{2}\ln\left(\dfrac{x+1}{x-1}\right)=$; $|x|>1$

(a) $\coth^{-1}(x)$
(b) $\operatorname{sech}^{-1}(x)$
(c) $\tanh^{-1}(x)$
(d) $\operatorname{cosech}^{-1}(x)$

(7) Which of the following is explicit function?

(a) $x^2+xy+y^2=7$
(b) $\dfrac{xy^2-y+1}{xy}=9$
(c) $x^2+y+2x-1=0$
(d) $x^2-xy+y^2=0$

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(8) The parametric equations of hyperbola are;

(a) $x = \sec\theta, y = \tan\theta$

(b) $x = 2\sec t, y = 4\tan t$

(c) $x = a\sec\theta, y = b\tan\theta$

(d) All of these

(9) $x = 2\cos\theta, y = 3\cos\theta$ **are parametric equation of;**

(a) Circle

(b) Straight line

(c) Ellipse

(d) Parabola

(10) Which of the following is an odd function?

(a) $f(x) = \cos x$

(b) $f(x) = 3x^2 + 7x + 9$

(c) $f(x) = e^x$

(d) None of these

(11) Which of the following is NOT true?

(a) $\sinh^2 x + \cosh^2 x = 1$

(b) $1 + \tanh^2 x = \operatorname{sech}^2 x$

(c) $\cosh^2 x - \sinh^2 x = \operatorname{sech} 2x$

(d) All of these

(12) If $g(x) = \frac{2}{x}$, **then** $g^2(x) =$

(a) $\frac{4}{x^2}$

(b) $\frac{2}{x}$

(c) x

(d) $\frac{4}{x}$

(13) Given that $gf(x) = 4x^2 + 4x$ **if** $f(x) = 2x + 1$ **then** $g(x) =$

(a) $4x^2$

(b) $x^2 - 1$

(c) $x^2 + 1$

(d) $x + x^2$

(14) Which of the following is NOT true generally?

(a) $f^2 = fof$

(b) $fof^{-1} = f^{-1}of$

(c) $fog = gof$

(d) $fofof = f^3$

(15) If $f(x) = 4x + 1$ **then** $f^{-1}(x) =$

(a) $\frac{1}{4x+1}$

(b) $\frac{x-1}{4}$

(c) $\frac{x+1}{2}$

(d) $x - 4$

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(16) If $f(x)=\frac{2x+1}{x-1}$, then $f^{-1}(3)=$

(a) 4
(b) -3
(c) 3
(d) -1

(17) What is domain of $f^{-1}(x)$ if $f(x)=\frac{x-1}{x-4}$?

(a) $\{x \mid x \in \mathbb{R} \wedge x \neq 4\}$
(b) $\{x \mid x \in \mathbb{R} \wedge x \neq 0\}$
(c) $\{x \mid x \in \mathbb{R} \wedge x \neq 1\}$
(d) $\{x \mid x \in \mathbb{R} \wedge x \neq 2\}$

(18) If $h(x)=\log_2 x$ then $h^{-1}(x)=$

(a) x^2
(b) 10^{2x}
(c) 2^x
(d) 10^{x^2}

(19) $\lim_{x\to 0}\left(\frac{4x+3}{x-3}\right)=$

(a) -1
(b) 0
(c) 2
(d) 1

(20) $\lim_{x\to 0}\frac{2^x-1}{x}=$

(a) ℓna
(b) $\log_e 2$
(c) $\log_e a$
(d) $\ln x$

(21) $\lim_{t\to 0}\frac{e^t-1}{t}=$

(a) e
(b) 1
(c) e^2
(d) e^{-1}

(22) $\lim_{x\to 4}\frac{x-4}{\sqrt{x}-\sqrt{4}}=$

(a) $2\sqrt{2}$
(b) 0
(c) $\frac{1}{4}$
(d) 4

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(23) $\lim_{x\to a}\frac{x^m-a^m}{x-a}=$

(a) mx^{m-1}

(b) $(m-1)a^m$

(c) ma^{m-1}

(d) $(m-1)x^m$

(24) **Which of the following is true?**

(a) $\lim_{x\to\infty}\frac{1}{x}=\infty$

(b) $\lim_{x\to-\infty}\frac{1}{x}=\infty$

(c) $\lim_{x\to 0^+}\frac{1}{x}=-\infty$

(d) $\lim_{x\to 0^-}\frac{1}{x}=-\infty$

(25) $\lim_{x\to\infty}\left(1+\frac{2}{n}\right)^{\frac{n}{3}}=$

(a) $e^{\frac{3}{2}}$

(b) e^2

(c) e^3

(d) $e^{\frac{2}{3}}$

(26) **If** $f:(-\infty,1]\to(-\infty,1]$ **such that** $f(x)=x(2-x)$, **then** $f^{-1}(x)=$

(a) $1+\sqrt{1-x}$

(b) $1-\sqrt{1-x}$

(c) $\sqrt{1-x}$

(d) $\sqrt{x-1}+1$

(27) **Which of the following is not true?**

(a) $\lim_{x\to\infty}e^x=\infty$

(b) $\lim_{x\to\infty}e^{-x}=0$

(c) $\lim_{x\to-\infty}e^x=0$

(d) $\lim_{x\to\infty}\frac{7}{x}=\infty$

(28) **The range of** $y=\frac{x^2}{1+x^2}$ **is;**

(a) $[0,1)$

(b) $(0,1]$

(c) $(0,1)$

(d) $[0,1]$

(29) $\lim_{t\to\infty}\left(\frac{t}{t+1}\right)^t=$

(a) e

(b) e^{-2}

(c) e^{-1}

(d) e^2

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(30) $\lim_{t\to 0}\frac{e^{\frac{1}{t}}-1}{e^{\frac{1}{t}}+1}=?$, where $t<0$;

(a) 0

(b) -1

(c) 1

(d) Cannot be determine

(31) **Given that** $\lim_{x\to c} f(x)=2, \lim_{x\to c} g(x)=2$ **and** $f(x)>h(x)>g(x)$ **then** $\lim_{x\to c} h(x)=$

(a) Cannot be determine

(b) 2

(c) 0

(d) $\frac{1}{2}$

(32) $\lim_{\theta\to 0}\frac{\sin 9\theta}{2\theta}=$

(a) 9.5

(b) 5.4

(c) 4.5

(d) 4.4

(33) $\lim_{t\to 0}\frac{\sin t^{\circ}}{t}=$

(a) 1

(b) $\frac{\pi}{180}$

(c) $\frac{180}{\pi}$

(d) π

(34) $\lim_{x\to 0}\frac{\sin 7x}{\sin 3x}=$

(a) 1

(b) $\frac{3}{7}$

(c) $\frac{7}{3}$

(d) 0

(35) $\lim_{x\to \frac{\pi}{2}}\frac{\sin x}{x}=$

(a) 1

(b) $\frac{\pi}{2}$

(c) 0

(d) $\frac{2}{\pi}$

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(36) $\lim_{x\to 0}\frac{1-\cos 2x}{1-\cos 3x}=$

(a) 1

(b) $\frac{4}{9}$

(c) $\frac{2}{3}$

(d) 0

(37) $\lim_{x\to\infty}\frac{4x^4-5x^3+2x^2}{3x^5+2x^2+1}=$

(a) 0

(b) $\frac{4}{3}$

(c) ∞

(d) -1

(38) $\lim_{x\to\infty}\frac{2-3x}{\sqrt{3+25x^2}}=$

(a) 1

(b) $\frac{-3}{5}$

(c) 0

(d) -1

(39) **For what value of "m"** $\lim_{x\to -1} f(x)$ **exists, if** $f(x)=\begin{cases}x+4, x\le -1\\ m+2, x>-1\end{cases}$

(a) $m=2$

(b) $m=-1$

(c) $m=1$

(d) $m=-4$

(40) **For what value of "k" the function** $h(x)=\begin{cases}\frac{x^2-4}{x-2}, x\neq 2\\ k \quad , x=2\end{cases}$ **is continuous at** $x=2$

(a) 2

(b) 6

(c) 4

(d) 16

(41) **The graph of** $x^2+y^2=9$ **is symmetric about;**

(a) $x-axis$

(b) Origin

(c) $y-axis$

(d) All of these

(42) **Range of** $g(x)=\begin{cases}6x+7, x\le -2\\ 4x+3, x>2\end{cases}$ **is;**

(a) $\mathbb{R}$

(b) $(-\infty,-5)\cup(11,\infty)$

(c) $(-\infty,2)\cup(5,\infty)$

(d) $(-\infty,-5]\cup(11,\infty)$

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(43) The function $f(x)=\frac{x^2-1}{x-1}$ has a vertical asymptote at ;

(a) $x=0$

(b) $x=1$

(c) $x=-1$

(d) $x=\frac{1}{2}$

(44) Let the domain of f is $[0,5]$ and $g(x)=|3x+4|$, then domain of $f(g(x))$ is;

(a) $\left(-3,\frac{1}{3}\right)$

(b) $\left[-3,\frac{1}{3}\right]$

(c) $\left[-3,\frac{1}{3}\right)$

(d) $\left(-3,\frac{1}{3}\right]$

(45) $\lim\limits_{x\to 0}\frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}}=$

(a) 0

(b) 1

(c) -1

(d) 2

(46) $\lim\limits_{x\to 0}\frac{\log_a(x+1)}{x}=$ _____, $a>0\,a\neq 1$

(a) $\log_a e$

(b) $\log_e a$

(c) 1

(d) 0

(47) $\lim\limits_{x\to\infty}\left(\frac{3x^2+2x+1}{x^2+x+2}\right)^{\frac{6x+1}{3x+2}}=$

(a) 3

(b) 1

(c) 9

(d) Dose not exist

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(48) If a,b,c,d are positive, then $\lim_{x\to\infty}\left(1+\frac{1}{cx+d}\right)^{ax+b}=$

(a) $e^{\frac{b}{d}}$

(b) $e^{\frac{a}{d}}$

(c) $e^{\frac{a}{c}}$

(d) $e^{\frac{b}{c}}$

(49) $\lim_{x\to 0}\frac{x}{|x|}=$

(a) 0

(b) 1

(c) -1

(d) Does not exist

(50) If $g(x)=\begin{cases}|x-1| & if \quad x\le 0\\ |x|-1 & if \quad x>0\end{cases}$ then $\lim_{x\to 0} g(x)=$

(a) 0

(b) 1

(c) -1

(d) Dose not exist

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ECAT MATHEMATICS CHAPTER-2

TOPIC: **DIFFERENTIATION**

(1) **If** $f(x)=\tan x$ **then** $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{f(x)-f\left(\frac{\pi}{4}\right)}{x-\frac{\pi}{4}}=$

(a) 0

(b) 1

(c) 2

(d) 3

(2) **The difference quotient** $\dfrac{s(t_1)-s(t)}{t_1-t}$ **represents the** _____ **of "** s **" over the time interval** t_1-t **;**

(a) Average rate of change

(b) Instantaneous rate of change

(c) Derivative

(d) All of these

(3) **Lagrange used the notation** _____ **for derivative of** $y=f(x)$

(a) $\dfrac{dy}{dx}$

(b) $f'(x)$

(c) $f^{\cdot}(x)$

(d) $Df(x)$

(4) **If** $y=\left(2\sqrt{x}+2\right)\left(\sqrt{x}-1\right)$**, then** $\dfrac{dy}{dx}=$

(a) 1

(b) $\dfrac{2x-1}{\sqrt{x}}$

(c) $\sqrt{x}+\dfrac{1}{2\sqrt{x}}$

(d) 2

(5) **The rule of differentiation** $(fog)'(x)=f'(g(x)).g'(x)$ **is called;**

(a) Composition of functions

(b) Product rule

(c) Power rule

(d) Chain rule

(6) **Derivative of area of circle w.r.t. radius gives its;**

(a) Radius

(b) Diameter

(c) Circumference

(d) Area

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(7) $\frac{d}{dx}(x)^{\sqrt{3}} =$

(a) $x^{\sqrt{3}} \ln x$

(b) $\sqrt{3}x^{\sqrt{3}-1}$

(c) $\sqrt{3}x^{\sqrt{2}}$

(d) None of these

(8) If $y = x\cos y$ then $\frac{dx}{dy} =$

(a) $\frac{1+x\sin y}{\cos y}$

(b) $\frac{\cos y}{1+x\sin y}$

(c) $\frac{1-x\sin y}{\cos y}$

(d) $\frac{\cos y}{1-x\sin y}$

(9) $\frac{d}{d\sin x}(\cos^3 x) =$

(a) $-3\sin x\cos x$

(b) $3\sin x\cos x$

(c) $3\cos^2 x\sin x$

(d) $-3\cos^2 x\sin x$

(10) $\frac{d}{dx}(e^{\pi}) =$

(a) e^{π}

(b) $\pi e^{\pi-1}$

(c) 0

(d) 1

(11) If $\tan y = \frac{1+\tan x}{1-\tan x}$ then $\frac{dy}{dx} =$

(a) -1

(b) 1

(c) $\frac{y}{x}$

(d) $\frac{1+\sec^2 x}{1-\sec^2 x}$

(12) If $y = \ln\left(x\sqrt{x^2+1}\right)$ then $y' =$

(a) $1+\frac{x}{x^2+1}$

(b) $\frac{1}{x\sqrt{x^2+1}}$

(c) $\frac{2x^2+1}{x\sqrt{x^2+1}}$

(d) $\frac{2x^2+1}{x\left(x^2+1\right)}$

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(13) **While finding derivatives of trigonometric functions, we assume that the angle is measured in;**

(a) Degree

(b) Radians

(c) Minutes

(d) All of these

(14) $\frac{d}{dx}\left(x^{\frac{1}{4}}-x^{\frac{1}{2}}\right)=$

(a) $\frac{1}{4x^{\frac{3}{4}}}-\frac{1}{2\sqrt{x}}$

(b) $\frac{1}{x^{\frac{3}{4}}}-\frac{1}{2\sqrt{x}}$

(c) $\frac{0.25}{x^{0.75}}-\frac{0.5}{x^{0.5}}$

(d) Both (a) and (c)

(15) **If** $y=t^3-2t$ **and** $x=t^2-1$ **then** $\frac{dx}{dy}=$

(a) $\frac{2t}{3t^2-2}$

(b) $\frac{3t^2-2}{2t}$

(c) $\frac{3t^2-2}{2t-1}$

(d) $\frac{2t}{t^2-2}$

(16) **If** $f(x)=\frac{1}{x^2+1}$ **and** $g(x)=\sqrt{x}$ **then the derivative of** $f(g(x))$ **is;**

(a) $-\frac{\sqrt{x}}{(x^2+1)^2}$

(b) $\frac{1}{(x+1)^2}$

(c) $-(x+1)^{-2}$

(d) $\frac{1}{2\sqrt{x}(x+1)}$

(17) **Differential coefficient of** $\frac{1}{a}\sin^{-1}\frac{a}{x}$ **is;**

(a) $\frac{1}{x\sqrt{x^2-a^2}}$

(b) $-\frac{1}{x\sqrt{x^2-a^2}}$

(c) $\frac{x}{\sqrt{x^2-a^2}}$

(d) $\frac{1}{\sqrt{x^2-a^2}}$

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(18) If $y = \sinh^{-1}(\tan x)$ then $\frac{dy}{dx} =$

(a) $\frac{1}{\sqrt{1-\tan^2 x}}.\sec^2 x$

(b) $-\sec x$

(c) $\frac{1}{\cos x}$

(d) $\frac{1}{\sqrt{1+\tan^2 x}}$

(19) $\frac{d}{dx}\left(e^{3\ln x}\right) =$

(a) 3

(b) $e^{3\ln x}$

(c) x^3

(d) $3x^2$

(20) $\frac{d}{dx}\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots\right) =$

(a) e^x

(b) $\sin x$

(c) $\cos x$

(d) None of these

(21) $\frac{d}{dx}\left(4\sin^3 x - 3\sin x\right) =$

(a) $-3\cos 3x$

(b) $3\cos 3x$

(c) $\sin 3x$

(d) None of these

(22) **Derivative of** $\ln(\sec x + \tan x)$ **is;**

(a) $\frac{1}{\sec x + \tan x}$

(b) $\operatorname{cosec} x$

(c) $\sec x$

(d) $\frac{\sec^2 x + \tan x}{\sec x + \tan x}$

(23) **For** $f(x) = 5^{x^2+1}$, $f'(0) =$

(a) $2x.\ln 5.5^{x^2+1}$

(b) 0

(c) $5.\ln 5$

(d) $5^{x^2+1}.\ln 5$

(24) $-\frac{1}{x^2-1}$ **is the derivative of;**

(a) $\sin^{-1} x$

(b) $\tan^{-1} x$

(c) $\tanh^{-1} x^2$

(d) $\tanh^{-1} x$

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(25) Derivative of $\sin^{-1}\frac{x-1}{x+1}$ w.r.t. $\operatorname{cosec}^{-1}\frac{x+1}{x-1}$ is;

(a) 0

(b) 1

(c) $\frac{1}{x^2-1}$

(d) $\frac{x+1}{x-1}$

(26) $\forall x<3, \frac{d}{dx}|x-3| =$

(a) -1

(b) 0

(c) 1

(d) Undefined

(27) For $f(x)=(\ln x)^x$, $f'(x) =$

(a) $\ln(\ln x)+\frac{1}{\ln x}$

(b) $x(\ln x)^{x-1}$

(c) $(\ln x)^x\left[\ln x+\frac{1}{\ln x}\right]$

(d) $(\ln x)^x\left[\ln(\ln x)+\frac{1}{\ln x}\right]$

(28) $y=\operatorname{sech}^{-1}x$ is not differentiable at;

(a) $x=1$

(b) $x=2$

(c) $x=-2$

(d) All of these

(29) If $y=\frac{1}{x-1}$, then $y_5 =$

(a) $-\frac{5!}{(x-1)^6}$

(b) $\frac{5!}{(x-1)^6}$

(c) $\frac{5!}{(x-1)^5}$

(d) $-\frac{5!}{(x-1)^{-6}}$

(30) The function $y=ax^2+bx+c$ has a minimum value if;

(a) $a<0$

(b) $a=-1$

(c) $a>0$

(d) $a=0$

(31) For function $y=|x|$, $(0,0)$ is a;

(a) Point of inflection

(b) Stationary point

(c) Minimum point

(d) Maximum point

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(32) **Third term in the Taylor series expansion of $\ln(1+x)$ at $x=2$ is;**

(a) $\frac{(x-2)^2}{18}$

(b) $-\frac{(x-2)^2}{18}$

(c) $\frac{(x-2)^3}{81}$

(d) $\frac{(x-2)^3}{18}$

(33) **Equation of tangent to $y=x^3+4$ at $(0,4)$ is;**

(a) $y+4=0$

(b) $x+y+4=0$

(c) $y=0$

(d) $y=4$

(34) **A series of the form $f(x)=a_0+a_1x+a_2x^2+a_3x^3+\ldots+a_nx^n+\ldots$ is called a _____ expa**

of the function $f(x)$;

(a) Maclaurin series

(b) Taylor's series

(c) Power series

(d) Binomial series

(35) **If $f(x)=\ln\left(\frac{e^x}{e^x-1}\right)$ then $f'(x)=$**

(a) 0

(b) $-\frac{1}{e^x-1}$

(c) $1+\frac{e^x}{e^x-1}$

(d) $1-\frac{1}{e^x}$

(36) **The series $1-x+x^2-x^3+\ldots+(-1)^n x^n+\ldots$ is a;**

(a) Maclaurin series for $\frac{1}{1+x}$

(b) Geometric series

(c) Binomial series for $\frac{1}{1+x}$

(d) All of these

(37) $1+(2x)+\frac{4x^2}{2!}+\frac{8x^3}{3!}+\frac{16x^4}{4!}+\ldots$ **is the Maclaurin series expansion of;**

(a) e^{2x}

(b) e^x

(c) $\sin x$

(d) $\cos x$

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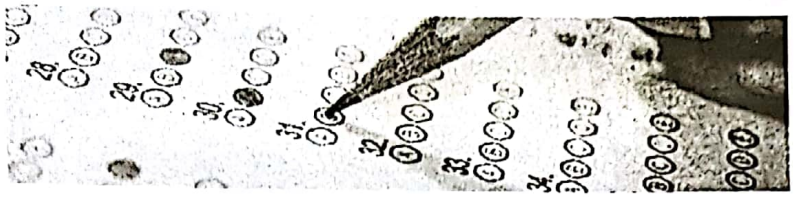

(38) The graph of the function $f(x)=(x-2)^{\frac{1}{3}}$ has vertical tangent at;

(a) $x=2$ (b) $x=0$

(c) $x=-2$ (d) None of these

(39) A function $f(x)$ increases on the interval (a,b) if $f(x_1)<f(x_2)$ whenever $\forall x_1,x_2\in(a,b)$;

(a) $x_2<x_1$ (b) $x_1<x_2$

(c) $x_1x_2>0$ (d) $x_1x_2<0$

(40) The function $f(x)=4x-x^2$ has the values as shown is table;

x	0	1	2	3	4	5
$f(x)$	0	3	4	3	0	−5

The average rate of change of $f(x)$ over the interval $[1,4]$ is;

(a) 1 (b) 9

(c) −1 (d) −9

(41) $\forall x\in(0,2\pi)$, $f(x)=\cos x$ is decreasing in the interval;

(a) $(0,\pi)$ (b) $\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$

(c) $\left(\pi,\frac{3\pi}{2}\right)$ (d) $\left(\frac{3\pi}{2},2\pi\right)$

(42) The minimum value of $f(x)=2\sin x-\cos x$ in the interval $[0,2\pi]$ is;

(a) $-\sqrt{3}$ (b) −1

(c) $-\sqrt{5}$ (d) −3

(43) Which of the following can be the dimensions of a rectangle whose perimeter is 120 with largest area?

(a) 60, 60 (b) 30, 30

(c) 40, 20 (d) 100, 20

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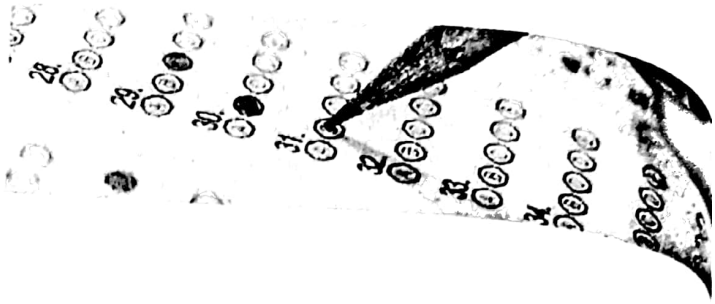

(44) $\frac{d}{dx}(4)^{\sqrt{x}} =$

(a) $4^{\sqrt{x}} \ln 4$

(b) $\frac{4^{\sqrt{x}} \ln 4}{\sqrt{x}}$

(c) $\frac{4^{\sqrt{x}} \ln 4}{4\sqrt{x}}$

(d) $\frac{4^{\sqrt{x}} \ln 4}{2\sqrt{x}}$

(45) **Derivative of** $\arccos x$ **w.r.t. 'x' is;**

(a) $-\sin x$

(b) $-\sin^{-1} x$

(c) $\frac{1}{\sqrt{1-x^2}}$

(d) $-\frac{1}{\sqrt{1-x^2}}$

(46) **If** $y = \ln \tanh \theta$ **then** $\frac{dy}{d\theta} =$

(a) $2\operatorname{sech} \theta$

(b) $2\cosh \theta$

(c) $2\operatorname{cosech}(2\theta)$

(d) $2\operatorname{sech}(2\theta)$

(47) $\frac{d}{dt}\sqrt{\tan t} =$

(a) $\frac{1}{2\sqrt{\tan t}}$

(b) $\frac{\sec^2 t}{2\sqrt{\tan t}}$

(c) $\frac{\sec^2 t}{\sqrt{\tan t}}$

(d) $\frac{1}{2t\sqrt{\tan t}}$

(48) **The minimum value of** $y = x^x$ **is;**

(a) $\frac{1}{\sqrt[e]{e}}$

(b) $\frac{1}{e}$

(c) e

(d) e^2

(49) **If** $y = \sin\left(x - \frac{\pi}{2}\right)$**, then** $y'' =$

(a) $-\sin x$

(b) $\sin x$

(c) $\cos x$

(d) $-\cos x$

(50) **The instantaneous rate of change of** $\frac{e^{2x} - e^{-2x}}{4}$ **w.r.t. 'x' is;**

(a) $\sinh 2x$

(b) $\cosh 2x$

(c) $2\cosh 2x$

(d) $-\cosh 2x$

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ECAT MATHEMATICS CHAPTER-3

TOPIC: INTEGRATION

(1) **Approximate increase in the volume of a cube if the length of its each edge changes from 5 to 5.5 is**

(a) 25.5
(b) 33.5
(c) 29.5
(d) 37.5

(2) $\int a^{ax}dx = ?$ $\quad a>0, a\neq 1$

(a) $\frac{a^{ax}}{\ln a}+c$
(b) $a^{ax}.\ln a+c$
(c) $\frac{a^{ax+1}}{\ln a}+c$
(d) $\frac{a^{ax+1}}{a(ax+1)}+c$

(3) $\int\frac{dx}{\sqrt{2-4x+x^2}}=$

(a) $\sinh^{-1}\left(\frac{x-2}{\sqrt{2}}\right)+c$
(b) $\cosh^{-1}\left(\frac{x-2}{\sqrt{2}}\right)+c$
(c) $\sin^{-1}\left(\frac{x-2}{\sqrt{2}}\right)+c$
(d) None of thsese

(4) **Given that** $y=x^2\sqrt{2x-1}$ **and** $\frac{dy}{dx}=\frac{x(5x-2)}{\sqrt{2x-1}}$, **then result of integration** $\int_1^5\frac{x(5x-2)}{\sqrt{2x-1}}dx$;

(a) 53
(b) 25
(c) 74
(d) 78

(5) $\int e^{\frac{1-\tan^2\theta}{1+\tan^2\theta}}.\sin 2\theta \quad d\theta =$

(a) $e^{\sec\theta}+c$
(b) $\frac{1}{2}e^{\sin\theta}+c$
(c) $e^{\tan\theta+\sec\theta}+c$
(d) $\frac{-1}{2}.e^{\cos 2\theta}+c$

(6) $\int\frac{xe^x}{(1+x)^2}dx=$

(a) $\frac{e^x}{(1+x)}+c$
(b) $e^x(1+x)+c$
(c) $\frac{e^x}{(1-x)}+c$
(d) $e^x(1-x)+c$

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(7) $\int \frac{\sin 2x}{\sin^2 x} dx =$

(a) $x + c$

(b) $\ln|\sin^2 x| + c$

(c) $\ln|\sin 2x| + c$

(d) $\sin^2 2x + c$

(8) $\int \left(x + \frac{1}{x}\right)\left(1 - \frac{1}{x^2}\right) dx =$

(a) $x^2 + \frac{1}{x^2}$

(b) $\frac{1}{2}\left(x^2 + \frac{1}{x^2} + 2\right) + c$

(c) $\left(x + \frac{1}{x}\right) + c$

(d) $\left(1 - \frac{1}{x^2}\right)^2 + c$

(9) $\int \sin\frac{x}{2} . \cos\frac{x}{2} dx$

(a) $1 - \cos x + c$

(b) $\frac{1}{4}(1 - \cos x) + c$

(c) $\frac{1}{2}(1 - \cos x) + c$

(d) None of these

(10) **Suitable substitution for the expression** $\sqrt{a^2 + x^2}$ **is;**

(a) $a\sec\theta$

(b) $a\sin\theta$

(c) $a\tan\theta$

(d) $a\,\text{cosec}\,\theta$

(11) $\int \frac{x}{e^x} dx =$

(a) $\frac{1 - x}{e^x} + c$

(b) $\frac{x - 1}{e^x} + c$

(c) $\frac{x + 1}{e^x} + c$

(d) $\frac{-x - 1}{e^x} + c$

(12) $\int \sqrt{a^2 + x^2}\, dx =$

(a) $\frac{x}{2}\sqrt{a^2 + x^2} + \frac{a^2}{2}\ln\left(x + \sqrt{a^2 + x^2}\right) + C$

(b) $\frac{a^2}{2}\sqrt{a^2 + x^2} + \frac{x}{2}\ln\left(x + \sqrt{a^2 + x^2}\right) + C$

(c) $\frac{x}{2}\sqrt{a^2 + x^2} - \frac{a^2}{2}\ln\left(x + \sqrt{a^2 + x^2}\right) + C$

(d) None of these

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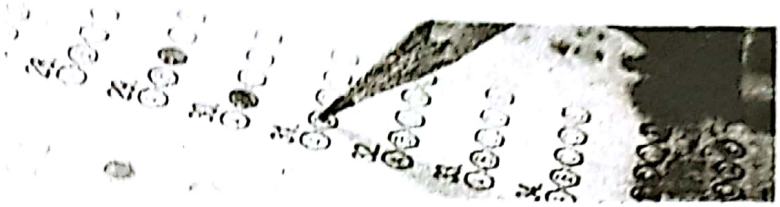

(13) $\int 3^{4x-9}dx = ?$

(a) $3^{4x-9}+c$

(b) $\frac{3^{4x-9}}{\ln 3}+c$

(c) $\frac{3^{4x-9}}{\ln 81}+c$

(d) $\frac{3^{4x-9}}{\ln 27}+c$

(14) $\int \frac{dx}{\sqrt{a^2-x^2}} =$

(a) $\frac{1}{a}\operatorname{cosec}^{-1}\frac{x}{a}+c$

(b) $\frac{1}{a}\sec^{-1}\frac{x}{a}+c$

(c) $\sin^{-1}\frac{x}{a}+c$

(d) $\sinh^{-1}\frac{x}{a}+c$

(15) $\int \sin\left(\tan^{-1}x\right)dx =$

(a) $-\cos\left(\tan^{-1}x\right)+c$j

(b) $\sqrt{1+x^2}+c$

(c) $x+\sqrt{1+x^2}+c$

(d) None of these

(16) $\int \frac{1}{x^2+2}dx =$

(a) $\tan^{-1}\frac{x}{\sqrt{2}}+c$

(b) $\frac{1}{\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}}+c$

(c) $\sqrt{2}\tan^{-1}\frac{x}{\sqrt{2}}+c$

(d) $\sinh^{-1}\left(\frac{x}{2}\right)+c$

(17) If $f(x)=\int_0^{x^2} te^{-t^2}dt$, then $f'(x)=$

(a) e^{-x^2}

(b) e^{-x^4}

(c) $-2x^3e^{-x^4}$

(d) xe^{-x^2}

(18) $\int \frac{x}{x-1}dx =$

(a) $\ln|x-1|+c$

(b) $x-\ln|x-1|+c$

(c) $\frac{x^2}{2}+\ln|x-1|+c$

(d) $x+\ln|x-1|+c$

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(19) $\int_0^1 |2x-1|\,dx =$

(a) 2

(b) 0

(c) $\frac{1}{2}$

(d) 1

(20) **If** $\int_1^6 f(x)\,dx = 3$ **then** $\int_1^6 [2f(x)+5]\,dx =$

(a) 11

(b) 31

(c) 6

(d) 17

(21) **The differential equation of the curve is** $\frac{dy}{dx} = 2x^2 - 5$, **if** $(3,8)$ **lies on the curve then equation of curve is;**

(a) $y = \frac{2}{3}x^3 - 5x + 8$

(b) $y = \frac{2}{3}x^3 - 5x + 11$

(c) $y = \frac{2}{3}x^3 - 5x + 3$

(d) $y = \frac{2}{3}x^3 - 5x + 5$

(22) **If** $\int_4^9 f(x)\,dx = 7$ **and** $\int_{13}^4 f(x)\,dx = 5$ **then** $\int_9^{13} f(x)\,dx =$

(a) 2

(b) 12

(c) -2

(d) -12

(23) $\int_0^{10\pi} |\sin x|\,dx =$

(a) 20

(b) 8

(c) 10

(d) 18

(24) **Area under the curve** $y = \sqrt{1-\cos 2x}$ **from** $x = 0$ **to** $x = 2\pi$

(a) $4\sqrt{2}$

(b) $2\sqrt{2}$

(c) $\sqrt{2}$

(d) 4

(25) $\int_{-e}^{e} x^7 \cos^{49} x\,dx =$

(a) $\frac{1}{e^2}$

(b) e^2

(c) $2e$

(d) 0

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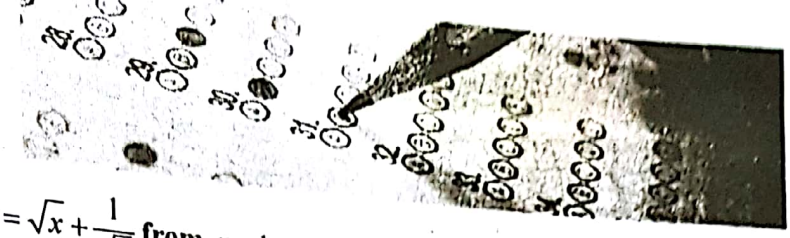

(26) The area under the curve $y=\sqrt{x}+\frac{1}{\sqrt{x}}$ from $x=1$ to $x=4$ is;

(a) $\frac{8}{3}$

(b) $\frac{20}{3}$

(c) $\frac{28}{3}$

(d) 12

(27) Which of the following has same value as that of $\int_1^2 x^2\,dx$;

(a) $\int_2^3 (x^2-1)\,dx$

(b) $\int_2^3 (x^2+1)\,dx$

(c) $\int_2^3 (x-1)^2\,dx$

(d) $\int_2^3 x^2 dx$

(28) Area enclosed by the curve $y^2=4x$ and line $x=3$;

(a) $2\sqrt{6}$

(b) $4\sqrt{3}$

(c) $8\sqrt{3}$

(d) $3\sqrt{2}$

(29) Solution of $xdy+y(x-1)dx=0$ is;

(a) $\ln y = cxe^{-x}$

(b) $y=cxe^{-x}$

(c) $\ln y = \ln x + c$

(d) None of these

(30) Order and degree of differential equation $2x\frac{dy}{dx}+3x^2\frac{d^2y}{dx^2}=4y\left[\frac{d^3y}{dx^2}\right]^{\frac{1}{3}}$ respectively are;

(a) $3,\frac{1}{3}$

(b) 2,1

(c) 3,1

(d) 3,3

(31) $\int\left(x^2-\frac{1}{x^4}\right)dx=0$ represents;

(a) Cubic equation

(b) Exponential equation

(c) Reciprocal equation

(d) Quadratic equation

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(32) $\int_0^{\alpha} \frac{dx}{x^2+9} = \frac{\pi}{9}$, then $\alpha =$

(a) 1

(b) $\frac{1}{\sqrt{3}}$

(c) $3\sqrt{3}$

(d) 3

(33) $\frac{d}{dx}\int_{\frac{-\pi}{3}}^{\frac{4\pi}{3}} 3x^2 \sin 3x\, dx =$

(a) $3x^2 \sin 3x$

(b) $\frac{-19\pi}{3}$

(c) 0

(d) Not possible

(34) $\sec x$ **is anti derivative of;**

(a) $\ln|\sec x + \tan x|$

(b) $\sec x . \tan x$

(c) $\sec^2 x$

(d) $\tan^2 x$

(35) $\int \tan^2 x\, dx =$

(a) $2\tan x \sec^2 x + c$

(b) $\tan x + x + c$

(c) $\tan x - x + c$

(d) $x \tan x - x + c$

(36) $\int \frac{e^{\omega x}}{e^{-\omega^2 x}} dx = ?$ **, where** ω **is cube root of unity;**

(a) $\frac{e^{\omega x}}{\omega} + c$

(b) $e^x + c$

(c) $\frac{e^{\omega^2 x}}{\omega^2} + c$

(d) $c - e^{-x}$

(37) $\int \sqrt{1+\cos\left(\frac{x}{4}\right)} dx = ?$

(a) $8\sqrt{2}\sin\left(\frac{x}{8}\right) + c$

(b) $4\sqrt{1+\sin\left(\frac{x}{4}\right)} + c$

(c) $\frac{8}{3}\left(1+\cos\left(\frac{x}{4}\right)\right)^{\frac{3}{2}} + c$

(d) None of these

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(38) $\int \frac{1}{1+5e^{-x}}dx = ?$

(a) $\frac{e^x}{x+5e^{-x}}+c$

(b) $\ln\left|1+5e^{-x}\right|$

(c) $\ln\left|5+e^x\right|$

(d) None of these

(39) $\int x\cot x^2\,dx$

(a) $\ln\left|\sin x\right|+c$

(b) $\ln\left|\sin x^2\right|+c$

(c) $2\ln\left|\sin x^2\right|+c$

(d) $\frac{1}{2}\ln\left|\sin x^2\right|+c$

(40) **If** $\int e^{\alpha x}(\cos x-\sin x)dx = e^{\alpha x}\sin x$ **then value of** α **is;**

(a) $\alpha = 1$

(b) $\alpha = -1$

(c) $\alpha = \frac{1}{2}$

(d) $\alpha = 0$

(41) $\int e^{\sin^2 x}\left(\sin^2 2x+2\cos 2x\right)dx = ?$

(a) $e^{\sin^2 x}.\sin 2x+c$

(b) $e^{\sin^2 x}.\sin^2 x+c$

(c) $e^{\sin^2 x}.\cos^2 2x+c$

(d) $e^{\sin^2 x}.\sin^2 2x+c$

(42) $\int \frac{2^x}{\sqrt{1-4^x}}dx = ?$

(a) $\sin^{-1}2^x+c$

(b) $\ln 2.\sin^{-1}2^x$

(c) $\frac{1}{\ln 2}\sin^{-1}2^x+c$

(d) $\sin^{-1}4^x+c$

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(43) If $\int_{-3}^{1} f(\theta)\,d\theta = 4$ and $\int_{-3}^{5} f(\lambda)\,d\lambda = -1$ then $\int_{1}^{5} f(t)\,dt = ?$

(a) 3

(b) 5

(c) -5

(d) Cannot be calculated

(44) **Area of region enclose by** $y = \sqrt{x}$ **and** $y = x^2$ **is;**

(a) $\frac{2}{3}$

(b) $\frac{10}{3}$

(c) $\frac{1}{3}$

(d) $\frac{8}{3}$

(45) **Area under the curve** $y = \tan x$ **from** $x = \frac{-\pi}{4}$ **to** $x = \frac{\pi}{4}$;

(a) 0

(b) 1

(c) $\ln 2$

(d) $\ln \sqrt{2}$

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ECAT MATHEMATICS CHAPTER-4

TOPIC: INTRODUCTION TO ANALYITIC GEOMETRY

(1) For what value of t, point $(3,t)$ lie on a unit circle centered at origin?

(a) $\sqrt{2}$ (b) 1

(c) $2\sqrt{2}$ (d) No value of t

(2) If distance between $(2,x)$ and $(5,y)$ is 5 then which of the following could be the value of x and y?

(a) $x=1, y=3$ (b) $x=-1, y=-3$

(c) $x=3, y=-1$ (d) $x=0, y=0$

(3) Which of the following point divides the join of $(-1,3)$ and $(1,2)$ in $m:n$? Such that $7m=3n$

(a) $(1.3, 2.3)$ (b) $(1.7, 2.7)$

(c) $(1.4, 2.5)$ (d) $(2.5, 1.9)$

(4) A triangle formed by joining the points $(1,5),(2,4)$ and $(3,0)$. Find a point that divides the joining from orthocenter to circumcenter in $2:1$;

(a) $(1,3)$ (b) $(3,2)$

(c) $(2,1)$ (d) $(2,3)$

(5) If translation of coordinate axes through $O'(h,k)$, then XY – coordinates of origin referred to the translated axes are;

(a) (x,y) (b) (X,Y)

(c) $(0,0)$ (d) $(-h,-k)$

(6) The segment from $(-1,4)$ to $(2,-2)$ is extended three times, then terminal point of segment is:

(a) $(11,-20)$ (b) $(9,-17)$

(c) $(5,13)$ (d) $(-7,12)$

(7) Gradient of a line whose intercepts are additive inverse of each other;

(a) 0 (b) -1

(c) 1 (d) Cannot be determine

(8) Product of slopes of lines $x=0$ and $y=0$ is;

(a) -1 (b) 0

(c) 1 (d) None of these

(9) The coordinates given in the table represents a line $y=mx+c$,

x	0	1	2	3
y	2	$\frac{10}{3}$	$\frac{14}{3}$	6

values of m and c are;

(a) $m=2, c=4$ (b) $m=2, c=\frac{4}{3}$

(c) $m=4, c=2$ (d) $m=\frac{4}{3}, c=2$

(10) The y – intercept and slope of line $3x+2y+6=0$ is;

(a) y – intercept $=\frac{-3}{2}$, slope $=-3$ (b) y – intercept $=-3$, slope $=\frac{-3}{2}$

(c) y – intercept $=\frac{-3}{2}$, slope $=\frac{-3}{2}$ (d) y – intercept $=-3$, slope $=-3$

(11) Equation of perpendicular bisector on line segment formed by joining $(-1,3)$ and $(4,-5)$ is:

(a) $10x-16y-31=0$ (b) $16x-10y+31=0$

(c) $8x+y-11=0$ (d) Both (a) and (c)

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(12) If area of triangle formed by the points $(2,\alpha),(1,\alpha^2)$ and $(2,1)$ is same as the area of square whose one side is 1m, then $\alpha =$

(a) 2 (b) -1
(c) 3 (d) 0

(13) Equation of a line parallel to the line $x+7y+7=0$, having $y-$intercept 2 is;

(a) $x-4y+7=0$ (b) $x+14y+2=0$
(c) $x+7y-2=0$ (d) $x+7y+2=0$

(14) For what value of t, lines $2x-3y+2=0, x-ty+5=0$ and $5x-y-1=0$ have a common point?

(a) $\frac{30}{7}$ (b) $\frac{26}{5}$
(c) $\frac{35}{6}$ (d) $\frac{27}{4}$

(15) Which of the following is true for the line $2x+7y-28=0$?

I. $x-$intercept is 14 II. It can never passes through origin
III. Slope of line is rational number IV. Inclination of a line is 45°
(a) I only (b) I, II and III only
(c) I and III only (d) I, II, III, and IV

(16) Circumcenter of ΔABC with vertices $A(8,6), B(8,-2)$ and $C(2,-2)$ is;

(a) $(-5,2)$ (b) $(5,-2)$
(c) $(3,-4)$ (d) $(5,2)$

(17) The centroid of ΔABC is $(1,4)$, if two of its vertices are $(4,-3)$ and $B(-9,7)$, then the area of triangle is;

(a) 183 (b) 91.5
(c) 61 (d) 29

(18) If points $A(k,-1), B(2,1)$ and $C(4,5)$ are collinear then value of k is;

(a) -1 (b) 1
(c) 3 (d) -3

(19) A line lying midway between two parallel lines ℓ_1 and ℓ_2 is $5x-3y+4=0$ then ℓ_1 and ℓ_2 are;

(a) $\ell_1: 5x+3y+6=0$, $\ell_2: 5x+3y+2=0$ (b) $\ell_1: 5x-3y+5=0$, $\ell_2: 5x+3y+3=0$
(c) $\ell_1: 5x-3y+5=0$, $\ell_2: 5x-3y+3=0$ (d) Not possible

(20) A point $(2t,3t-1)$ is at a distance of 4 units from line $3x-4y+5=0$, then point lie in;

(a) 1st and 4th Quad. (b) 1st and 3rd Quad.
(c) 2nd and 4th Quad. (d) 2nd and 3rd Quad.

(21) Which of the following point lie above the line $3x-5y+1=0$?

(a) $(1,1)$ (b) $(0,1)$
(c) Both (a) and (b) (d) (0, 0)

(22) Which of the following is homogeneous equation of degree 1?

(a) $ax^2+2hxy+by^2=0$ (b) $ax+by=0$
(c) $ax^2+2hxy+by^2=c$ (d) $ax+by+c=0$

(23) The value of h for which $2x^2-2hxy+3y^2=0$ represents a pair of coincident lines;

(a) $\pm\sqrt{6}$ (b) $\pm\sqrt{2}$
(c) ± 1 (d) $\pm 2\sqrt{3}$

(24) For what value of k, $3x^2-8xy+ky^2=0$ represents two mutually perpendicular lines?

(a) $k=1$ (b) $k=-3$
(c) $k=3$ (d) $k=\frac{3}{2}$

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(25) If $8x^2-9xy+2y^2=0$ represents a pair of straight lines, then geometric mean between their slopes is;

(a) ± 4
(b) $\pm\sqrt{2}$
(c) ± 2
(d) ± 3

(26) Point of intersection of two straight lines given by $4x^2-12xy+9y^2=0$ is;

(a) $(0,0)$
(b) $(3,2)$
(c) $(-9,-6)$
(d) All of these

(27) If joint equation of lines through origin perpendicular to the lines $ax^2+by^2+2hxy=0$ is same, then which of the following is true?

(a) $a-b=0$
(b) $a+b=0$
(c) $h=0$
(d) $\frac{a+b}{2}=h$

(28) Angle between pair of straight lines given by $x^2-xy-6y^2=0$ is;

(a) 120°
(b) 60°
(c) 45°
(d) None of these

(29) A point where the lines $x-3y+2=0$ and $5x-y-4=0$ meet each other?

(a) $(1,0)$
(b) $(2,1)$
(c) $(1,1)$
(d) No such point

(30) If the lines $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ are orthogonal to each other then which of the following is true;

(a) $a_1b_2-a_2b_1=0$
(b) $a_1b_1-a_2b_2=0$
(c) $a_1a_2-b_1b_2=0$
(d) $a_1b_1+a_2b_2=0$

(31) Equation of line bisecting 2nd and 4th Quadrant is;

(a) $x-y=0$
(b) $x+y=0$
(c) $x-y-1=0$
(d) $x+y+1=0$

(32) Angle between the lines $x-2y+3=0$ and $x-y+1=0$ is;

(a) $\tan^{-1}\left(\frac{1}{9}\right)$
(b) $\tan^{-1}\left(\frac{1}{3}\right)$
(c) $\tan^{-1}\left(\frac{3}{10}\right)$
(d) None of these

(33) The matrix form of system of linear equations $x-4y+9=0, x=0$ and $y=0$;

(a) $\begin{bmatrix}1&-4&9\\1&0&0\\0&0&1\end{bmatrix}\begin{bmatrix}x\\y\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$
(b) $\begin{bmatrix}1&-4&9\\1&0&0\\0&1&0\end{bmatrix}\begin{bmatrix}x\\y\\0\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$
(c) $\begin{bmatrix}1&-4&9\\0&1&0\\1&0&0\end{bmatrix}\begin{bmatrix}x\\y\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$
(d) $\begin{bmatrix}1&-4&9\\1&0&0\\0&1&0\end{bmatrix}\begin{bmatrix}x\\y\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$

(34) The area bounded by the lines $x=4$, $y=-3$ and $2x-y-4=0$ is;

(a) $\frac{49}{4}$
(b) $\frac{49}{2}$
(c) $\frac{81}{2}$
(d) $\frac{81}{4}$

(35) Which of the following is an equation of a straight line passing through $A(x_1,y_1)$ and $B(x_2,y_2)$?

(a) $\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}$
(b) $(y-y_1)(x_2-x_1)=(x-x_1)(y_2-y_1)$
(c) $\begin{vmatrix}x_1&x_2&x\\y_1&y_2&y\\1&1&1\end{vmatrix}=0$
(d) All of these

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(36) **Slope – intercept form of the line $ax+by+c=0$ is;**

(a) $y=mx+c$

(b) $y-y_1=m(x-x_1)$

(c) $\frac{x}{a}+\frac{y}{b}=1$

(d) None of these

(37) **Distance of the straight line $12x+5y+13=0$ from origin is;**

(a) $\frac{13}{\sqrt{5}}$

(b) 1

(c) $\frac{13}{12}$

(d) $\frac{5}{13}$

(38) **How many lines can pass through the point of intersection of $6x+3y+7=0$ and $2x+y+24=0$?**

(a) Infinite

(b) Only one

(c) Zero

(d) 2

(39) **The xy – coordinate axes are rotated about the origin through an angle of $\arctan\frac{4}{3}$. The coordinates of a point P referred to the new axes are $P(-1,-7)$. The coordinates of the original point are;**

(a) $(5,-5)$

(b) $(-5,-5)$

(c) $(5,5)$

(d) $(-5,5)$

(40) **The average entry test score of medical colleges was 592 in the year 1998 while the score was 564 in 2002. Assuming the relationship between the score and time is linear. The average score in 2006:**

(a) 500

(b) 536

(c) 589

(d) 570

(41) **If $A(0,b).B(0,0)$ and $C(a,0)$ are vertices of a triangle, then the condition for which median AD and BE are mutually perpendicular?**

(a) $a=\sqrt{2}b$

(b) $b=-\sqrt{2}a$

(c) $a=\pm\sqrt{2}b$

(d) $a=\frac{b}{\sqrt{2}}$

(42) **Equation of line passing through the point of intersection of $x-3y+4=0$ and $9y-3x+1=0$ is;**

(a) $3x-y+1=0$

(b) $2x-3y+5=0$

(c) $5x-4y+9=0$

(d) None of these

(43) **Points $A(0,0),B(-1,1)$ and $C(1,-1)$ form;**

(a) Isosceles triangle

(b) Equilateral triangle

(c) Scalene triangle

(d) None of these

(44) **Intercept form of line $2x+3y-5=0$;**

(a) $\frac{2x}{5}+\frac{3y}{5}=1$

(b) $\frac{x}{\frac{5}{2}}+\frac{y}{\frac{5}{3}}=1$

(c) $\frac{x}{\frac{5}{2}}+\frac{y}{\frac{5}{3}}=1$

(d) $\frac{x}{2}+\frac{y}{3}=5$

(45) **Equation of a line passing through $(4,3)$ and making intercept on coordinate axes whose sum is –1 is;**

(a) $4x+y-19=0$

(b) $x-3y+2=0$

(c) $3x-2y-6=0$

(d) Not possible

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ECAT MATHEMATICS CHAPTER-5+7

TOPIC: LINEAR PROGRAMMING, VECTORS

(1) **Solution set of the inequality $x \le \frac{7}{2}$ is _____**

(a) $[-\infty, 3.5]$ (b) $[-\infty, 7]$

(c) $(-\infty, 3.5)$ (d) $(-\infty, 3.5]$

(2) **Associated equation of the inequality $2x+3y<6$ is _____**

(a) $2x+3y-6=0$ (b) $2x+3y+6=0$

(c) $2x+3y=-6$ (d) $2x+3y=0$

(3) **$x+y+3=0$ is associated equation of _____**

(a) $x+y+3<0$ (b) $x+y+3>0$

(c) $x+y<-3$ (d) All of these

(4) **Which of the following is solution of the inequality $x+2y<6$?**

(a) $(1,1)$ (b) $(2,2)$

(c) $(3,2)$ (d) $(3,3)$

(5) **How many points can be the solution of the inequality $x+4y<-8$?**

(a) Only one (b) All points in 4th quadrant

(c) Infinite (d) Only $(0,0)$

(6) **The graph of the inequality $x+2>0$ is _____**

(a) Open right half plane (b) Closed right half plane

(c) Open left half plane (d) Closed left half

(7) **Which of the following cannot be a test point of the inequality $3x+y \ge 2$?**

(a) $(0,0)$ (b) (1, 1)

(c) (2, 1) (d) (1, -1)

(8) **Each point of the feasible region is called _____**

(a) Feasible solution (b) Feasible solution set

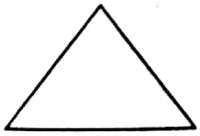
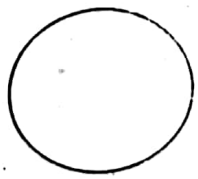
(c) Feasible solution region (d) All of these

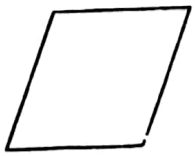
(9) **Which of the following is a corner point of the feasible region of $2x-3y<6$, $2x+y \ge 2$, $x \ge 0, y \ge 0$?**

(a) $(1,0)$ (b) $(3,0)$

(c) $(0,2)$ (d) All of these

(10) **Which of the following region is NOT convex?**

(a)  (b) 

(c) (d) 

(11) **Values of x and y which maximize the objective function $f(x,y)=x+3y$ subject to the constraints $2x+5y \le 30, 5x+4y \le 20, x \ge 0 y \ge 0$ are _____ respectively.**

(a) 0, 0 (b) 0, 5

(c) 5, 0 (d) 5, 5

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(12) **Which of the following is NOT scalar?**
(a) Speed
(b) Work
(c) Potential Energy
(d) Force

(13) **Norm of the vector** $2\underline{i}-7\underline{k}=?$
(a) 49
(b) 53
(c) 7
(d) 7.28

(14) **If** $A(1,-1), B(2,0), C(-1,3)\, D(-2,2)$ **then** $\overrightarrow{AB}-\overrightarrow{CD}=?$
(a) $2\underline{i}+3\underline{j}$
(b) $4\underline{j}+\underline{i}$
(c) $\underline{i}-4\underline{j}$
(d) $2\underline{i}+2\underline{j}$

(15) **If** $\overrightarrow{AB}=4\underline{i}-2\underline{j}$ **and** $B(-2,5)$ **then a vector from A to origin is**
(a) $-6\underline{i}+7\underline{j}$
(b) $6\underline{i}+7\underline{j}$
(c) $6\underline{i}-7\underline{j}$
(d) $-6\underline{i}-7\underline{j}$

(16) **Which of the following is a unit vector** $\|$ **to** $3\underline{i}-4\underline{j}$?
(a) $3\underline{i}-4\underline{j}$
(b) $\frac{3}{5}i-\frac{4}{5}\underline{j}$
(c) $\frac{3}{5}j-\frac{4}{5}i$
(d) $\frac{4}{5}i-\frac{3}{5}\underline{j}$

(17) **If P.V of a point C is** $2\underline{i}-3\underline{j}$ **and that of D is** $3\underline{i}+2\underline{j}$ **then the P.V of a point which divides the** $\overrightarrow{CD}$ **in ratio 4:3 is;**
(a) $\frac{18}{7}\underline{i}-\frac{1}{7}\underline{j}$
(b) $\frac{18}{7}\underline{i}+\frac{1}{7}\underline{j}$
(c) $\frac{1}{7}\underline{i}-\frac{18}{7}\underline{j}$
(d) $\frac{1}{7}\underline{i}+\frac{18}{7}\underline{j}$

(18) **Which of the following is NOT a unit vector?**
(a) $\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$
(b) (1, 0, 0)
(c) (1, 1, 1)
(d) $(\sin\alpha, \cos\alpha, 0)$

(19) **Distance of the point** $P(x_1, y_1, z_1)$ **from z – axis = ?**
(a) $\sqrt{x_1^2+z_1^2}$
(b) $\sqrt{x_1^2+y_1^2}$
(c) $\sqrt{x_1^2+y_1^2+z_1^2}$
(d) z_1

(20) **If** $\underline{a}=2\underline{i}+3\underline{j}+\underline{k}$ **and** $\underline{b}=4\underline{i}+6\underline{j}+2\underline{k}$ **then** $|\underline{a}+2\underline{b}|=?$
(a) $10\underline{i}+15\underline{j}+\underline{k}$
(b) $-10\underline{i}+15\underline{j}+\underline{k}$
(c) $10\underline{i}-5\underline{j}+3\underline{k}$
(d) None of these

(21) **If** α, β, γ **are direction angle of a unit vector then which of the following must be true?**
(a) $\alpha^2+\beta^2+\gamma^2=1$
(b) $\cos\alpha+\cos\beta+\cos\gamma=1$
(c) $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$
(d) All of these

(22) **For a unit vector** $2\underline{i}+3\underline{j}+4\underline{k}$, $\cos\beta$ **= ?**
(a) $\frac{2}{\sqrt{29}}$
(b) $\frac{3}{\sqrt{29}}$
(c) $\frac{4}{\sqrt{29}}$
(d) $\frac{1}{\sqrt{29}}$

(23) **For what value of "m"** $|m\underline{i}+(m+1)\underline{j}+2\underline{k}|=3$?
(a) 1
(b) -2
(c) 3
(d) Both a and b

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(24) **Which of the following is a vector of norm 2 and parallel to $-\underline{i}+\underline{j}+\underline{k}$?**

(a) $\frac{-1}{\sqrt{3}}\underline{i}-\frac{2}{\sqrt{3}}\underline{j}+\frac{2}{\sqrt{3}}\underline{k}$

(b) $\frac{2}{\sqrt{3}}\underline{i}-\frac{2}{\sqrt{3}}\underline{j}-\frac{2}{\sqrt{3}}\underline{k}$

(c) $-\frac{1}{\sqrt{3}}\underline{i}-\frac{1}{\sqrt{3}}\underline{j}+\frac{1}{\sqrt{3}}\underline{k}$

(d) Both b and c

(25) **For what value of "m" $2\underline{i}+3\underline{j}+4\underline{k}$, $-\underline{i}+3\underline{j}-\underline{k}$ and $\underline{i}+6\underline{j}+m\underline{k}$ represents two sides of triangle?**

(a) $m=?$

(b) $m=-2$

(c) $m=0$

(d) $m=1$

(26) **Given that $3\underline{i}-\underline{j}+4\underline{k}$ and $m\underline{i}+n\underline{j}-2\underline{k}$ are parallel vectors then which of the following is NOT true?**

(a) $m+n=-1$

(b) $m-n=1$

(c) $\frac{m}{n}=-3$

(d) $mn=\frac{-3}{4}$

(27) **Direction cosines of the vector $6\underline{i}-2\underline{j}+\underline{k}$ are _____ respectively.**

(a) $\frac{6}{\sqrt{41}},\frac{-2}{\sqrt{41}},\frac{1}{\sqrt{41}}$

(b) $\frac{-6}{\sqrt{41}},\frac{2}{\sqrt{41}},\frac{-1}{\sqrt{41}}$

(c) $-\frac{6}{\sqrt{41}},\frac{2}{\sqrt{41}},\frac{1}{\sqrt{41}}$

(d) Both (a) and (b)

(28) **Which of the following triples can be the direction angles of a single vector?**

(a) $45^\circ, 45^\circ, 60^\circ$

(b) $30^\circ, 45^\circ, 60^\circ$

(c) $60^\circ, 60^\circ, 90^\circ$

(d) $45^\circ, 60^\circ, 60^\circ$

(29) **For two vector $a_1\underline{i}+a_2\underline{j}+a_3\underline{k}$, $b_1\underline{i}+b_2\underline{j}+b_3\underline{k}$ $\underline{a}.\underline{b}=?$**

(a) $a_1a_2+b_1b_2+c_1c_2$

(b) $a_1b_1+a_2b_2+a_3b_3$

(c) $|\underline{a}||b|\cos\theta$

(d) Both (b) and (c)

(30) **The angle between $\underline{a}=2\underline{i}-\underline{j}+\underline{k}$ and $\underline{b}=-\underline{i}+\underline{j}$ is _____**

(a) 30°

(b) 120°

(c) 150°

(d) 60°

(31) **For what value of "m" the vector $2\underline{i}+m\underline{j}+5\underline{k}$ and $3\underline{i}+\underline{j}+m\underline{k}$ are perpendicular?**

(a) 1

(b) -1

(c) 0

(d) 3

(32) **Projection of $\underline{a}=3\underline{i}+\underline{j}-\underline{k}$ along $\underline{b}=-2\underline{i}-\underline{j}+\underline{k}$ is _____**

(a) $\frac{-8}{\sqrt{6}}$

(b) $\frac{8}{\sqrt{6}}$

(c) $\frac{8}{\sqrt{11}}$

(d) $\frac{-8}{\sqrt{11}}$

(33) **Which of following is NOT true?**

(a) $b=c\cos A+a\cos C$

(b) $c=a\cos A+b\cos B$

(c) $c=a\cos B+b\cos A$

(d) $a=b\cos C+c\cos B$

(34) $\underline{a}\times\underline{b}=?$ **by definition**

(a) $|a||b|\sin\theta$

(b) $|\underline{a}\times\underline{b}|\sin\theta\hat{n}$

(c) $|\underline{a}||\underline{b}|\sin\theta\hat{n}$

(d) All of these

(35) **A vector which is $\perp$ to both $\underline{a}$ and $\underline{b}$ is _____**

(a) $\underline{a}\times\underline{b}$

(b) $\frac{\underline{a}\cdot\underline{b}}{|\underline{a}\times\underline{b}|}$

(c) $|\underline{a}\times\underline{b}|$

(d) $\frac{|\underline{a}\times\underline{b}|}{\underline{a}\times\underline{b}}$

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(36) **Which of the following is NOT true?**

(a) $\hat{i}\hat{j}=0$ (b) $\underline{u}\times\underline{u}=0$

(c) $\underline{a}.\underline{b}=\underline{b}.\underline{a}$ (d) $\underline{u}\times\underline{v}=-\underline{v}\times\underline{u}$

(37) **If** $\underline{a}=2\underline{i}+\underline{j}+\underline{k}$ **and** $\underline{b}=4\underline{i}+2\underline{j}-\underline{k}$ **then** $\underline{a}.\underline{b}\times\underline{a}=?$

(a) $-\underline{i}+6\underline{j}+8\underline{k}$ (b) $4\underline{i}-2\underline{j}+3\underline{k}$

(c) 0 (d) 1

(38) **For the three vector** $\underline{u},\underline{v}$ and $\underline{\omega}$ **if** $\underline{u}+\underline{v}+\underline{w}=0$ **then** $u\times v=?$

(a) $\underline{v}\times\underline{u}$ (b) $\underline{v}\times\underline{\omega}$

(c) $\underline{\omega}\times\underline{v}$ (d) $\underline{u}\times\underline{w}$

(39) **Which of the following is same as** $\underline{a}.\underline{b}\times\underline{c}$?

(a) $\underline{a}\times\underline{b}.\underline{c}$ (b) $\underline{b}.\underline{c}\times\underline{a}$

(c) $\underline{b}\times\underline{c}.\underline{a}$ (d) All of these

(40) **For what value of "P"** $P\underline{i}+\underline{j},\ \underline{i}+\underline{j}+3\underline{k}$ **and** $2\underline{i}+\underline{j}-2\underline{k}$ **are coplanar vectors?**

(a) $\frac{8}{5}$ (b) $\frac{5}{8}$

(c) $-\frac{5}{8}$ (d) 0

(41) $\left[3\underline{j}\ \ \underline{k}\ \ \underline{i}\right]=?$

(a) 3 (b) –3

(c) 2 (d) 0

(42) **Volume of parallelepiped determine by**

$\underline{a}=\underline{i}+2\underline{j}-\underline{k}$, $\underline{b}=\underline{i}-2\underline{j}+3\underline{k}$ and $c=\underline{i}-7\underline{j}-4\underline{k}$ is;

(a) 24 (b) 48

(c) 64 (d) 8

(43) **Volume of a tetrahedron whose verti. s are (0, 1, 2), (3, 2, 1), (1, 2, 1) and (5, 5, 6) is**

(a) 3.472 (b) 2.66

(c) 3.676 (d) 4.321

(44) **If a particle is displaced from A (3, 1, -2) to B (2, 4, 6) by a force** $\underline{F}=4\underline{i}+3\underline{j}+5\underline{k}$ **then work done?**

(a) 45 units (b) 54 units

(c) 60 units (d) 70 units

(45) **A force** $\underline{A}=7\underline{i}+4\underline{j}-3\underline{k}$ **is applied at** $P(1,-2,3)$**. Its moment about the point** $G(2,1,1)$ *is;*

(a) $\underline{i}+11\underline{j}+17\underline{k}$ (b) $\underline{i}+11\underline{j}+17\underline{k}$

(c) $\underline{i}+\underline{j}+11\underline{k}$ (d) None of these

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ECAT MATHEMATICS CHAPTER-6

TOPIC: **CONIC SECTIONS**

Self assessment Test Part A (Circle)

(1) **If the cutting plane is parallel to the axis of the cone and intersects both of its nappes then the curve of intersection is _____**

(a) Parabola
(b) Hyperbola
(c) Ellipse
(d) Point Circle

(2) **Which of the following is parametric equation of circle?**

(a) $x = a\cos\alpha$, $y = a\sin\alpha$
(b) $x = r\cos\theta$, $y = r\sin\theta$
(c) $x = \cos t$, $y = \sin t$
(d) All of these

(3) **Equation of a circle whose centre is $(5,-2)$ and radius is 5;**

(a) $(x-5)^2+(y-2)^2=25$
(b) $(x+5)^2+(y+2)^2=25$
(c) $(x-5)^2+(y+2)^2=25$
(d) $(x+5)^2+(y-2)^2=25$

(4) **Equation of circle for which ends of a diameter are $(0,-2)$ and $(0,2)$.**

(a) $x^2+y^2=2$
(b) $x^2+y^2=4$
(c) $x^2+y^2+4=0$
(d) $x^2+y^2-2=0$

(5) **If $ax^2+2hxy+y^2+16x-9y-13=0$ represents a circle then $a+h=?$**

(a) 1
(b) 2
(c) 0
(d) -1

(6) **For what values of 'h' and "k" the circles $x^2+y^2+6x+8y-3=0$ and $(x-h)^2+(y-k)^2=16$ concentric?**

(a) $h=6, k=8$
(b) $h=3, k=4$
(c) $h=-3, k=-4$
(d) $h=3, k=-4$

(7) **If $3x-2y+k=0$ is tangent to the circle $x^2+y^2+6x-4y=0$ then $k=?$**

(a) 1
(b) -1
(c) 0
(d) $\frac{7}{2}$

(8) **Which one of the following points lies inside the circle $x^2+y^2=100$?**

(a) $(9,14)$
(b) $(9,8)$
(c) $(9,1)$
(d) $(10,3)$

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(9) How many real tangents can be drawn from a point $P(-5,6)$ to the circle

$x^2+y^2+4x-6y-12=0$

(a) 2 (b) 1

(c) 0 (d) Infinite

(10) Length of tangent drawn from $(0,5)$ to the circle $x^2+y^2-16=0$

(a) 9 (b) 3

(c) 2 (d) 1

(11) Which of the following can be a normal line of the circle $x^2+y^2=25$?

(a) $3x+2y+1=0$ (b) $x+y+1=0$

(c) $3x+2y=0$ (d) $x+3y-10=0$

(12) The length of the chord intercepted by the line $3x+4y=10$ to the circle $x^2+y^2=25$ is;

(a) $\sqrt{13}$ (b) 23

(c) $2\sqrt{21}$ (d) $\frac{\sqrt{3}}{2}$

(13) The tangential distance from point $P(-1,2)$ to the circle $x^2+y^2+4x+2y=0$ is

(a) $\sqrt{10}$ (b) $\sqrt{5}$

(c) $2\sqrt{5}$ (d) $\sqrt{21}$

(14) Equation of the circle which touches the line $5x+12y=11$ and has centre at $(3,4)$ is

(a) $(x+3)^2-(y+4)^2=16$ (b) $(x-3)^2+(y-4)^2=(4)^2$

(c) $(x-3)^2-(y-4)^2=4$ (d) $(x+3)^2-(y+4)^2=4$

(15) Two circles $S(C_1,r_1)$ and $S(C_2,r_2)$ touch each other internally if and only if

(a) $|C_1C_2|=r_1+r_2$ (b) $|C_1C_2|=r_1r_2$

(c) $|C_1C_2|=r_1-r_2$ (d) $|C_1C_2|=|r_1-r_2|$

(16) The line $y=mx+c$ intersects the circle $x^2+y^2=a^2$ at the two real distinct points if

(a) $a^2(1+m^2)-c^2=0$ (b) $a^2(1+m^2)-c^2>0$

(c) $-a\sqrt{1+m^2}<c<a\sqrt{1+m^2}$ (d) $a\sqrt{1+m^2}<c<-a\sqrt{1+m^2}$

(17) If the equation of circle is $3x^2+3y^2+kxy+9x+(k-6)y+3=0$ then its centre is;

(a) $\left(-\frac{9}{2},3\right)$ (b) $\left(\frac{-3}{2},1\right)$

(c) (3,1) (d) Cannot be determined

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(18) **If the equation of circle is $(4a-3)x^2+ay^2+6x-2y+2a=0$, then its radius is**

(a) $\sqrt{10}$
(b) $2\sqrt{3}$
(c) $2\sqrt{2}$
(d) 8

(19) **The area of a circle centered at $(1,2)$ and passing through $(4,6)$ is**

(a) 30π sq. units
(b) 5π .sq. units
(c) 15π sq. units
(d) 25π squints

(20) **Which one is not the feature of the equation of a circle $x^2+y^2-2ax-2by+a^2+b^2-r^2=0$?**

(a) It is an equation of second degree x and y
(b)The coefficients of x^2 and y^2 are equal
(c) There is no term involving the product xy
(d) None

(21) **The value of eccentricity of the circle is**

(a) 0
(b) 1
(c) -1
(d) 4

(22) **The equation of normal to the circle $x^2+y^2+2gx+2fy=c$ from the point $P(x_1,y_1)$ is**

(a) $xx_1+yy_1=a^2$
(b) $x_1y=xy$
(c) $y-y_1=-\dfrac{(x_1+y)}{y_1+f}(x-x_1)$
(d) None

(23) **Mid point of hypotenuse of a right triangle is _____ of the circle**

(a) In-center
(b) Centroid
(c) Circumcenter
(d) Ex-center

(24) **Perpendicular bisector of any chord of a circle passes through**

(a) Point of tangency
(b) Centre of circle
(c) Origin
(d) End points of diameter

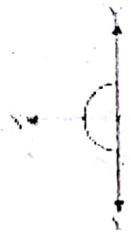
(25) **Tangent are drawn from $(-3,4)$ to the circle $x^2+y^2=21$ then equation of chord of contact is;**

(a) $3x+4y=21$
(b) $3x-4y=21$
(c) $3x+4y=21$
(d) $3x+4y=-21$

(26) **$y=\sqrt{a^2-x^2}$ represents;**

(a)

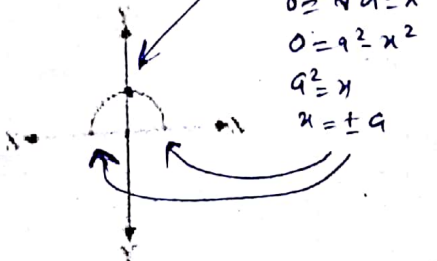
(b)



Put $x=0$
$y=\sqrt{a^2}$
$y=a$
Now
Put $y=0$
$0=\sqrt{a^2-x^2}$
$0=a^2-x^2$
$a^2=x$
$x=\pm a$

(c)

(d)



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(27) Area of a circle whose centre is $(\cos\theta, \sin\theta)$ and passes through the origin is _____

(a) π (b) 1

(c) πr^2 (d) 2π

(28) Angle subtended in the major arc of circle $x^2 + y^2 = 18$, whose end points are $(3,3),(-3,3)$;

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$

(29) For what value of m the circle $3x^2 + 3y^2 + 60x + 180y + 75m = 0$ become a point circle;

(a) 20 (b) 75

(c) 40 (d) 3

Self assessment Test Part B (Conics)

(1) Vertex of $x^2 = 4(y-1)$ is;

(a) $(0,1)$ (b) $(1,0)$

(c) $(0,0)$ (d) $(1,-1)$

(2) Length of latus rectum of $x^2 - 4x - 8y + 4 = 0$ is;

(a) 4 (b) 8

(c) 2 (d) 1

(3) Eccentricity of the Cone $y = 6x^2 - 1$ is;

(a) 0 (b) 1

(c) $\frac{1}{2}$ (d) 2

(4) Which of the following points is closest to the focus of the parabola $y^2 = -8(x-3)$?

(a) $(0,0)$ (b) $(3,0)$

(c) $(0,3)$ (d) $(3,3)$

(5) Length of latus rectum of the parabola $y = 16x^2$ is;

(a) 16 (b) 8

(c) 4 (d) None of these

(6) One of the foci of $\frac{x^2}{16} + \frac{y^2}{7} = 1$ is;

(a) $(0,9)$ (b) $(0,3)$

(c) $(1,-3)$ (d) $(3,0)$

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(7) **Length of latusrectum of the conic $4x^2+9y^2=36$ is;**

(a) $\frac{8}{3}$ (b) $\frac{3}{4}$

(c) $\frac{15}{4}$ (d) $\frac{3}{14}$

(8) **Centre of ellipse $\frac{(2x-1)^2}{4}+\frac{(y+2)^2}{16}=1$ is;**

(a) (1, -2) (b) $\left(\frac{1}{2},2\right)$

(c) $\left(\frac{1}{2},-2\right)$ (d) (-1, 2)

(9) **Distance between foci of the ellipse $25x^2+9y^2=225$ is;**

(a) 4 (b) 8

(c) 16 (d) 32

(10) **Sum of the focal distances of a point $\left(\frac{8}{3},1\right)$ on the ellipse $\frac{x^2}{8}+\frac{y^2}{9}=1$ is;**

(a) 8 (b) 16

(c) 9 (d) 6

(11) **Minimum area of the circle which can contain the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is;**

(a) 9π (b) 4π

(c) 3π (d) 2π

(12) **If the vertices of an ellipse are $(0,\pm5)$ and foci are $(0,\pm3)$ then its eccentricity is;**

(a) $\frac{5}{3}$ (b) $\frac{3}{5}$

(c) 5 (d) $\frac{3}{2}$

(13) **Equation of directrices of $a^2x^2+b^2y^2=a^2b^2$ are;**

(a) $x=\pm\frac{c}{e^2}$ (b) $y=\pm\frac{c}{e^2}$

(c) $x=\pm\frac{c}{a}$ (d) $y=\pm\frac{c}{a}$

(14) **Centre of the hyperbola $\frac{(y+h)^2}{9}-\frac{(x-k)^2}{16}=1$ is $(2,-2)$ then**

(a) $h=2,k=2$ (b) $h=-2,k=2$

(c) $h=2,k=-2$ (d) $h=-2,k=-2$

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(15) **Foci of hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ are;**

(a) $(\pm 7, 0)$

(b) $\left(\pm\sqrt{7}, 0\right)$

(c) $(\pm 25, 0)$

(d) $(\pm 5, 0)$

(16) **Eccentricity of hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is;**

(a) $e=\sqrt{\frac{1-a^2}{b}}$

$e = \frac{c}{a} = \frac{\sqrt{a^2+b^2}}{a} = \frac{\sqrt{a^2\left(1+\frac{b^2}{a^2}\right)}}{a}$

(b) $e=\sqrt{1+\frac{b^2}{a^2}}$

(c) $e=\sqrt{1+\frac{a^2}{b^2}}$

(d) None of these

(17) **Which of the following are asymptotes of the hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$?**

(a) $y=\pm\frac{3}{4}x$

(b) $x=\pm\frac{4}{3}y$

(c) $y=\pm\frac{4}{3}x$ $y=\pm\frac{b}{a}$

(d) $x=\pm\frac{16}{9}y$

(18) **Distance of a focus of the hyperbola $y^2-4x^2=4$ from its centre is;**

$\frac{y^2}{4}-\frac{x^2}{1}=1$ centre $=0$

(a) $\sqrt{5}$

(b) $\sqrt{3}$

(c) $2\sqrt{3}$

$c^2=1+4$, $c^2=5$, $c=\sqrt{5}$, $(0,\pm\sqrt{5})$ $(0,0)$

(d) $2\sqrt{5}$

(19) **A straight line $y=mx+c$ is tangent to the hyperbola $\frac{x^2}{a^2}-\frac{x^2}{b^2}=1$ if**

(a) $c=\pm\sqrt{a^2m^2+b^2}$

(b) $c=\pm\sqrt{a^2m^2-b^2}$

(c) $c=\pm\sqrt{b^2m^2+a^2}$

(d) $c=\pm\sqrt{b^2m^2-a^2}$

(20) **The product of the distances from the foci to a tangent to the hyperbola $\frac{x^2}{16}-\frac{y^2}{25}=1$ is:**

$\frac{x^2}{16}-\frac{y^2}{25}=1$, $c^2=a^2+b^2$

(a) 16

(b) 25

(c) 4

(d) 5

(21) **The number of points of intersection of $3x^2+4y^2=43$ and $2x^2-y^2=14$ are;**

$c^2=41$, $c=\sqrt{41}$, $2c=2\sqrt{41}$

(a) 2

(b) 4

(c) 1

(d) Zero

(22) $7x^2-6\sqrt{3}xy+13y^2-6=0$ **is** _____

(a) Parabola

(b) Ellipse

(c) Hyperbola

(d) Circle

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(23) $\ell x + my + n = 0$ is tangent to the parabola $y^2 = 4ax$ then which of the following must be true?

(a) $n = \frac{a}{m}$

(b) $am^2 = n\ell$

(c) $\ell = a^2 m$

(d) $m = \ell a^2$

(24) Locus of the centre of the circle $x^2 + y^2 - 2\cos\theta y + 4\sin\theta x + 7 = 0$ is _____

(a) Circle

(b) Parabola

(c) Ellipse

(d) Hyperbola

(25) The turning point of the parabola $y = ax^2 + bx + c$ is _____

(a) $\left(-\frac{b}{2a}, \frac{b^2}{4a}\right)$

(b) $\left(\frac{-b}{2a}, \frac{c-b^2}{4a}\right)$

(c) $\left(\frac{-b}{2a}, \frac{c}{2a}\right)$

(d) None of these

(26) The graph of $x^2 + y^2 + 7 = 0$ is _____

(a) Circle

(b) Ellipse

(c) Parabola

(d) None of these

(27) Equation of normal at point $p(x_1, y_1)$ to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is;.

(a) $\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$

(b) $\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 - b^2$

(c) $\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 + b^2$

(d) $\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$

(28) The equation $\frac{x^2}{6-k} + \frac{y^2}{k-2} = 1$ represents an ellipse if;

(a) $k > 2$

(b) $k > 6$

(c) $2 < k < 6$

(d) $2 \le k \le 6$

(29) If in an ellipse the major axis is equal to twice the distance between foci, then eccentricity is;

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{1}{\sqrt{3}}$

(30) The line through the focus and $\perp$ to the directrix of the parabola is _____

(a) Latus Rectum

(b) Tangent

(c) Axis

(d) None of these

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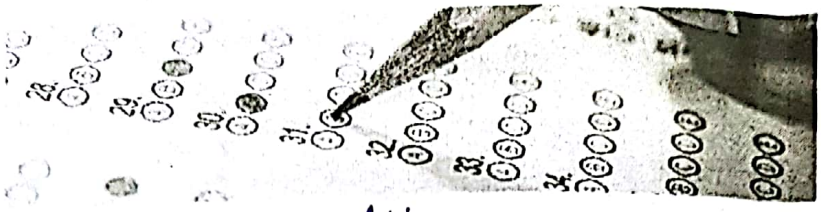
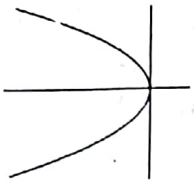
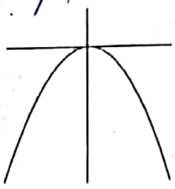
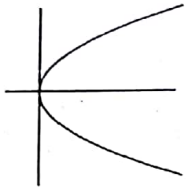
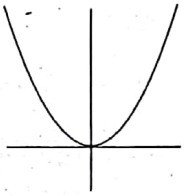


(31) **Which of the following is a graph of $y = 16x^2$?**

(a) 

(b) 

(c) 

(d) 

(32) $2x^2 - xy + 5x - 2y = 0$ **represents;**

(a) Parabola

(b) Hyperbola

(c) Circle

(d) Two intersecting lines

(33) **At what angle the axes are rotated so the xy-term is eliminated from** $4x^2 - 4xy + 7y^2 + 12x + 6y - 9 = 0$;

(a) $\tan^{-1}\left(\frac{4}{3}\right)$

(b) $2\tan^{-1}\left(\frac{4}{3}\right)$

(c) $\frac{1}{2}\tan^{-1}\left(\frac{4}{3}\right)$

(d) $\tan^{-1}\left(\frac{3}{4}\right)$

(34) **When xy-axes are rotated through an angle 45°, then $xy = 1$, is transformed to which of the following?**

(a) $X^2 + Y^2 = 1$

(b) $X^2 - Y^2 = 1$

(c) $X^2 + Y^2 = 2$

(d) $X^2 - Y^2 = 2$

(35) **The equations of latera recta of** $25x^2 - 16y^2 = 400$ **are;**

(a) $x = \pm\sqrt{34}$

(b) $y = \pm\sqrt{34}$

(c) $x = \pm\sqrt{41}$

(d) $y = \pm\sqrt{41}$

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ANSWER KEY

CHAPTER-1

1	c	12	c	23	c	34	c	45	a
2	a	13	b	24	d	35	d	46	a
3	d	14	c	25	d	36	b	47	c
4	c	15	b	26	a	37	a	48	c
5	b	16	a	27	d	38	b	49	d
6	a	17	a	28	a	39	c	50	d
7	c	18	c	29	c	40	c		
8	d	19	a	30	b	41	d		
9	c	20	b	31	b	42	d		
10	d	21	b	32	c	43	b		
11	d	22	d	33	b	44	b		

CHAPTER-2

1	c	12	d	23	b	34	c	45	d
2	a	13	b	24	d	35	b	46	c
3	b	14	d	25	b	36	d	47	b
4	d	15	a	26	a	37	a	48	a
5	d	16	c	27	d	38	a	49	c
6	c	17	b	28	d	39	b	50	b
7	b	18	c	29	a	40	c		
8	b	19	d	30	c	41	a		
9	a	20	a	31	c	42	c		
10	c	21	a	32	b	43	b		
11	b	22	c	33	d	44	d		

CHAPTER-3

1	d	12	a	23	a	34	b	45	c
2	c	13	c	24	a	35	c		
3	a	14	b	25	d	36	d		
4	c	15	b	26	b	37	a		
5	d	16	b	27	c	38	c		
6	a	17	c	28	c	39	d		
7	b	18	d	29	b	40	b		
8	b	19	c	30	c	41	a		
9	c	20	b	31	c	42	c		
10	c	21	d	32	c	43	c		
11	d	22	d	33	c	44	c		

CHAPTER-4

1	d	12	a	23	a	34	a	45	c
2	c	13	c	24	b	35	d		
3	b	14	c	25	c	36	d		
4	d	15	b	26	d	37	b		
5	d	16	d	27	b	38	c		
6	a	17	b	28	d	39	a		
7	b	18	b	29	c	40	b		
8	d	19	c	30	a	41	c		
9	d	20	b	31	b	42	d		
10	b	21	c	32	d	43	d		
11	a	22	b	33	d	44	c		

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CHAPTER-5+7

1	d	12	d	23	d	34	c	45	b
2	a	13	d	24	b	35	a		
3	d	14	d	25	a	36	a		
4	a	15	c	26	b	37	c		
5	c	16	b	27	d	38	b		
6	a	17	a	28	d	39	d		
7	d	18	c	29	d	40	a		
8	a	19	b	30	c	41	a		
9	d	20	d	31	b	42	b		
10	c	21	c	32	a	43	b		
11	b	22	b	33	b	44	a		

CHAPTER-6 Part A

1	b	12	c	23	c
2	d	13	b	24	b
3	c	14	b	25	a
4	b	15	d	26	d
5	a	16	c	27	a
6	c	17	b	28	a
7	c	18	c	29	c
8	c	19	d		
9	c	20	d		
10	b	21	a		
11	c	22	c		

CHAPTER-6 Part B

1	a	12	b	23	b	34	d
2	b	13	b	24	c	35	c
3	b	14	a	25	b		
4	b	15	d	26	d		
5	a	16	b	27	c		
6	d	17	c	28	d		
7	a	18	a	29	a		
8	c	19	d	30	c		
9	b	20	b	31	d		
10	d	21	a	32	b		
11	a	22	b	33	c		

$\sin\alpha\sin\beta = 2s$

$\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha$

$\cos 4\alpha = 4\cos^3\alpha - 3\cos\alpha$ $\quad \frac{a^2\sin\beta\sin\gamma}{2\sin\alpha}$

$\tan 3\alpha = \frac{3\tan\alpha - 3\tan^3\alpha}{1 - 3\tan^2\alpha}$ $\quad R = \frac{abc}{4\Delta}$

$R = \frac{a}{2\sin\alpha}$ $\quad \boxed{\gamma = \frac{\Delta}{s}}$

$\tan\alpha$

$\cos\alpha\cos\beta = 2\cos\frac{(\alpha+\beta)}{2} + \cos\frac{(\alpha-\beta)}{2}$

$\gamma_1 = \frac{\Delta}{s-a}$ $\quad \gamma = \Delta$

$\tan 3\alpha = \frac{3\tan\alpha - 3\tan^3\alpha}{}$

$\gamma_1 = \frac{3\tan\alpha - \tan^3\alpha}{1 - 3\tan^2\alpha}$

$\sin\frac{\alpha}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$

$\sin\frac{\beta}{2} = \sqrt{\frac{(s-a)(s-c)}{}}$

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